List R6 — Liquidity and risk management

Worked solutions

Statements are in `Listas/lista-r6-liquidity-risk-management.md`; each exercise is cited by number and printed page and its full text is reproduced right before its solution. Exercises 10 and 11 (Tirole 5.7 and 5.9) are the deepened solutions of the Topic 5 list (its exercises 8 and 9), included here from that file. Editions are the local copies: Tirole, *The Theory of Corporate Finance* (2005); Berk = Berk, DeMarzo & Harford, *Fundamentals of Corporate Finance*, 2nd ed. (2011); Grinblatt–Titman (G&T), *Financial Markets and Corporate Strategy*, 2nd ed. (2002); Ross = Ross, Westerfield, Jordan & Lamb, *Fundamentos de Administração Financeira*, 9th ed. (2013). Throughout the Tirole exercises, $\rho_1 \equiv p_H R$ is the expected income per continuation and $\rho_0 \equiv p_H(R - B/\Delta p)$ the expected pledgeable income per continuation.

Contents

1. Can a firm hold too much cash? — Ross CQ 19.1, p. 690 **1**

2. Excess cash and agency problems — Ross CQ 19.10, p. 690 **3**

3. Small firms hedge less — Grinblatt & Titman 21.1, p. 771 **4**

4. Partial hedging by oil firms — Grinblatt & Titman 21.7, p. 771 **6**

5. Entering the year with a cash buffer — Berk 20.6, p. 613 **7**

6. Long-term contract and loan commitment — Tirole Exercise 5.1, p. 229 **9**

7. Credit rationing, predation, and liquidity shocks — Tirole Exercise 5.2, pp. 229–230 **14**

8. Liquidity needs and pricing of liquid assets — Tirole Exercise 5.5, p. 231 **20**

9. Should debt contracts be indexed to output prices? — Tirole Exercise 5.11, pp. 234–235 **24**

10. Decreasing returns to scale and debt maturity — Tirole Exercise 5.7, p. 232 **29**

11. The priority game: uncoordinated lending and short-term bias — Tirole Exercise 5.9, p. 233 **35**

12. Two-factor hedge with no cash outlay — Grinblatt & Titman 22.12, p. 816 **38**

13. Continuous entrepreneurial effort and liquidity needs — Tirole Exercise 5.6, pp. 231–232 **39**

14. Liquidity and deepening investment — Tirole Exercise 5.10, p. 234 **43**

15. Multistage investment with interim news about prospects — Tirole Exercise 5.8, pp. 232–233 **48**

1. Can a firm hold too much cash? — Ross CQ 19.1, p. 690

Statement.

Source: Ross, Ch. 19, Exercise CQ 19.1 (Revisão de conceitos e questões instigantes 1), p. 690 (PDF 714) · Answer in book: no

Administração de caixa [OA3] É possível que uma empresa tenha muito caixa? Por que os acionistas se importariam se uma empresa acumulasse grandes quantidades de caixa?

What is asked. Two questions. First, whether a firm can hold *too much* cash, i.e. whether there is a level of cash beyond which more cash lowers the value of the firm. Second, why shareholders should care if the firm piles up large cash balances.

Approach. Use the cost–benefit view of cash in Ross §19.1: cash has benefits (transaction, precautionary and speculative motives; compensating balances) and a cost (the opportunity cost of funds earning little or nothing). Add the agency view of Ross §1.4 (managers like to control resources) and Tirole's liquidity view (cash has value only up to the liquidity needs the firm cannot cover by returning to the market). The optimal cash level balances the two; beyond it cash destroys value.

Notation.

- C: cash held above the amount needed for transactions (excess cash), \$.
- k: shareholders' required return on the firm's assets; r_c: pre-tax yield on cash or money-market assets, $r_c < k$; T: corporate tax rate on interest income.

Step 1 — Benefits of cash. Ross lists three motives (p. 662): *transaction* (pay bills when inflows and outflows are not synchronised), *precautionary* (a safety reserve), *speculative* (exploit bargains or attractive rates); plus compensating balances required by banks. Ross stresses that the precautionary and speculative motives call for *liquidity*, not necessarily cash: unused borrowing capacity and money-market securities serve them as well (pp. 662–663). In Tirole's model (Ch. 5) liquidity is valuable to a firm whose optimal cutoff ρ^* exceeds what it can raise by diluting, ρ_0: it must hoard $\rho^* - \rho_0$ (or hold a credit line) to survive shocks in $(\rho_0, \rho^*]$.

Step 2 — The cost of excess cash. Each dollar beyond those needs earns r_c, taxed at T at the corporate level, while shareholders require k. The annual cost of carrying C is

$$\text{cost} = \big[k - r_c(1-T)\big]\, C. \tag{1}$$

Illustration (own numbers): $C = 100$ (\$ millions), $k = 10\%$, $r_c = 4\%$, $T = 34\%$. The after-tax yield is $r_c(1-T) = 0.04(1-0.34) = 0.0264$, so

$$\text{cost} = (0.10 - 0.0264)(100) = 7.36 \text{ (\$ millions per year)}.$$

Shareholders could hold the same money-market assets themselves without the corporate tax on the interest, so the tax wedge alone is a loss to them.

Step 3 — Agency cost. Ross §1.4 (pp. 12–13): managers, left alone, tend to maximise the resources under their control, which leads to overemphasis on size and growth, perks (the "jatinho luxuoso"), and overpaying for acquisitions. Large cash balances are exactly the resources that let managers do this without the scrutiny of the capital market.

Answer (a). Yes. Beyond the amount needed for transactions, precaution and the firm's liquidity needs, extra cash lowers firm value.

Answer (b). Shareholders care because excess cash (i) earns less than their required return, after a corporate tax on the interest (\$7.36M a year in the illustration), and (ii) invites agency costs: managers may waste it on perks, empire building or overpriced acquisitions.

Intuition. Cash is insurance against having to raise funds at a bad moment. Like any insurance it is worth buying only up to the risk it covers; the premium is the opportunity cost of idle funds plus the room it gives managers to spend money that is not theirs.

Common mistakes.

- Saying "more cash is always safer, so it is always better": the question is about the *marginal* dollar, whose benefit falls to zero once liquidity needs are covered.

- Confusing cash with liquidity: a credit line or Treasury bills meet the precautionary motive without idle cash (Ross pp. 662–663; Tirole §5.2.3).
- Ignoring taxes: interest on corporate cash is taxed before shareholders get it.

Ref: Ross §19.1 "Motivos para manter saldos de caixa" and "Custos de manter caixa", pp. 662–663; §1.4 "O problema de agência e o controle da empresa", pp. 12–13; Tirole §5.2.3, pp. 205–206.

2. Excess cash and agency problems — Ross CQ 19.10, p. 690

Statement.

Source: Ross, Ch. 19, Exercise CQ 19.10 (Revisão de conceitos e questões instigantes 10), p. 690 (PDF 714) · Answer in book: no

Problemas de agência [OA3] Às vezes, afirma-se que o caixa excedente retido por uma empresa pode agravar os problemas de agência (discutidos no Capítulo 1) e, de modo geral, reduzir os incentivos para a maximização do valor para o acionista. Como você analisaria essa questão?

Context needed to solve it *From Ross, Ch. 1, §1.4 "O problema de agência e o controle da empresa", pp. 12–13 (PDF 36–37):*

O relacionamento entre acionistas e administradores é chamado de relacionamento de agência. Tal relacionamento existe sempre que alguém (o principal) contrata outra pessoa (o agente) para representar seus interesses. Por exemplo, você pode contratar alguém (um agente, representante seu) para vender um carro seu enquanto você está na escola. Nesses relacionamentos, existe a possibilidade de conflitos de interesses entre principal e agente. Tal conflito é um problema de agência.

[. . .]

Para ver como os interesses de administradores e acionistas podem ser diferentes, imagine que a empresa esteja considerando um novo investimento, que deve ter impacto favorável sobre o valor das ações mas que também é um empreendimento relativamente arriscado. Os proprietários da empresa vão querer fazer o investimento (porque o valor da ação subirá), mas os administradores podem não querer, porque existe a possibilidade de que as coisas não deem certo e eles percam seus empregos. Se a administração não aceitar o investimento, os acionistas podem perder uma oportunidade valiosa. Este é um exemplo de um custo de agência.

De modo mais geral, o termo custo de agência se refere aos custos do conflito de interesses entre acionistas e administradores. [. . .]

Os custos de agência podem ser diretos ou indiretos. Um custo de agência indireto é uma oportunidade perdida, como a que acabamos de descrever.

Os custos diretos de agência são de dois tipos. O primeiro é um gasto da empresa que beneficia a administração, mas que custa para os acionistas. A compra de um jatinho luxuoso e desnecessário pode ser vista como um gasto desse tipo. O segundo tipo é uma despesa que surge da necessidade de monitorar as ações dos administradores. O pagamento de auditores externos para avaliar a exatidão das informações das demonstrações financeiras seria um exemplo disso.

Às vezes, argumenta-se que, se deixados por conta própria, os administradores tenderiam a maximizar os recursos sobre os quais têm controle ou, de maneira geral, o poder ou a riqueza da empresa. Esse objetivo levaria a uma ênfase excessiva no tamanho ou no crescimento da empresa. Por exemplo, são comuns os casos em que uma administração é acusada de realizar gastos muito superiores ao valor de mercado na compra de outra empresa, apenas para aumentar o tamanho da empresa que controlam ou para demonstrar poder empresarial. Obviamente, se o gasto exagerado ocorrer realmente, tal compra não beneficiará os acionistas da empresa compradora.

What is asked. Evaluate the claim that excess cash retained by a firm worsens agency problems between managers and shareholders and weakens the incentive to maximise shareholder value: when is it right, why, and what can be done about it.

Approach. The agency view of Ross §1.4 and Jensen's free-cash-flow argument, formalised in Tirole §5.6.1: when the firm's internal cash exceeds what efficient reinvestment needs (the *free-cash-flow assumption* $r > \rho^*$), an entrepreneur left with the cash reinvests too much, and the optimal contract must "pump out" the excess through payouts or short-term debt. The opposing force is Tirole's liquidity-hoarding result: cash-poor firms need liquidity, and taking it away from them is costly.

Notation.

- Tirole §5.6.1 model (continuous investment I): rI short-term (date-1) income; ρI reinvestment need, $\rho \sim F$; ρ^* optimal continuation cutoff; P_1 date-1 payout to investors.

Step 1 — Why excess cash aggravates agency problems. Managers with discretionary cash need not justify spending to outside investors; they can (i) take negative-NPV projects that enlarge the firm (Ross p. 13: spending "muito superiores ao valor de mercado na compra de outra empresa, apenas para aumentar o tamanho"), (ii) consume perks, (iii) shirk on cost control because a cash cushion hides poor performance. Berk makes the same point about conservative financing: excess cash "increases the possibility that managers of the firm will use this excess cash nonproductively" (p. 599).

Step 2 — Tirole's formal version. Under the free-cash-flow assumption $r > \rho^*$, an entrepreneur who keeps the date-1 income reinvests whenever $\rho \leq r$, i.e. also for $\rho \in (\rho^*, r]$, where continuation destroys investors' value. The optimum requires pumping out

$$P_1 = (r - \rho^*)\, I \tag{2}$$

at date 1 and denying the entrepreneur the right to dilute initial investors (Tirole p. 226). Illustration (own numbers): $r = 0.6$, $\rho^* = 0.4$, $I = 100$ give $P_1 = (0.6 - 0.4)(100) = 20$ that must leave the firm, e.g. as short-term debt repayment.

Step 3 — Remedies. (i) Pay out excess cash (dividends, repurchases); (ii) commit to payouts with debt, whose repayment is not discretionary (short-term debt in Tirole §5.6.1); (iii) align compensation with value (stock-based pay, Ross §1.4); (iv) monitoring by the board, large shareholders, and the takeover market.

Step 4 — The other side. The claim does not mean cash is bad: a firm whose pledgeable income is below its efficient reinvestment needs ($\rho_0 < \rho^*$) must hoard liquidity or hold a credit line (Tirole §5.2.3). The agency cost bites only on *excess* cash, beyond those needs. The analysis must therefore distinguish cash-rich firms with few investment opportunities (Jensen's case) from cash-poor growth firms (Tirole's liquidity case).

Answer. The claim is correct for cash in excess of the firm's liquidity needs: it gives managers resources free of capital-market discipline, leading to overinvestment and perks. The cure is to commit to pay it out (payouts, debt), to align pay with value and to monitor; for firms whose liquidity needs are high, cash is valuable and the claim does not apply.

Intuition. Cash is a substitute for going back to investors. Not having to go back is valuable when going back is hard (liquidity), but harmful when going back is exactly the discipline managers need (agency). Debt maturity in Tirole §5.6 is the tool that trades the two off.

Common mistakes.

- Treating all retained cash as excess: liquidity hoarding by constrained firms is value-creating.
- Thinking dividends solve it once and for all: a one-off payout leaves future discretion intact; debt commits.
- Forgetting that the agency cost is borne by shareholders even if no money is "stolen": negative-NPV investment is the main channel.

Ref: Ross §1.4, pp. 12–13; §19.1, pp. 662–663; Berk §20.2 "Conservative Financing Policy", p. 599; Tirole §5.6.1 "Optimal Claim Maturity", p. 226.

3. Small firms hedge less — Grinblatt & Titman 21.1, p. 771

Statement.

Source: Grinblatt–Titman, Ch. 21, Exercise 21.1, p. 771 (PDF 784) · Answer in book: no

Small firms currently hedge less than large firms. Why is this? Do you expect smaller firms to start hedging more in the future? Explain.

What is asked. (a) Why do small firms currently hedge less than large firms? (b) Should we expect small firms to hedge more in the future, and why?

Approach. G&T §21.9 confronts the theory of why firms hedge (Results 21.3–21.6: taxes, financial distress, financing of investment, management evaluation) with the evidence. The theory says small firms have *more* to gain; the evidence that they hedge less is explained by the *costs* of hedging, which have a large fixed component. A fixed cost against a benefit proportional to size gives a size threshold.

Notation.

- S: size of the firm's hedgeable exposure (\$); b: value gain from hedging per \$ of exposure (lower expected distress costs, taxes, underinvestment); K: fixed cost of setting up a hedging operation (staff, systems, expertise, legal).

Step 1 — Benefits are larger for small firms. Small firms face higher bankruptcy costs relative to their size and harder access to external finance, so the distress and financing motives (Results 21.4 and 21.5, pp. 747–748) are stronger for them. Theory alone would predict *more* hedging by small firms (G&T p. 768).

Step 2 — Costs have a fixed component. G&T explain the evidence by "the fixed costs of setting up a hedging operation and their lower level of sophistication" (p. 768), and note the economies of scale in consolidating trades and in risk-management expertise (p. 755). With a fixed cost K and a benefit bS, the firm hedges iff

$$bS \geq K \quad \Longleftrightarrow \quad S \geq \frac{K}{b}. \tag{3}$$

Illustration (own numbers): $K = 0.5$ (\$ millions) and $b = 0.2\%$ give a threshold $S = 0.5/0.002 = 250$ (\$ millions): firms with smaller exposures do not hedge even though b may be larger for them.

Step 3 — Consistent evidence. Among firms that *do* hedge, smaller firms hedge more completely (Dolde, 1993, cited p. 768): once the fixed cost is paid, the larger marginal benefit of small firms shows up. "Size is a barrier to setting up a hedging operation."

Step 4 — The future. As derivatives markets become more liquid and risk-management expertise spreads (G&T p. 755: "As risk management expertise becomes more widespread and the futures, forward, and swap markets become more liquid…"), K falls and the threshold K/b falls with it. Banks also package hedges for small clients (e.g. swaps bundled with loans), which spreads the fixed cost.

Answer (a). Because hedging has fixed set-up costs and requires expertise, which small firms find relatively more expensive; this outweighs their larger benefit from hedging.

Answer (b). Yes: falling transaction costs and wider expertise lower the fixed cost, so smaller firms, which gain more from hedging, should hedge more.

Intuition. Hedging is like any technology with a fixed cost: it is adopted first by those who can spread the cost over a large base, and diffuses down as it gets cheaper, even though the late adopters may benefit most.

Common mistakes.

- Arguing that small firms hedge less because they face less risk: they usually face *more* distress risk.
- Ignoring the cost side: the benefit argument alone predicts the opposite of the evidence.
- Forgetting the intensive margin: small hedgers hedge *more completely*.

Ref: G&T §21.4 Results 21.4–21.5, pp. 745–748; §21.5 on where hedging is organised, p. 755; §21.9 "Which Firms Hedge? The Empirical Evidence", pp. 767–769.

4. Partial hedging by oil firms — Grinblatt & Titman 21.7, p. 771

Statement.

Source: Grinblatt–Titman, Ch. 21, Exercise 21.7, p. 771 (PDF 784) · Answer in book: no

Oil firms hedge only part of their exposure to oil price movements. Why might that be a good idea?

What is asked. Explain why it can be optimal for an oil firm to hedge only part of its exposure to oil prices instead of all of it.

Approach. G&T's "Desirability of Partial Hedging" (p. 748) and Example 21.3 (p. 749), and Tirole's incomplete-hedging result with positively correlated investment needs (§5.4.2): when the firm's investment opportunities and its access to external finance rise with the hedged risk factor, the firm wants cash to be high exactly when the factor is high. Full hedging removes that covariance. The firm hedges just enough to avoid distress in bad states.

Notation.

- G&T Example 21.3 (all in $ millions, 1 million barrels a year): spot price S at extraction, forward price 24/bbl, extraction cost 6/bbl, fixed obligations $3 + 8 = 11$ a year; $h \in [0, 1]$: fraction of output sold forward.

Step 1 — Why not hedge fully. Oil firms want to explore more when oil prices are high, and they can raise external capital on better terms after high earnings. Full hedging gives them the same cash whether prices are high or low, so they lack internal funds exactly when the best investments appear (G&T p. 748; Tirole §5.4.2: "attractive-reinvestment-opportunities effect"). Also, historically the forward price was often below spot, so hedging had a positive expected cost (G&T p. 749).

Step 2 — Why hedge at all. In low-price states, distress costs bite: the firm must still cover overhead and interest. The firm hedges to guarantee those fixed obligations. Annual operating cash with fraction h sold forward:

$$CF(S) = 24h + (1 - h)S - 6. \tag{4}$$

Guaranteeing 11 even if $S \to 0$ requires

$$\begin{aligned} 24h - 6 &= 11 \\ h &= \frac{17}{24} = 0.708333. \end{aligned} \tag{5}$$

Step 3 — Cash in each state. Substituting (5) in (4), $CF(S) = 17 + \frac{7}{24}S - 6 = 11 + \frac{7}{24}S$:

Policy	$S = 0$	$S = 24$	$S = 30$
No hedge, $S - 6$	-6	18	24
Full hedge, $24 - 6$	18	18	18
Partial, $11 + \frac{7}{24}S$	11	18	19.75

At $S = 30$ the unhedged 7/24 of output adds $\frac{7}{24}(30) = 8.75$. The partial hedge keeps the floor of 11 (no distress) and keeps exposure to the high-price states, where National would restart exploration ($35M investment if $S > 30$).

Answer. Partial hedging is optimal because the firm's investment needs and its ability to raise outside funds rise with oil prices; it hedges just enough to cover fixed obligations in low-price states (in Example 21.3, $h = 17/24$ of output) and keeps the upside to fund investment when prices are high. Options achieve the same floor-plus-upside shape.

Intuition. The aim of hedging is not to minimise variance but to have cash when it is most valuable. For an oil firm, cash is most valuable both in a price collapse (to avoid distress) and in a boom (to invest); a partial hedge, or a put option, serves both.

Common mistakes.

- Equating optimal hedging with minimum variance: here the variance-minimising full hedge is suboptimal.
- Forgetting that the guaranteed floor must cover *fixed* costs (overhead plus interest), not total costs.
- Ignoring the expected cost of hedging when forwards trade below spot (backwardation).

Ref: G&T §21.4 "The Desirability of Partial Hedging", p. 748, Example 21.3, p. 749; §21.9 (oil and gas evidence), p. 769; Tirole §5.4.2, pp. 216–219.

5. Entering the year with a cash buffer — Berk 20.6, p. 613

Statement.

Source: Berk, Ch. 20, Exercise 20.6 (with the shared table for Problems 3–7), p. 613 (PDF 644) · Answer in book: no

The following table includes quarterly working capital levels for your firm for the next year. Use it to answer Problems 3–7.

($000) Quarter	1	2	3	4
Cash	$100	$100	$100	$100
Accounts Receivable	200	100	100	600
Inventory	200	500	900	50
Accounts Payable	100	100	100	100

6. If you choose to enter the year with $400 total in cash, what is your maximum short-term borrowing (see MyFinanceLab for the data in Excel format)?

Context needed to solve it *From Berk, Ch. 20, §20.2 "The Matching Principle", pp. 598–599 (PDF 629–630):*

In a perfect capital market, the choice of financing is irrelevant; thus, how the firm chooses to finance its short-term cash needs cannot affect the firm's value. In reality, important market frictions exist, including transaction costs. For example, one transaction cost is the opportunity cost of holding cash in accounts that pay little or no interest. Firms also face high transaction costs if they need to negotiate a loan on short notice to cover a cash shortfall. Firms can increase their value by adopting a policy that minimizes these kinds of costs. One such policy is known as the matching principle. The matching principle states that short-term cash needs should be financed with short-term debt and long-term cash needs should be financed with long-term sources of funds.

Permanent Working Capital. Permanent working capital is the amount that a firm must keep invested in its short-term assets to support its continuing operations. Because this investment in working capital is required so long as the firm remains in business, it constitutes a long-term investment. The matching principle indicates that the firm should finance this permanent investment in working capital with long-term sources of funds. Such sources have lower transaction costs than short-term sources of funds that have to be replaced more often.

Temporary Working Capital. Another portion of a firm's investment in its accounts receivable and inventory is temporary and results from seasonal fluctuations in the firm's business or unanticipated shocks. This temporary working capital is the difference between the firm's actual level of investment in short-term working capital needs and its permanent working capital investment. Because temporary working capital represents a short-term need, the firm should finance this portion of its investment with short-term financing.

[. . .]

Conservative Financing Policy. Alternatively, a firm could finance its short-term needs with long-term debt, a practice known as a conservative financing policy. For example, when following such a policy, a firm would use long-term sources of funds to finance its fixed assets, permanent working capital, and some of its seasonal needs. The firm would use short-term debt very sparingly to meet its peak seasonal needs. To implement such a policy effectively, there will necessarily be periods when excess cash is available—those periods when the firm requires little or no investment in temporary working capital. In an imperfect capital market, this cash will earn a below-market interest rate, thereby reducing the firm's value. It also increases the

possibility that managers of the firm will use this excess cash nonproductively—for example, on perquisites for themselves.

What is asked. Given the quarterly working-capital forecast, find the largest short-term loan the firm will need during the year if it starts the year holding \$400 (thousand) of cash in total.

Approach. Berk's matching principle (§20.2): permanent working capital (the minimum level over the year) is financed long term; temporary working capital (the seasonal excess) is financed short term. Cash held above the required minimum is excess cash that funds temporary needs before any borrowing.

Notation.

- Quarter $t = 1, \dots, 4$; all amounts in \$000. $NWC_t = \text{Cash}_t + AR_t + \text{Inv}_t - AP_t$, where the table's cash row is the required (minimum) cash balance, as in Berk's Table 20.6.
- $P = \min_t NWC_t$: permanent working capital; $\tau_t = NWC_t - P$: temporary working capital.
- E_0: excess cash at the start of the year; STB_t: short-term borrowing in quarter t.

Assumption. \$400 "total in cash" includes the \$100 minimum balance of the table, so excess cash is $E_0 = 400 - 100$. The firm follows the matching principle: permanent working capital and the excess cash are financed long term.

Step 1 — Net working capital each quarter.

$$\begin{aligned} NWC_1 &= 100 + 200 + 200 - 100 = 400, \\ NWC_2 &= 100 + 100 + 500 - 100 = 600, \\ NWC_3 &= 100 + 100 + 900 - 100 = 1000, \\ NWC_4 &= 100 + 600 + 50 - 100 = 650. \end{aligned} \tag{6}$$

Step 2 — Permanent and temporary needs. The minimum is $P = \min(400, 600, 1000, 650) = 400$. Temporary needs are

$$\tau_t = NWC_t - 400 = (0,\ 200,\ 600,\ 250). \tag{7}$$

Step 3 — Excess cash and short-term borrowing. Excess cash is $E_0 = 400 - 100 = 300$. It is drawn down first, so

$$\begin{aligned} STB_t &= \max(\tau_t - E_0,\, 0) \\ &= \big(\max(0 - 300, 0),\ \max(200 - 300, 0),\ \max(600 - 300, 0),\ \max(250 - 300, 0)\big) \\ &= (0,\ 0,\ 300,\ 0). \end{aligned} \tag{8}$$

The cash balance is $100 + \max(E_0 - \tau_t, 0) = (400,\ 200,\ 100,\ 150)$: excess cash of $300, 100, 0, 50$. Long-term financing covers $P + E_0 = 400 + 300 = 700$.

Answer. Maximum short-term borrowing is \$300 thousand, in quarter 3 (versus \$600 thousand in Problem 5, when the firm holds only the \$100 minimum).

Intuition. Every dollar of excess cash carried into the year replaces a dollar of peak short-term borrowing. That is Berk's conservative policy: lower funding risk and transaction costs, paid for with idle cash at a below-market return in quarters 1, 2 and 4, and the agency risk of Exercise 2.

Common mistakes.

- Treating all \$400 as excess cash (borrowing $600 - 400 = 200$): \$100 is the required operating balance.
- Financing the permanent \$400 short term, then reading the peak need as 1000.
- Forgetting to subtract accounts payable when computing working capital.

Ref: Berk §20.2 "The Matching Principle", pp. 597–600 (Table 20.6, p. 598; conservative policy, p. 599).

6. Long-term contract and loan commitment — Tirole Exercise 5.1, p. 229

Statement.

Source: Tirole, Ch. 5, Exercise 5.1, p. 229 (PDF 242) · Answer in book: yes, p. 591

Exercise 5.1 (long-term contract and loan commitment). Consider the two-project, two-period version of the fixed-investment model of Section 3.2 and a unit discount factor. Assume, say, that the borrower initially has no equity ($A = 0$). Show the following.

(i) If $p_H(p_H R - I) + (p_H R - I - p_H B/\Delta p) \geqslant 0$, then the optimal long-term contract specifies a loan commitment in which the second-period project is financed at least if the first-period project is successful. Show that if $p_H(p_H R - I) + (p_H R - I - p_H B/\Delta p) > 0$, then the optimal long-term contract specifies that the second-period project is implemented with probability 1 in the case of first-period success, and with probability $\xi \in (0, 1)$ in the case of failure.

(ii) In question (i), look at how ξ varies with various parameters.

(iii) Is the contract "renegotiation proof," that is, given the first-period outcome, would the parties want to modify the contract to their mutual advantage?

(iv) Investigate whether the long-term contract outcome can be implemented through a sequence of short-term contracts where the first-period contract specifies that the borrower receives $\bar{A} = I - p_H(R - B/\Delta p)$ with probability 1 in the case of success and with probability ξ in the case of failure.

Context needed to solve it *From Tirole §3.2.1–3.2.2, pp. 115–117 (PDF 128–130):*

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. [...] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. [...]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit"); [...]

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. That is, we have in mind that several prospective lenders compete for issuing a loan to the borrower, and that, if the most attractive loan offer made a positive profit, the borrower could turn to an alternative lender and offer to switch for a slightly lower interest rate. [...]

Let us turn to the loan contract. A contract first stipulates whether the project is financed. If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure (the gross payoffs are the *ex post* monetary payoffs and take no account of past investments and private benefit). Intuitively, there is no point in specifying a positive transfer from the lenders to the borrower, as such a transfer can only weaken incentives, while it has no insurance benefit under risk neutrality. This property will be proved more rigorously and is here taken for granted. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

The zero-profit constraint for the lenders can be written as

$$p_H R_l = I - A,$$

assuming that the loan agreement induces the borrower to behave (which under our assumptions will be the case). [...]

We summarize the timing in Figure 3.1.

Figure 3.1 (timing, transcribed): Loan agreement (sharing rule) → Investment → Moral hazard → Outcome.

We assume that the project is viable only in the absence of moral hazard. That is, the project has positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \qquad (3.1)$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \qquad (3.2)$$

3.2.2 The Lenders' Credit Analysis. Because the project has negative NPV in the case of misbehavior, the loan agreement must be careful to preserve enough of a stake for the borrower in the enterprise. The borrower faces the following tradeoff once the financing has been secured: by misbehaving, she obtains private benefit B, but she reduces the probability of success from p_H to p_L. Because she has stake R_b in the firm's income (she receives R_b in the case of success and 0 in the case of failure), the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\text{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_H \left(R - \frac{B}{\Delta p} \right) \geqslant I - A, \qquad (\text{IR}_l)$$

where "IR_l" stands for the lenders' individual rationality constraint (which we will also often call the "breakeven constraint" or the "participation constraint"). Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_H \frac{B}{\Delta p} - (p_H R - I). \qquad (3.3)$$

To make things interesting, we will assume that

$$\bar{A} > 0 \iff p_H R - I < p_H \frac{B}{\Delta p}, \qquad (3.4)$$

otherwise even a borrower with no wealth of her own would be able to obtain credit. Condition (3.4) says that the NPV is smaller than the minimum expected rent that must be left to the borrower to provide her with an incentive to behave. [. . .]

Conversely, if $A \geqslant \bar{A}$, the entrepreneur is able to secure financing, and so condition (3.3) is both a necessary and a sufficient condition for financing.

From Tirole §4.2.4 "Sequential Projects: The Build-up of Net Worth", p. 163 (PDF 176):

Section 4.7, in the supplementary section, investigates the case of a sequence of two projects, project 1 at date 1 and project 2 at date 2. The key difference with the case of two "simultaneous projects" analyzed in Section 4.2.1 is that the outcome (success or failure) in the first project is realized before the investment in the second project needs to be sunk. The new feature is that the investment in the second project can be made contingent on the first project's outcome. In particular, the optimal contract may threaten the entrepreneur with nonrefinancing if the first project fails even though the projects are independent and so there is no learning about the second project's profitability from first-period performance.

From Tirole §4.7.2 "Long-Term Finance and the Build-up of Net Worth", p. 184 (PDF 197):

Let us now consider the sequential case, in which the outcome of the first project is known before investment is sunk in the second project: project 1 and its realization occur at date 1, project 2 and its

realization at date 2. To make the simultaneous and sequential cases comparable, we assume that there is no discounting between the two periods. We initially assume that the first loan agreement covers only the first project, and study how the build-up of equity motivates the entrepreneur. We then analyze the optimal long-term contract and ask whether there is scope for lender commitment of future financing.

What is asked. (i) With two sequential fixed-size projects and no equity, show that if a condition on pledgeable income holds the optimal long-term contract funds the second project at least after a first success, and that with strict inequality it funds it for sure after success and with probability $\xi \in (0,1)$ after failure. (ii) How ξ changes with the parameters. (iii) Whether the contract is renegotiation-proof after the first outcome. (iv) Whether the outcome can be obtained with a sequence of short-term contracts in which the borrower receives $\bar{A}$ after success (and with probability ξ after failure).

Approach. Tirole's sequential-projects model (§4.7.2) with the fixed-investment technology of §3.2. Continuation of the second project is an incentive device: the threat of not refinancing after failure disciplines first-period effort (the "loan commitment" is a promise to refinance). Write the borrower's rents as continuation values, find the cheapest way to satisfy both incentive constraints, then maximise NPV subject to investors' breakeven. For (iv) use the date-2 credit market of §3.2: a borrower is funded iff her cash is at least $\bar{A}$.

Notation.

- Each project costs I, returns R (success) or 0; success probability p_H (behave) or p_L (misbehave, private benefit B), $\Delta p = p_H - p_L$. $A = 0$; discount factor 1; investors competitive.
- $N \equiv p_H R - I > 0$: NPV of one project. $M \equiv p_H B/\Delta p$: minimum expected rent of a funded project. Assumption (3.4) with $A = 0$: $N < M$ (a single project cannot be financed).
- $\bar{A} \equiv I - p_H(R - B/\Delta p) = M - N > 0$: cash needed to finance one project.
- $x_k \in [0,1]$: probability of funding project 2 after first-period outcome $k \in \{S, F\}$.
- R_b^1: payment to the borrower after a first success; w_k: payment after a second success following outcome k; failures pay 0.
- $V_k \equiv x_k p_H w_k$: borrower's date-2 continuation value after outcome k.

Step 1 — Date-2 incentive constraint. If project 2 is funded after outcome k, the borrower behaves iff $p_H w_k \geq p_L w_k + B$, i.e.

$$(\Delta p)\, w_k \geq B \quad \Longrightarrow \quad V_k \geq x_k\, p_H \frac{B}{\Delta p} = x_k M. \tag{9}$$

Step 2 — Date-1 incentive constraint. Behaving at date 1 gives $p_H(R_b^1 + V_S) + (1 - p_H)V_F$; misbehaving gives $p_L(R_b^1 + V_S) + (1 - p_L)V_F + B$. Subtracting,

$$\begin{aligned} p_H(R_b^1 + V_S) + (1 - p_H)V_F - p_L(R_b^1 + V_S) - (1 - p_L)V_F &\geq B \\ (\Delta p)\,(R_b^1 + V_S - V_F) &\geq B. \end{aligned} \tag{10}$$

Step 3 — Objective and breakeven. Investors break even, so the borrower's utility is the NPV:

$$U_b = N\,[1 + p_H x_S + (1 - p_H)x_F]. \tag{11}$$

The borrower's expected rent is $\mathcal{R} = p_H(R_b^1 + V_S) + (1 - p_H)V_F$. Investors receive the NPV minus the rent, so breakeven (with $A = 0$) is

$$\mathcal{P} \equiv N\,[1 + p_H x_S + (1 - p_H)x_F] - \mathcal{R} \geq 0. \tag{12}$$

Step 4 — Minimum rent for given (x_S, x_F). Rewrite the rent as

$$\mathcal{R} = V_F + p_H\,(R_b^1 + V_S - V_F).$$

The first term is at least $x_F M$ by (9); the bracket is at least $B/\Delta p$ by (10). Both bounds are attained with $w_F = B/\Delta p$ and (10) binding, so

$$\begin{aligned}\mathcal{R}_{\min} &= x_F M + p_H \frac{B}{\Delta p} \\ &= M\,(1 + x_F).\end{aligned} \tag{13}$$

The minimum rent does not depend on x_S: refinancing after success is a free carrot. How the bracket is split between R_b^1 and V_S does not matter for the rent; take $R_b^1 = 0$ (reward only after two successes). With $x_S = 1$ and $R_b^1 = 0$, binding (10) means $p_H w_S = B/\Delta p + x_F M$, i.e.

$$w_S = \frac{B}{\Delta p}\,\frac{1 + x_F p_H}{p_H} \;\geq\; \frac{B}{\Delta p}, \tag{14}$$

so (9) holds after success too.

Step 5 — (i) $x_S = 1$. Substituting (13) into (12),

$$\mathcal{P}(x_S, x_F) = N\left[1 + p_H x_S + (1 - p_H) x_F\right] - M(1 + x_F). \tag{15}$$

$\partial \mathcal{P}/\partial x_S = p_H N > 0$ and $\partial U_b/\partial x_S = p_H N > 0$: raising x_S raises both NPV and pledgeable income, so $x_S = 1$. Then

$$\mathcal{P}(1, x_F) = N(1 + p_H) - M + x_F\big[(1 - p_H)N - M\big]. \tag{16}$$

The coefficient of x_F is negative because $N < M$: refinancing after failure raises NPV but costs pledgeable income.

Step 6 — (i) The two claims. With $x_F = 0$, (16) is nonnegative iff

$$(1 + p_H)N - M \geq 0 \quad\Longleftrightarrow\quad p_H(p_H R - I) + \Big(p_H R - I - \frac{p_H B}{\Delta p}\Big) \geq 0, \tag{17}$$

which is the condition of the statement: the contract "fund project 2 after success, pay $w_S = B/(p_H \Delta p)$ after two successes" is then feasible. If (17) is strict, breakeven is slack at $x_F = 0$, and since U_b rises with x_F at rate $(1 - p_H)N > 0$, the optimum raises x_F until (16) binds:

$$\begin{aligned} N(1 + p_H) - M + \xi\big[(1 - p_H)N - M\big] &= 0 \\ \xi\big[M - (1 - p_H)N\big] &= (1 + p_H)N - M \\ \xi &= \frac{(1 + p_H)N - M}{M - (1 - p_H)N}. \end{aligned} \tag{18}$$

$\xi > 0$ by the strict (17). $\xi < 1$ iff $(1 + p_H)N - M < M - (1 - p_H)N$, i.e. $2N < 2M$, which is (3.4).

Numerical example (own numbers): $p_H = 0.8$, $p_L = 0.3$, $B = 1$, $R = 5$, $I = 2.8$. Then $\Delta p = 0.5$, $N = 0.8(5) - 2.8 = 1.2$, $M = 0.8(1)/0.5 = 1.6$, $\bar{A} = 1.6 - 1.2 = 0.4$, and (3.2) holds: $p_L R - I + B = 1.5 - 2.8 + 1 = -0.3 < 0$. Condition (17): $1.8(1.2) - 1.6 = 0.56 > 0$. By (18),

$$\xi = \frac{0.56}{1.6 - 0.2(1.2)} = \frac{0.56}{1.36} = \frac{7}{17} = 0.411765.$$

By (14), $w_S = 2\,(1 + 0.411765 \times 0.8)/0.8 = 3.323529$, so $V_S = 0.8\, w_S = 2.658824$ and $V_F = \xi M = 0.658824$; check of (10): $0.5\,(2.658824 - 0.658824) = 1 = B$. Investors: $N + p_H(N - V_S) + (1 - p_H)\xi(N - M) = 1.2 - 1.167059 - 0.032941 = 0$. NPV $= 1.2\,(1 + 0.8 + 0.2 \times 0.411765) = 2.258824$, equal to the borrower's utility $0.8(2.658824) + 0.2(0.658824)$.

Answer (i). If $p_H(p_H R - I) + (p_H R - I - p_H B/\Delta p) \geq 0$, the optimal contract commits to fund project 2 after a first success ($x_S = 1$, reward $B/(p_H \Delta p)$ after two successes). If the inequality is strict, $x_S = 1$ and $x_F = \xi = \dfrac{(1 + p_H)N - M}{M - (1 - p_H)N} \in (0, 1)$ (0.4118 in the example).

Step 7 — (ii) Comparative statics. Write $D \equiv M - (1-p_H)N > 0$. Differentiating (18) (quotient rule) holding the other arguments fixed:

$$\frac{\partial \xi}{\partial N} = \frac{(1+p_H)D + (1-p_H)\big[(1+p_H)N - M\big]}{D^2} = \frac{(1+p_H)M - (1-p_H)M}{D^2} = \frac{2p_H M}{D^2} > 0, \tag{19}$$

$$\frac{\partial \xi}{\partial M} = \frac{-D - \big[(1+p_H)N - M\big]}{D^2} = \frac{-M + (1-p_H)N - (1+p_H)N + M}{D^2} = -\frac{2p_H N}{D^2} < 0, \tag{20}$$

$$\frac{\partial \xi}{\partial p_H}\Big|_{N,M} = \frac{N\,D - \big[(1+p_H)N - M\big]N}{D^2} = \frac{2N(M-N)}{D^2} > 0. \tag{21}$$

Chain rule: R enters through N ($\partial N/\partial R = p_H$), so ξ rises with R; I lowers N, so ξ falls with I; B raises M ($\partial M/\partial B = p_H/\Delta p$), so ξ falls with B; p_L raises M ($\partial M/\partial p_L = p_H B/\Delta p^2$), so ξ falls with p_L; p_H raises N, lowers M ($\partial M/\partial p_H = -p_L B/\Delta p^2$) and has the direct effect (21), all positive, so ξ rises with p_H. In the example, $\partial \xi/\partial N = 2(0.8)(1.6)/1.36^2 = 1.384083$.

Answer (ii). ξ increases with R and p_H, and decreases with I, B and p_L (it reaches 1 when $N \geq M$, and 0 when (17) binds).

Step 8 — (iii) Renegotiation after the first outcome. *After success:* the contract funds project 2, which is efficient ($N > 0$); any change is a pure transfer, so no change benefits both parties. *After failure, lottery says "stop"*: funding project 2 would give investors at most $N - M < 0$ (they must leave the borrower $B/\Delta p$ in case of success), and the borrower has no cash to compensate them (she received nothing). No deal. *After failure, lottery says "continue"*: stopping saves investors $M - N$ but costs the borrower her rent M; since $M - N < M$, investors cannot compensate her. No deal.

Answer (iii). Yes, the contract is renegotiation-proof (given that the lottery after failure is public and enforceable).

Step 9 — (iv) The proposed short-term sequence. At date 2, a borrower with cash A_2 is funded iff $A_2 \geq \bar{A}$ and then gets $A_2 + N$ (competitive investors, §3.2); with $A_2 = \bar{A}$ this is $\bar{A} + N = M$. The proposed date-1 contract gives $\bar{A}$ after success and $\bar{A}$ with probability ξ after failure, so $V_S = M$, $V_F = \xi M$, and the date-1 incentive constraint (10) (with R_b^1 absorbed in V_S) reads

$$(\Delta p)\,(M - \xi M) = (1-\xi)\,p_H B \;\geq\; B\;?$$

Since $(1-\xi)p_H < 1$, it fails: the borrower would shirk in period 1 (example: $0.5 \times (1 - 0.411765) \times 1.6 = 0.470588 < 1$). The long-term contract needs $V_S - V_F = B/\Delta p$, and $\bar{A}$ after success delivers only $(1-\xi)M < B/\Delta p$.

Step 10 — (iv) A sequence that works. Pay instead, after success, the cash A^S that gives the long-term continuation value $V_S = p_H w_S$:

$$\begin{aligned} A^S + N &= \frac{B}{\Delta p}(1 + \xi p_H) \\ A^S &= \frac{B}{\Delta p}(1 + \xi p_H) - N = \bar{A} + \frac{B}{\Delta p}(1 - p_H + \xi p_H). \end{aligned} \tag{22}$$

(The last step uses $\bar{A} = M - N = p_H B/\Delta p - N$.) After failure keep $\bar{A}$ with probability ξ. Then V_S and V_F equal the long-term values, (10) binds, date-2 investments coincide, and date-1 investors get $p_H R - I - p_H A^S - (1-p_H)\xi\bar{A}$, which equals (16) at $x_F = \xi$, i.e. zero. In the example $A^S = 2.658824 - 1.2 = 1.458824$ instead of $\bar{A} = 0.4$.

Answer (iv). Not with $\bar{A}$ after success: that leaves too little stake, $(1-\xi)p_H B < B$, and the borrower shirks at date 1. A sequence of short-term contracts does implement the long-term outcome

if the date-1 contract leaves her $A^S = \bar{A} + \frac{B}{\Delta p}(1 - p_H + \xi p_H)$ after success (and $\bar{A}$ with probability ξ after failure): with risk neutrality the loan commitment can be replaced by retained earnings.

Intuition. Refinancing after success is the cheapest carrot: it adds NPV and costs no extra rent, because the reward for two successes does double duty for both periods' incentives. Refinancing after failure is insurance; it is bought only as far as pledgeable income allows, because each unit of it raises the borrower's rent by M while adding only $(1 - p_H)N$ of value.

Common mistakes.

- Forgetting that the date-1 incentive constraint is in *continuation values*: $V_S - V_F \geq B/\Delta p$, not $w_S \geq B/\Delta p$.
- Setting $x_F = 1$ "to insure the borrower": it violates breakeven when $N < M$.
- Using the date-2 financing threshold $\bar{A}$ as if it were enough to motivate date-1 effort.

Book answer. (i) Matches the book's answer (p. 591), including the reward $B/(p_H \Delta p)$ after two successes. (ii) Same signs (the book adds "as long as p_L is not too large"; in the interior we find ξ decreasing in p_L without restriction). The book's equation for ξ has $(1 - \xi)(\Delta p)p_H B/\Delta p$ where the rent computation gives $(1 - \xi)p_H\, p_H B/\Delta p$; as printed it does not reduce to the condition of (i) at $\xi = 0$ (it differs from (16) by $(1 - \xi)p_L M$, 0.48 at $\xi = 0$ in the example), so we read it as a typo; (18) is the corrected solution. (iii) Matches. (iv) The book calls the sequence "behaviorally equivalent"; Step 9 shows that with exactly $\bar{A}$ after success the period-1 incentive constraint fails, and Step 10 gives the payout that makes the equivalence hold.

Ref: Tirole §3.2.1–3.2.2, pp. 115–117; §4.7.2 "Long-Term Finance and the Build-up of Net Worth", pp. 184–186; Exercise 5.1, p. 229; answer p. 591.

7. Credit rationing, predation, and liquidity shocks — Tirole Exercise 5.2, pp. 229–230

Statement.

Source: Tirole, Ch. 5, Exercise 5.2, p. 229 (PDF 242) · Answer in book: yes, p. 591

Exercise 5.2 (credit rationing, predation, and liquidity shocks). (i) Consider the fixed-investment model. An entrepreneur has cash A and can invest $I_1 > A$ in a project. The project's payoff is R_1 in the case of success and 0 otherwise. The entrepreneur can work, in which case her private benefit is 0 and the probability of success is p_H, or shirk, in which case her private benefit is B_1 and the probability of success p_L. The project has positive NPV ($p_H R_1 > I_1$), but will not be financed if the contract induces the entrepreneur to shirk. The (expected) rate of return demanded by investors is 0.

What is the threshold value of A such that the project is financed?

In the following, let

$$\rho_0^1 \equiv p_H \left(R_1 - \frac{B_1}{\Delta p} \right).$$

The next three questions add a prior period, period 0, in which the entrepreneur's equity A is determined. The discount factor between dates 0 and 1 is equal to 1.

(ii) In this question, the entrepreneur's date-1 (entire) equity is determined by her date-0 profit. This profit can take one of two values, a or A, such that

$$a < I_1 - \rho_0^1 < A.$$

At date 0, the entrepreneur faces a competitor in the product market. The competitor can "prey" or "not prey." The entrepreneur's date-0 profit is a in the case of predation and A in the absence of predation. Preying reduces the competitor's profit at date 0, but by an amount smaller than the competitor's date-1 gain from the entrepreneur's date-1 project not being funded.

- What happens if the entrepreneur waits until date 1 to go to the capital market?

- Can the entrepreneur avoid this outcome? You may want to think about a credit line from a bank. Would such a credit line be credible, that is, would it be renegotiated to the mutual advantage of the entrepreneur and his investors at the end of date 0?

(iii) Forget about the competitor, but keep the assumption that the entrepreneur's date-0 profit can take the same two values, a and A. We now introduce a date-0 moral-hazard problem on the entrepreneur's side.

Assume that the entrepreneur's date-0 production involves an investment cost I_0 and that the entrepreneur initially has no cash. The entrepreneur can work or shirk at date 0. Working yields no private benefit and probability of profit A equal to q_H (and probability $1 - q_H$ of obtaining profit a). Shirking yields private benefit B_0 to the entrepreneur, but reduces the probability of profit A to $q_L = q_H - \Delta q$ ($0 < q_L < q_H < 1$). Assume that

$$I_1 + I_0 - (q_L A + (1 - q_L)a) > \rho_0^1.$$

- Interpret this condition.

Consider the following class of long-term contracts between the entrepreneur and investors. "The date-1 project is financed with probability 1 if the date-0 profit is A and with probability $x < 1$ if this profit is a. The entrepreneur receives $R_b = B_1/\Delta p$ if the date-1 project is financed *and* succeeds, and 0 otherwise." Assume that such contracts are not renegotiated.

- What is the optimal probability x^*? (Assume that $(\Delta q)p_H B_1 \geqslant (\Delta p)B_0$.)
- Assuming that $\rho_0^1 > I_1$, is the previous contract robust to (a mutually advantageous) renegotiation?

(iv) Show that the entrepreneur cannot raise sufficient funds at date 0 if renegotiation at the end of date 0 cannot be prevented, if $\rho_0^1 > I_1$, and if

$$I_0 + I_1 - (q_H A + (1 - q_H)a) > \rho_0^1 - \left(\frac{q_H B_0}{\Delta q}\right).$$

Context needed to solve it *From Tirole §3.2.1, pp. 115–116 (PDF 128–129):*

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. [...] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. [...]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit"); [...]

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Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

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The zero-profit constraint for the lenders can be written as

$$p_H R_l = I - A,$$

assuming that the loan agreement induces the borrower to behave (which under our assumptions will be the case). [...]

We summarize the timing in Figure 3.1.

Figure 3.1 (timing, transcribed): Loan agreement (sharing rule) → Investment → Moral hazard → Outcome.

We assume that the project is viable only in the absence of moral hazard. That is, the project has positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \qquad (3.1)$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \qquad (3.2)$$

From Tirole §5.2.1, p. 202 (PDF 215):

Consider the setup of Section 3.2, except that there is an intermediate date at which income accrues and some reinvestment need is realized. As indicated in Figure 5.1, the entrepreneur at date 0 has wealth A and borrows $I - A$, where I is the fixed cost of investment.

At date 1, the investment yields deterministic and verifiable income $r \geqslant 0$. Continuation, though, requires reinvesting an amount ρ, where ρ is *ex ante* unknown and has cumulative distribution function $F(\rho)$ with density $f(\rho)$ on $[0, \infty)$. The realization of ρ is learned at date 1. Note that we here assume that the date-1 income is deterministic while the reinvestment need is random. The important assumption is that at least one of the two is random.

If the firm does not reinvest ρ, then the firm is liquidated. The liquidation value is 0. If the firm reinvests ρ, then the firm yields, at date 2, R with probability p and 0 with probability $1 - p$, where $p = p_H$ if the entrepreneur behaves (and then gets no private benefit) and $p = p_L = p_H - \Delta p$ if the entrepreneur misbehaves (in which case she receives private benefit B).

The entrepreneur and the investors are risk neutral, the entrepreneur is protected by limited liability, and the investors demand a rate of return equal to 0.

Thus, the model is nothing but an extension of the basic fixed-investment one in Section 3.2. We have just added an intermediate income r and a reinvestment need ρ (the bold type in Figure 5.1). (Put differently, the model of Section 3.2 corresponds to the special case $r = 0$ and F being a spike at $\rho = 0$.) We assume that there exists in the economy a store of value that yields the consumers' rate of interest (0 here). That is, 1 unit invested at date 0 delivers a return of 1 unit at date 1 (Chapter 15 will investigate the reasonableness of this assumption). We now give a *heuristic* description of the optimal contract.

Figure 5.1 (timing, transcribed): **0** — Entrepreneur has wealth A and fixed-investment project costing $I > A$. **1** — **Short-term income** $r > 0$**. Reinvestment need** ρ **(drawn from** $F(\cdot)$**).** Then, (if reinvestment): Moral hazard ($p = p_H$ or p_L). **2** — Success (profit R) with probability p, failure (profit 0) with probability $1 - p$.

Suppose in a first step that the initial contract can specify whether the firm continues or liquidates for each value of ρ (as we will see, it actually does not matter whether the realized value of ρ is verifiable, as long as there is no use that can be made of the date-1 cash flow besides reinvesting it and distributing it to investors). Intuitively, it is optimal to continue whenever it is cheap to do so:

$$\rho \leqslant \rho^*,$$

where ρ^* is a cutoff.

From Tirole §5.2.3 "Credit Lines for Cash-Poor Firms", pp. 205–206 (PDF 218–219):

Suppose in contrast that the investment "takes a long time" to produce income. At the extreme, there is no short-term profit: $r = 0$.

Can the entrepreneur just "wait and see," that is, borrow I at date 0 in exchange for shares in the firm and return to the capital market at date 1 if need occurs? Let us thus assume that the entrepreneur does not plan her liquidity in advance and that the liquidity shock occurs at date 1. To raise cash on the capital market to pay ρ, the entrepreneur must issue new shares and thereby dilute historical investors. Letting $\rho_0 \equiv p_H(R - B/\Delta p)$, and to illustrate this dilution, suppose that the entrepreneur faces a liquidity shock $\rho = \frac{1}{2}\rho_0$. The value of external shares held by initial investors is equal to ρ_0. Suppose that the number of shares is doubled. That is, as many shares are sold to new investors as already exist. So the value of each share is halved. The firm thereby raises $\frac{1}{2}\rho_0 = \rho$ in cash and can withstand the liquidity shock. Are initial

investors willing to let themselves be diluted? The value of their shares is, of course, reduced to $\frac{1}{2}\rho_0$. But contemplate the alternative of liquidating the investment, under which the initial investors receive nothing! Thus, initial investors are willing to accept the dilution.

Similarly, to meet a liquidity shock equal to $\frac{3}{4}\rho_0$, the firm must quadruple the number of shares, and so on. But there is an upper bound to this process: investors will never pay more than the firm is worth to them. Hence, even in a frictionless capital market, the firm cannot raise more than ρ_0. Going back to the capital market at date 1 then at best allows the firm to withstand a liquidity shock of magnitude

$$\rho \leqslant \rho_0 = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the optimal financing arrangement specifies

$$p_H \left(R - \frac{B}{\Delta p} \right) < \rho^* \leqslant p_H R,$$

the entrepreneur must secure a line of credit or hoard liquidity in order to face the date-1 liquidity shock. We will shortly describe how to do so, but there are basically two alternatives and combinations thereof: a credit line or liquid assets of magnitude ρ^* with no right to dilute existing claimholders by issuing new claims at date 1 (so the entrepreneur borrows $I + \rho^*$); or a smaller credit line or amount of liquid assets, equal to $[\rho^* - p_H(R - B/\Delta p)]$ with a right to dilute claimholders as needed to ensure continuation. Either way, the entrepreneur must plan liquidity management.

The optimum can be implemented by a nonrevokable line of credit granted by, say, one of the lenders (a bank) at level ρ^*. It is important that this line of credit be nonrevokable (in a broad sense: see below). Otherwise the lender would have an incentive not to abide by his promise to rescue the firm if $\rho > \rho_0$, that is, if the liquidity shock exceeds the date-1 pledgeable income ρ_0. [...]

Remark (renegotiation). Could this line of credit be renegotiated to the parties' mutual advantage once the fraction ρ is realized? First, note that if $\rho \leqslant \rho^*$, then *a fortiori* $\rho < p_H R$ and therefore it is *ex post* efficient to continue; so there is no scope for a renegotiation in which the lender would compensate the borrower for not using the credit line, as this renegotiation would reduce total surplus and therefore at least one of the parties would be strictly better off not renegotiating. Second, could the two parties both benefit from an increase in the line of credit to ρ when $\rho^* < \rho \leqslant p_H R$? Even though this increase would yield the ex post efficient reinvestment policy, there is no way for the borrower to compensate the lender, again because the borrower's stake is incompressible. One can show that the lender will turn down any request for an increase in the credit line. So will any alternative lender (other lenders may have even less incentives to refinance, because unlike the initial lender they do not have a vested stake to lose).

What is asked. (i) The minimum cash A for the date-1 project to be financed. (ii) With a predatory competitor at date 0: what happens if the entrepreneur waits until date 1 to raise funds, how a credit line can avoid it, and whether that credit line would be renegotiated. (iii) With date-0 moral hazard: interpret the stated condition, find the optimal refinancing probability x^* after a low profit, and check robustness to renegotiation when $\rho_0^1 > I_1$. (iv) Show that funding fails if renegotiation cannot be prevented, under the stated condition.

Approach. (i) is the fixed-investment credit analysis of §3.2. (ii) uses the logic of credit lines (§5.2.3): a commitment signed before the shock removes the competitor's incentive to prey, and it is credible because continuation is ex post efficient. (iii)–(iv) use termination as an incentive device (as in Exercise 6) and show that renegotiation destroys it when refinancing is ex post profitable for investors, so incentives must be given with costlier money rewards.

Notation.

- Date-1 project: cost I_1, return R_1 or 0, success p_H / p_L, private benefit B_1, $\Delta p = p_H - p_L$; $p_H R_1 > I_1$. $\rho_0^1 \equiv p_H(R_1 - B_1/\Delta p)$: date-1 pledgeable income.
- Date-0 profit A (high) or a (low); I_0: date-0 cost; q_H/q_L: probability of A if the entrepreneur works/shirks at date 0, $\Delta q = q_H - q_L$; B_0: date-0 private benefit.
- x: probability of funding the date-1 project after profit a; $\mathcal{R}_b$: date-1 stake after profit A in (iv).

Step 1 — (i) Credit analysis. The borrower behaves iff $p_H R_b \geq p_L R_b + B_1$, i.e. $(\Delta p)R_b \geq B_1$. Investors get $R_1 - R_b$ in case of success and must recover $I_1 - A$:

$$p_H(R_1 - R_b) \geq I_1 - A.$$

The left side is largest at the smallest incentive-compatible stake $R_b = B_1/\Delta p$:

$$p_H\Big(R_1 - \frac{B_1}{\Delta p}\Big) = \rho_0^1 \geq I_1 - A \quad \Longleftrightarrow \quad A \;\geq\; I_1 - \rho_0^1. \tag{23}$$

Answer (i). The project is financed iff $A \geq I_1 - \rho_0^1$.

Step 2 — (ii) Waiting until date 1. If the entrepreneur waits, the competitor compares its date-0 cost of preying with its date-1 gain from the entrepreneur's project not being funded; the gain is larger by assumption. If it preys, date-0 profit is $a < I_1 - \rho_0^1$, so by (23) the date-1 project is not funded, which is what made preying profitable. So the competitor preys and the entrepreneur loses the NPV $p_H R_1 - I_1$. Illustration (own numbers): $p_H = 0.9$, $p_L = 0.1$, $R_1 = 10$, $B_1 = 2$, so $\Delta p = 0.8$, $B_1/\Delta p = 2.5$ and $\rho_0^1 = 0.9(10 - 2.5) = 6.75$; with $I_1 = 8$ the threshold is $8 - 6.75 = 1.25$, and $a = 0.5 < 1.25 < A = 2$ (shirking has negative NPV: $p_L R_1 + B_1 = 3 < I_1$).

Step 3 — (ii) Credit line. At date 0, before the competitor moves, the entrepreneur secures a commitment to fund the date-1 project whatever her date-0 profit: a credit line covering the shortfall in the low state,

$$L = I_1 - \rho_0^1 - a \quad (= 8 - 6.75 - 0.5 = 0.75 \text{ in the illustration}), \tag{24}$$

on top of the ρ_0^1 she can raise against the project. Funding is then guaranteed, predation has no date-1 payoff, so the competitor does not prey; the line is not drawn on the equilibrium path.

Step 4 — (ii) Credibility. Suppose profit is a and the bank would like to cancel. Cancelling saves the lenders $I_1 - a - \rho_0^1$ (their loss on the funded project) and costs the entrepreneur her rent net of her cash, $p_H B_1/\Delta p - a$. The difference

$$\Big(\frac{p_H B_1}{\Delta p} - a\Big) - (I_1 - a - \rho_0^1) = \frac{p_H B_1}{\Delta p} + p_H\Big(R_1 - \frac{B_1}{\Delta p}\Big) - I_1 = p_H R_1 - I_1 > 0$$

is the NPV that cancelling destroys: the lenders cannot compensate the entrepreneur (illustration: $2.25 - 0.5 = 1.75$ versus 0.75; the gap is $p_H R_1 - I_1 = 1$). Renegotiation never benefits both sides.

Answer (ii). Waiting invites predation and the project is not funded. A date-0 credit line of $I_1 - \rho_0^1 - a$ (or a guarantee of date-1 funding) deters predation; it is credible because funding is ex post efficient ($p_H R_1 > I_1$), so it is not renegotiated.

Step 5 — (iii) Interpreting the condition. $I_1 + I_0 - (q_L A + (1 - q_L)a) > \rho_0^1$ says: if the entrepreneur shirks at date 0 and the date-1 project is funded unconditionally, investors' outlay $I_0 + I_1$ exceeds what they can recover, expected date-0 profit under shirking plus date-1 pledgeable income. So the contract must induce date-0 effort.

Step 6 — (iii) Optimal x^*. Under the contract class (stake $B_1/\Delta p$ only on date-1 success), the entrepreneur's expected utility is the probability of funding times $p_H B_1/\Delta p$. Working beats shirking iff

$$\begin{aligned} [q_H + (1 - q_H)x]\frac{p_H B_1}{\Delta p} &\geq [q_L + (1 - q_L)x]\frac{p_H B_1}{\Delta p} + B_0 \\ (\Delta q)(1 - x)\frac{p_H B_1}{\Delta p} &\geq B_0. \end{aligned} \tag{25}$$

NPV rises with x (each funding adds $p_H R_1 - I_1 > 0$), so x is set as high as (25) allows:

$$1 - x^* = \frac{(\Delta p)B_0}{(\Delta q)p_H B_1}$$
$$x^* = 1 - \frac{(\Delta p)B_0}{(\Delta q)p_H B_1} \in [0, 1), \tag{26}$$

nonnegative by the assumption $(\Delta q)p_H B_1 \geq (\Delta p)B_0$. Investors must break even, $q_H A + (1 - q_H)a + [q_H + (1 - q_H)x^*](\rho_0^1 - I_1) \geq I_0$; when $\rho_0^1 < I_1$ this may instead cap x below (26). Illustration (own numbers, now $I_1 = 5 < \rho_0^1 = 6.75$): $q_H = 0.75$, $q_L = 0.25$, $B_0 = 0.45$, $A = 4$, $a = 0$, $I_0 = 4.2$. The assumption holds ($0.5 \times 0.9 \times 2 = 0.9 \geq 0.8 \times 0.45 = 0.36$); $x^* = 1 - 0.36/0.9 = 0.6$; the condition of (iii) holds ($5 + 4.2 - 1 = 8.2 > 6.75$); breakeven: $3 + (0.75 + 0.25 \times 0.6)(6.75 - 5) = 3 + 0.9 \times 1.75 = 4.575 \geq 4.2$.

Step 7 — (iii) Renegotiation when $\rho_0^1 > I_1$. In the state "profit a, project not funded" (probability $(1 - q_H)(1 - x^*)$), investors can fund the project and earn $\rho_0^1 - I_1 > 0$, while the entrepreneur earns $p_H B_1/\Delta p > 0$: both gain. The threat of non-refinancing is not credible.

Answer (iii). The condition means that unconditional funding with date-0 shirking cannot repay investors. $x^* = 1 - (\Delta p)B_0/[(\Delta q)p_H B_1]$ (date-0 incentive constraint binding). If $\rho_0^1 > I_1$, the contract is not renegotiation-proof: after a low profit both sides gain from funding anyway.

Step 8 — (iv) Incentives without the termination threat. If renegotiation cannot be prevented and $\rho_0^1 > I_1$, the date-1 project is always funded. Date-0 effort must be rewarded with money: stake $\mathcal{R}_b$ after profit A and $R_b = B_1/\Delta p$ after a. Working pays iff

$$q_H p_H \mathcal{R}_b + (1 - q_H)p_H R_b \geq q_L p_H \mathcal{R}_b + (1 - q_L)p_H R_b + B_0$$
$$(\Delta q)\, p_H(\mathcal{R}_b - R_b) \geq B_0. \tag{27}$$

The extra stake is paid with probability q_H, so at the minimum ($p_H(\mathcal{R}_b - R_b) = B_0/\Delta q$) expected date-1 pledgeable income falls to

$$\rho_0^1 - q_H p_H(\mathcal{R}_b - R_b) = \rho_0^1 - \frac{q_H B_0}{\Delta q}. \tag{28}$$

Funding with effort requires $q_H A + (1 - q_H)a + \rho_0^1 - q_H B_0/\Delta q \geq I_0 + I_1$, which the stated condition violates; funding with shirking is ruled out by the condition of (iii). A date-0 cash bonus instead of $\mathcal{R}_b$ costs the same, $q_H B_0/\Delta q$. Illustration: with the numbers of Step 6, $q_H B_0/\Delta q = 0.75 \times 0.45/0.5 = 0.675$ and $\rho_0^1 - 0.675 = 6.075$; the condition of (iv) holds ($4.2 + 5 - 3 = 6.2 > 6.075$), and investors get $3 + 6.075 = 9.075 < 9.2 = I_0 + I_1$ (with shirking, $1 + 6.75 = 7.75$), although the committed contract of Step 6 was fundable.

Answer (iv). With renegotiation, incentives must come from a higher stake after a good date-0 profit, which cuts pledgeable income by $q_H B_0/\Delta q$; under the stated condition investors cannot break even, so no funds are raised at date 0.

Intuition. The threat to cut credit is the cheapest incentive device, but it is only as good as the lender's commitment to carry it out. When refinancing is ex post profitable for lenders, they will not carry it out, and the firm must pay with money, which uses up pledgeable income. The same commitment, in (ii), is valuable for the opposite reason: a credit line is credible because carrying it out is ex post efficient.

Common mistakes.

- In (ii), saying the credit line is costly because it is drawn: in equilibrium the competitor does not prey and the line is not used.
- In (iii), writing the date-0 incentive constraint without the factor $p_H B_1/\Delta p$: what the entrepreneur loses when not refinanced is her rent, not the project's NPV.
- In (iv), forgetting that the money reward is paid with probability q_H, not 1.

Book answer. Matches the book's answer (pp. 591–592): threshold $\rho_0^1 \geq I_1 - A$; credit line $I_1 - \rho_0^1 - a$ not renegotiated because $p_H R_1 > I_1$; x^* from $(\Delta q)(1-x^*)(p_H B_1/\Delta p) \geq B_0$; renegotiation gains $\rho_0^1 - I_1$ and $p_H B_1/\Delta p$; pledgeable income $\rho_0^1 - q_H B_0/\Delta q$ in (iv).

Ref: Tirole §3.2, pp. 115–117; §5.2.3 "Credit Lines for Cash-Poor Firms", pp. 205–206; Exercise 5.2, pp. 229–230; answer pp. 591–592.

8. Liquidity needs and pricing of liquid assets — Tirole Exercise 5.5, p. 231

Statement.

Source: Tirole, Ch. 5, Exercise 5.5, p. 231 (PDF 244) · Answer in book: yes, p. 592

Exercise 5.5 (liquidity needs and pricing of liquid assets). Consider the liquidity-needs model with a *fixed investment* and *two possible liquidity shocks*. The borrower has cash A and wants to finance a fixed-size investment $I > A$ at date 0. At date 1, a cash infusion equal to ρ is needed in order for the project to continue. If ρ is not invested at date 1, the project stops and yields nothing. If ρ is invested, the borrower chooses between working (no private benefit, probability of success p_H) and shirking (private benefit B, probability of success $p_L = p_H - \Delta p$). The project then yields, at date 2, R in the case of success and 0 in the case of failure.

The liquidity shock is equal to ρ_L with probability $(1-\lambda)$ and to ρ_H with probability λ, where

$$\rho_L < \rho_0 < \rho_H < \rho_1,$$

where $\rho_1 \equiv p_H R$ and $\rho_0 \equiv p_H(R - B/\Delta p)$. Assume further that

$$\rho_0 - \rho_L > I - A. \tag{1}$$

There is a single liquid asset, Treasury bonds. A Treasury bond yields 1 unit of income for certain at date 1 (and none at dates 0 and 2). It is sold at date 0 at price $q \geqslant 1$. (The investors' rate of time preference is equal to 0.)

(i) Suppose that the firm has the choice between buying enough Treasury bonds to withstand the high liquidity shock and buying none. Show that it chooses to hoard liquidity if

$$(q-1)(\rho_H - \rho_0) \leqslant (1-\lambda)(\rho_0 - \rho_L) - \lambda(\rho_H - \rho_0) - I + A \tag{2}$$

and

$$(q-1)(\rho_H - \rho_0) \leqslant \lambda(\rho_1 - \rho_H). \tag{3}$$

(ii) Suppose that the economy is composed of a continuum, with mass 1, of identical firms with characteristics as described above. The liquidity shocks of the firms are perfectly correlated. There are T Treasury bonds in the economy, with $T < \rho_H - \rho_0$. Show that when λ is small, the liquidity premium $(q-1)$ commanded by Treasury bonds is proportional to the probability of a high liquidity shock. (Hint: show that either (2) or (3) must be binding, and use (1) to conclude that (3) is binding.)

(iii) Suppose that, in the economy considered in the previous subquestion, the government issues at date 0 not only the T Treasury bonds, but also a security that yields at date 1 a payoff equal to 1 in the good state (the firms experience liquidity shock ρ_L) and 0 in the bad state (the firms experience liquidity shock ρ_H). What is the equilibrium date-0 price q' of this new asset? (Prices of the Treasury bonds and of this new asset are market clearing prices.)

Context needed to solve it *From Tirole §5.2.1, pp. 202–204 (PDF 215–217):*

Consider the setup of Section 3.2, except that there is an intermediate date at which income accrues and some reinvestment need is realized. As indicated in Figure 5.1, the entrepreneur at date 0 has wealth A and borrows $I - A$, where I is the fixed cost of investment.

At date 1, the investment yields deterministic and verifiable income $r \geqslant 0$. Continuation, though, requires reinvesting an amount ρ, where ρ is *ex ante* unknown and has cumulative distribution function $F(\rho)$ with density $f(\rho)$ on $[0, \infty)$. The realization of ρ is learned at date 1. Note that we here assume that the date-1 income is deterministic while the reinvestment need is random. The important assumption is that at least one of the two is random.

If the firm does not reinvest ρ, then the firm is liquidated. The liquidation value is 0. If the firm reinvests ρ, then the firm yields, at date 2, R with probability p and 0 with probability $1-p$, where $p = p_H$ if the entrepreneur behaves (and then gets no private benefit) and $p = p_L = p_H - \Delta p$ if the entrepreneur misbehaves (in which case she receives private benefit B).

The entrepreneur and the investors are risk neutral, the entrepreneur is protected by limited liability, and the investors demand a rate of return equal to 0.

Thus, the model is nothing but an extension of the basic fixed-investment one in Section 3.2. We have just added an intermediate income r and a reinvestment need ρ (the bold type in Figure 5.1). (Put differently, the model of Section 3.2 corresponds to the special case $r = 0$ and F being a spike at $\rho = 0$.) We assume that there exists in the economy a store of value that yields the consumers' rate of interest (0 here). That is, 1 unit invested at date 0 delivers a return of 1 unit at date 1 (Chapter 15 will investigate the reasonableness of this assumption). We now give a *heuristic* description of the optimal contract.

Figure 5.1 (timing, transcribed): **0** — Entrepreneur has wealth A and fixed-investment project costing $I > A$. **1** — **Short-term income** $r > 0$**. Reinvestment need** ρ **(drawn from** $F(\cdot)$**).** Then, (if reinvestment): Moral hazard ($p = p_H$ or p_L). **2** — Success (profit R) with probability p, failure (profit 0) with probability $1-p$.

Suppose in a first step that the initial contract can specify whether the firm continues or liquidates for each value of ρ (as we will see, it actually does not matter whether the realized value of ρ is verifiable, as long as there is no use that can be made of the date-1 cash flow besides reinvesting it and distributing it to investors). Intuitively, it is optimal to continue whenever it is cheap to do so:

$$\rho \leqslant \rho^*,$$

where ρ^* is a cutoff.

As is now familiar to the reader, competition among investors deprives them of a surplus, and so the borrower's utility is equal to the NPV. Assuming, as usual, that the optimal contract induces the high effort in the case of continuation and noting that the probability of continuation is $\Pr(\rho \leqslant \rho^*) = F(\rho^*)$, the borrower's net utility is

$$U_b(\rho^*) = [r + F(\rho^*)p_H R] - \left[I + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho\right],$$

where the first bracket represents expected revenue and the second bracket total investment (initial investment plus expected reinvestment).

Ensuring good behavior in the case of continuation suggests giving to the entrepreneur, at date 2, R_b in the case of success and 0 in the case of failure, where

$$(\Delta p)R_b \geqslant B.$$

Furthermore, there is no loss of generality in assuming that the entrepreneur receives nothing at date 1. Suppose she receives $r_b > 0$. Then the contract could eliminate this short-term compensation and increase R_b by δR_b, so that the expected total reward remains constant: $F(\rho^*)p_H\delta R_b = r_b$. If anything, this substitution alleviates moral hazard in the case of continuation. [...]

The pledgeable income, $\mathcal{P}$, deflated by the investors' initial outlay, $I - A$, is therefore

$$\mathcal{P}(\rho^*) - (I - A) = \left[r + F(\rho^*)\left[p_H\left(R - \frac{B}{\Delta p}\right)\right]\right] - \left[I + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho - A\right],$$

since the entrepreneur no longer has cash and so the reinvestment must be paid out of either the investors' pocket or date-1 revenue.

Taking derivatives in U_b and $\mathcal{P}$, the key insights are as follows:

- The NPV (U_b) is increasing in the cutoff ρ^* as long as $\rho^* < p_H R$, and is decreasing thereafter. Intuitively, one would want to salvage an investment when the cost, ρ, of a rescue is smaller than the expected payoff, $p_H R$, of continuing.
- By contrast, the pledgeable income increases with ρ^* for $\rho^* < p_H(R - B/\Delta p)$ and decreases thereafter. This is again intuitive: investors have to bear the cost, ρ, of salvaging the investment and can put their hands on at most $p_H(R - B/\Delta p)$ given that the entrepreneur must be given incentives to behave in the case of continuation.

We are then led to consider three cases. Depending on the strength of the balance sheet, there may be (i) an efficient amount of liquidation, (ii) an overoptimal amount of liquidation to satisfy investors, or (iii) no funding at all:

(i) $\mathcal{P}(p_H R) \geqslant I - A$.

In this case, the "first-best cutoff" $\rho^* = p_H R$, which maximizes U_b, leaves sufficient income to investors. The contract then specifies, say, no compensation r_b at date 1 for the entrepreneur, and a reward R_b in the case of continuation and success at date 2.

(ii) $\mathcal{P}(p_H R) < I - A \leqslant \mathcal{P}\left(p_H\left(R - \dfrac{B}{\Delta p}\right)\right)$.

The optimal contract then specifies $r_b = 0$ and $R_b = B/\Delta p$. The entrepreneur receives nothing at the intermediate date and, in the case of continuation, receives the lowest compensation, $R_b = B/\Delta p$, that is incentive compatible. [...] The cutoff $\rho^* \in [p_H(R - B/\Delta p), p_H R]$ is then given by

$$r + F(\rho^*)\left[p_H\left(R - \frac{B}{\Delta p}\right)\right] = I + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho - A.$$

[...] In order to be able to invest more *ex ante*, the borrower accepts a level of reinvestment below the *ex post* efficient level ($\rho^* < p_H R$). The intuition is that, because incentives must be preserved, the borrower cannot pledge to the lenders the entire benefit of the reinvestment decision. Also, ρ^* exceeds the per-unit pledgeable income $p_H(R - B/\Delta p)$, which is the level that maximizes the borrowing capacity. [...]

(iii) $\mathcal{P}\left(p_H\left(R - \dfrac{B}{\Delta p}\right)\right) < I - A$.

In this case, funding is not feasible. The value ρ^* that maximizes the pledgeable income ($\rho^* = p_H(R - B/\Delta p)$) does not suffice to compensate the investors for their initial outlay.

From Tirole §5.2.3 "Credit Lines for Cash-Poor Firms", pp. 205–206 (PDF 218–219):

Suppose in contrast that the investment "takes a long time" to produce income. At the extreme, there is no short-term profit: $r = 0$.

Can the entrepreneur just "wait and see," that is, borrow I at date 0 in exchange for shares in the firm and return to the capital market at date 1 if need occurs? Let us thus assume that the entrepreneur does not plan her liquidity in advance and that the liquidity shock occurs at date 1. To raise cash on the capital market to pay ρ, the entrepreneur must issue new shares and thereby dilute historical investors. Letting $\rho_0 \equiv p_H(R - B/\Delta p)$, and to illustrate this dilution, suppose that the entrepreneur faces a liquidity shock $\rho = \frac{1}{2}\rho_0$. The value of external shares held by initial investors is equal to ρ_0. Suppose that the number of shares is doubled. That is, as many shares are sold to new investors as already exist. So the value of each share is halved. The firm thereby raises $\frac{1}{2}\rho_0 = \rho$ in cash and can withstand the liquidity shock. Are initial investors willing to let themselves be diluted? The value of their shares is, of course, reduced to $\frac{1}{2}\rho_0$. But contemplate the alternative of liquidating the investment, under which the initial investors receive nothing! Thus, initial investors are willing to accept the dilution.

Similarly, to meet a liquidity shock equal to $\frac{3}{4}\rho_0$, the firm must quadruple the number of shares, and so on. But there is an upper bound to this process: investors will never pay more than the firm is worth to them. Hence, even in a frictionless capital market, the firm cannot raise more than ρ_0. Going back to the capital market at date 1 then at best allows the firm to withstand a liquidity shock of magnitude

$$\rho \leqslant \rho_0 = p_H\left(R - \frac{B}{\Delta p}\right).$$

Because the optimal financing arrangement specifies

$$p_H\left(R - \frac{B}{\Delta p}\right) < \rho^* \leqslant p_H R,$$

the entrepreneur must secure a line of credit or hoard liquidity in order to face the date-1 liquidity shock. We will shortly describe how to do so, but there are basically two alternatives and combinations thereof: a credit line or liquid assets of magnitude ρ^* with no right to dilute existing claimholders by issuing new claims at date 1 (so the entrepreneur borrows $I + \rho^*$); or a smaller credit line or amount of liquid assets, equal to $[\rho^* - p_H(R - B/\Delta p)]$ with a right to dilute claimholders as needed to ensure continuation. Either way, the entrepreneur must plan liquidity management.

The optimum can be implemented by a nonrevokable line of credit granted by, say, one of the lenders (a bank) at level ρ^*. It is important that this line of credit be nonrevokable (in a broad sense: see below). Otherwise the lender would have an incentive not to abide by his promise to rescue the firm if $\rho > \rho_0$, that is, if the liquidity shock exceeds the date-1 pledgeable income ρ_0. [...]

Remark (renegotiation). Could this line of credit be renegotiated to the parties' mutual advantage once the fraction ρ is realized? First, note that if $\rho \leqslant \rho^*$, then *a fortiori* $\rho < p_H R$ and therefore it is *ex post* efficient

to continue; so there is no scope for a renegotiation in which the lender would compensate the borrower for not using the credit line, as this renegotiation would reduce total surplus and therefore at least one of the parties would be strictly better off not renegotiating. Second, could the two parties both benefit from an increase in the line of credit to ρ when $\rho^* < \rho \leqslant p_H R$? Even though this increase would yield the ex post efficient reinvestment policy, there is no way for the borrower to compensate the lender, again because the borrower's stake is incompressible. One can show that the lender will turn down any request for an increase in the credit line. So will any alternative lender (other lenders may have even less incentives to refinance, because unlike the initial lender they do not have a vested stake to lose).

What is asked. (i) Show that a firm hoards enough Treasury bonds to withstand the high liquidity shock if conditions (2) and (3) hold. (ii) In an economy of identical firms with perfectly correlated shocks and a scarce supply T of bonds, show that for small λ the liquidity premium $q-1$ is proportional to λ. (iii) Price a new security that pays 1 only in the good state.

Approach. Tirole's fixed-investment liquidity model (§5.2) with two shocks. A shock below ρ_0 can be met by diluting investors at date 1; a shock above ρ_0 needs liquidity hoarded at date 0, $\rho_H - \rho_0$. Bonds bought at $q > 1$ cost a liquidity premium. The firm's choice compares NPV gain and pledgeable-income cost; the market price adjusts until firms are indifferent. Assets are valued for liquidity only if they pay in the state where liquidity is needed.

Notation.

- $\rho_L < \rho_0 < \rho_H < \rho_1$: low/high shock, probabilities $1-\lambda$ and λ; $I - A$: investors' date-0 outlay; (1): $\rho_0 - \rho_L > I - A$.
- $q \geq 1$: date-0 price of a bond paying 1 at date 1; $x_L, x_H \in \{0, 1\}$: continuation in each state.
- $T < \rho_H - \rho_0$: bond supply; θ: fraction of firms that hoard; q': price of the good-state security.

Step 1 — NPV and pledgeable income. In state L the firm raises $\rho_L < \rho_0$ by issuing claims, so it needs no liquidity. In state H it can raise at most ρ_0, so continuing requires $\rho_H - \rho_0$ bonds bought at date 0 for $q(\rho_H - \rho_0)$; they return $\rho_H - \rho_0$ at date 1 in both states, so their net cost is $(q-1)(\rho_H - \rho_0)$. Hence the NPV (the borrower's utility)

$$U_b = (1-\lambda)(\rho_1 - \rho_L)x_L + \lambda(\rho_1 - \rho_H)x_H - I - (q-1)(\rho_H - \rho_0)x_H, \tag{29}$$

and investors' breakeven

$$(1-\lambda)(\rho_0 - \rho_L)x_L + \lambda(\rho_0 - \rho_H)x_H - (q-1)(\rho_H - \rho_0)x_H \geq I - A. \tag{30}$$

$x_L = 1$: it raises both (29) (as $\rho_L < \rho_1$) and the left side of (30) (as $\rho_L < \rho_0$).

Step 2 — (i) Hoard or not. Going from $x_H = 0$ to $x_H = 1$ changes the NPV by

$$\Delta U_b = \lambda(\rho_1 - \rho_H) - (q-1)(\rho_H - \rho_0) \ \geq 0 \quad \Longleftrightarrow \quad (q-1)(\rho_H - \rho_0) \leq \lambda(\rho_1 - \rho_H),$$

which is (3). Hoarding is feasible iff (30) holds with $x_L = x_H = 1$:

$$\begin{aligned}(1-\lambda)(\rho_0 - \rho_L) - \lambda(\rho_H - \rho_0) - (q-1)(\rho_H - \rho_0) &\geq I - A \\ (q-1)(\rho_H - \rho_0) &\leq (1-\lambda)(\rho_0 - \rho_L) - \lambda(\rho_H - \rho_0) - I + A,\end{aligned}$$

which is (2).

Answer (i). Under (2) hoarding is fundable and under (3) it raises the NPV, so the firm hoards $\rho_H - \rho_0$ bonds.

Step 3 — (ii) Market clearing. If every firm hoards, demand is $\rho_H - \rho_0 > T$: excess demand. If no firm hoards, demand $0 < T$, the price falls; at $q = 1$ condition (3) holds strictly ($\lambda(\rho_1 - \rho_H) > 0$) and (2) holds for small λ by (1), so firms would hoard. Hence in equilibrium firms are indifferent, a fraction θ hoards, and

$$\theta(\rho_H - \rho_0) = T \quad \Longrightarrow \quad \theta = \frac{T}{\rho_H - \rho_0} \in (0, 1).$$

Indifference means that the tighter of (2), (3) binds. Their right sides differ by

$$\begin{aligned}\text{RHS}(2) - \text{RHS}(3) &= (1-\lambda)(\rho_0 - \rho_L) - \lambda(\rho_H - \rho_0) - (I - A) - \lambda(\rho_1 - \rho_H) \\ &= (\rho_0 - \rho_L) - (I - A) - \lambda\big[(\rho_0 - \rho_L) + (\rho_H - \rho_0) + (\rho_1 - \rho_H)\big] \\ &= (\rho_0 - \rho_L) - (I - A) - \lambda(\rho_1 - \rho_L),\end{aligned}$$

positive iff $\lambda < \bar{\lambda} \equiv [(\rho_0 - \rho_L) - (I - A)]/(\rho_1 - \rho_L)$, and $\bar{\lambda} > 0$ by (1). So for small λ, (3) binds:

$$\begin{aligned}(q-1)(\rho_H - \rho_0) &= \lambda(\rho_1 - \rho_H) \\ q - 1 &= \lambda \frac{\rho_1 - \rho_H}{\rho_H - \rho_0}.\end{aligned} \tag{31}$$

Non-hoarding firms are still funded, since slack (2) implies $(1-\lambda)(\rho_0 - \rho_L) > I - A$. Illustration (own numbers): $\rho_L = 0.2$, $\rho_0 = 1$, $\rho_H = 1.5$, $\rho_1 = 2$, $I - A = 0.5$, $\lambda = 0.1$, $T = 0.3$. Then $\bar{\lambda} = 0.3/1.8 = 0.166667 > 0.1$; $q - 1 = 0.1 \times 0.5/0.5 = 0.1$; (2): $0.05 \leq 0.72 - 0.05 - 0.5 = 0.17$; $\theta = 0.3/0.5 = 0.6$.

Answer (ii). $q - 1 = \lambda(\rho_1 - \rho_H)/(\rho_H - \rho_0)$, proportional to the probability λ of the high shock.

Step 4 — (iii) The good-state security. It pays 1 when firms face ρ_L, which they meet by dilution, and 0 when they face ρ_H, the only state where liquidity has value. Firms therefore value it at its expected payoff, like consumers with zero time preference:

$$q' = (1-\lambda) \cdot 1 + \lambda \cdot 0 = 1 - \lambda. \tag{32}$$

Answer (iii). $q' = 1 - \lambda$ (0.9 in the illustration): no liquidity premium.

Intuition. A Treasury bond is a good-state claim plus a bad-state claim. The whole premium sits on the bad-state part: its implied price is $q - q' = \lambda + (q-1) = \lambda(\rho_1 - \rho_0)/(\rho_H - \rho_0)$ (here $1.1 - 0.9 = 0.2$), above its expected payoff λ, because it delivers cash when every firm needs it at once. With correlated shocks, firms cannot insure each other, and the scarcity of assets that pay in the bad state is priced.

Common mistakes.

- Counting the full price $q(\rho_H - \rho_0)$ as the cost of hoarding: the bonds return $\rho_H - \rho_0$, so only the premium $(q-1)(\rho_H - \rho_0)$ is a cost.
- Hoarding ρ_H instead of $\rho_H - \rho_0$: the firm can still dilute investors up to ρ_0.
- In (iii), giving the asset a premium because it is "safe government paper": what matters is *when* it pays.

Book answer. Matches the book's answer (pp. 592–593): $x_L = 1$; hoarding iff (2) and (3); for small λ, (3) binds and $q - 1 = \lambda(\rho_1 - \rho_H)/(\rho_H - \rho_0)$; $q' = 1 - \lambda$. The book's U_b subtracts $(I - A)$, i.e. it is gross of A; (29) is net. Our $\bar{\lambda}$ makes "λ small" explicit.

Ref: Tirole §5.2.1, pp. 201–204; §5.2.3, pp. 205–206; §5.3.1 "The Two-Shock Case", p. 207; Exercise 5.5, p. 231; answer pp. 592–593.

9. Should debt contracts be indexed to output prices? — Tirole Exercise 5.11, pp. 234–235

Statement.

Source: Tirole, Ch. 5, Exercise 5.11, p. 234 (PDF 247) · Answer in book: yes, p. 595

Exercise 5.11 (should debt contracts be indexed to output prices?). This exercise returns to optimal corporate risk management when profits are positively serially correlated (see Section 5.4.2). The source of serial correlation is now a permanent shift in the market price of output, as summarized in Figure 5.15. The

model is the fixed-investment model, except that the date-1 and date-2 incomes depend on an exogenous market price P, with mean $\bar{P}$, that is realized at date 1. The realizations of P and ρ are independent.

Figure 5.15 (timing, transcribed; the time line carries no date numbers): Entrepreneur has wealth A and fixed-investment project costing $I > A$. → (If investment) → Short-term income $Pr > 0$. Reinvestment need ρ (drawn from $F(\cdot)$). → (If reinvestment) → Moral hazard ($p = p_H$ or p_L). → Success (profit PR) with probability p, failure (profit 0) with probability $1 - p$.

The rest of the model is otherwise the same as in Section 5.2. Following the steps of Section 5.4.2:

(i) Determine the optimal reinvestment policy $\rho^*(P)$.

(ii) Show that, accounting for seasoned offerings, the optimal debt is fully indexed debt:

$$d(P) = Pr - \ell_0,$$

where ℓ_0 is a positive constant.

Context needed to solve it *From Tirole §5.2.1, pp. 202–204 (PDF 215–217):*

Consider the setup of Section 3.2, except that there is an intermediate date at which income accrues and some reinvestment need is realized. As indicated in Figure 5.1, the entrepreneur at date 0 has wealth A and borrows $I - A$, where I is the fixed cost of investment.

At date 1, the investment yields deterministic and verifiable income $r \geqslant 0$. Continuation, though, requires reinvesting an amount ρ, where ρ is *ex ante* unknown and has cumulative distribution function $F(\rho)$ with density $f(\rho)$ on $[0, \infty)$. The realization of ρ is learned at date 1. Note that we here assume that the date-1 income is deterministic while the reinvestment need is random. The important assumption is that at least one of the two is random.

If the firm does not reinvest ρ, then the firm is liquidated. The liquidation value is 0. If the firm reinvests ρ, then the firm yields, at date 2, R with probability p and 0 with probability $1 - p$, where $p = p_H$ if the entrepreneur behaves (and then gets no private benefit) and $p = p_L = p_H - \Delta p$ if the entrepreneur misbehaves (in which case she receives private benefit B).

The entrepreneur and the investors are risk neutral, the entrepreneur is protected by limited liability, and the investors demand a rate of return equal to 0.

Thus, the model is nothing but an extension of the basic fixed-investment one in Section 3.2. We have just added an intermediate income r and a reinvestment need ρ (the bold type in Figure 5.1). (Put differently, the model of Section 3.2 corresponds to the special case $r = 0$ and F being a spike at $\rho = 0$.) We assume that there exists in the economy a store of value that yields the consumers' rate of interest (0 here). That is, 1 unit invested at date 0 delivers a return of 1 unit at date 1 (Chapter 15 will investigate the reasonableness of this assumption). We now give a *heuristic* description of the optimal contract.

Figure 5.1 (timing, transcribed): **0** — Entrepreneur has wealth A and fixed-investment project costing $I > A$. **1** — **Short-term income** $r > 0$**. Reinvestment need** ρ **(drawn from** $F(\cdot)$**).** Then, (if reinvestment): Moral hazard ($p = p_H$ or p_L). **2** — Success (profit R) with probability p, failure (profit 0) with probability $1 - p$.

Suppose in a first step that the initial contract can specify whether the firm continues or liquidates for each value of ρ (as we will see, it actually does not matter whether the realized value of ρ is verifiable, as long as there is no use that can be made of the date-1 cash flow besides reinvesting it and distributing it to investors). Intuitively, it is optimal to continue whenever it is cheap to do so:

$$\rho \leqslant \rho^*,$$

where ρ^* is a cutoff.

As is now familiar to the reader, competition among investors deprives them of a surplus, and so the borrower's utility is equal to the NPV. Assuming, as usual, that the optimal contract induces the high effort in the case of continuation and noting that the probability of continuation is $\Pr(\rho \leqslant \rho^*) = F(\rho^*)$, the borrower's net utility is

$$U_b(\rho^*) = [r + F(\rho^*)p_H R] - \left[I + \int_0^{\rho^*} \rho f(\rho)\, \mathrm{d}\rho \right],$$

where the first bracket represents expected revenue and the second bracket total investment (initial investment plus expected reinvestment).

Ensuring good behavior in the case of continuation suggests giving to the entrepreneur, at date 2, R_b in the case of success and 0 in the case of failure, where

$$(\Delta p)R_b \geqslant B.$$

Furthermore, there is no loss of generality in assuming that the entrepreneur receives nothing at date 1. Suppose she receives $r_b > 0$. Then the contract could eliminate this short-term compensation and increase R_b by δR_b, so that the expected total reward remains constant: $F(\rho^*)p_H\delta R_b = r_b$. If anything, this substitution alleviates moral hazard in the case of continuation. [...]

The pledgeable income, $\mathcal{P}$, deflated by the investors' initial outlay, $I - A$, is therefore

$$\mathcal{P}(\rho^*) - (I - A) = \left[r + F(\rho^*)\left[p_H\left(R - \frac{B}{\Delta p}\right)\right]\right] - \left[I + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho - A\right],$$

since the entrepreneur no longer has cash and so the reinvestment must be paid out of either the investors' pocket or date-1 revenue.

Taking derivatives in U_b and $\mathcal{P}$, the key insights are as follows:

- The NPV (U_b) is increasing in the cutoff ρ^* as long as $\rho^* < p_H R$, and is decreasing thereafter. Intuitively, one would want to salvage an investment when the cost, ρ, of a rescue is smaller than the expected payoff, $p_H R$, of continuing.
- By contrast, the pledgeable income increases with ρ^* for $\rho^* < p_H(R - B/\Delta p)$ and decreases thereafter. This is again intuitive: investors have to bear the cost, ρ, of salvaging the investment and can put their hands on at most $p_H(R - B/\Delta p)$ given that the entrepreneur must be given incentives to behave in the case of continuation.

We are then led to consider three cases. Depending on the strength of the balance sheet, there may be (i) an efficient amount of liquidation, (ii) an overoptimal amount of liquidation to satisfy investors, or (iii) no funding at all:

(i) $\mathcal{P}(p_H R) \geqslant I - A$.

In this case, the "first-best cutoff" $\rho^* = p_H R$, which maximizes U_b, leaves sufficient income to investors. The contract then specifies, say, no compensation r_b at date 1 for the entrepreneur, and a reward R_b in the case of continuation and success at date 2.

(ii) $\mathcal{P}(p_H R) < I - A \leqslant \mathcal{P}\left(p_H\left(R - \dfrac{B}{\Delta p}\right)\right)$.

The optimal contract then specifies $r_b = 0$ and $R_b = B/\Delta p$. The entrepreneur receives nothing at the intermediate date and, in the case of continuation, receives the lowest compensation, $R_b = B/\Delta p$, that is incentive compatible. [...] The cutoff $\rho^* \in [p_H(R - B/\Delta p), p_H R]$ is then given by

$$r + F(\rho^*)\left[p_H\left(R - \frac{B}{\Delta p}\right)\right] = I + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho - A.$$

[...] In order to be able to invest more *ex ante*, the borrower accepts a level of reinvestment below the *ex post* efficient level ($\rho^* < p_H R$). The intuition is that, because incentives must be preserved, the borrower cannot pledge to the lenders the entire benefit of the reinvestment decision. Also, ρ^* exceeds the per-unit pledgeable income $p_H(R - B/\Delta p)$, which is the level that maximizes the borrowing capacity. [...]

(iii) $\mathcal{P}\left(p_H\left(R - \dfrac{B}{\Delta p}\right)\right) < I - A$.

In this case, funding is not feasible. The value ρ^* that maximizes the pledgeable income ($\rho^* = p_H(R - B/\Delta p)$) does not suffice to compensate the investors for their initial outlay.

From Tirole §5.4.2 "When Is Incomplete Hedging Optimal?", item (b), pp. 217–219 (PDF 230–232):

(b) *Serial correlation of profits.* An important assumption behind the full-hedging result of Section 5.4.1 is that the date-1 profit realization conveys no information about the firm's prospects: it is a transitory shock. Suppose in contrast that a high date-1 profit is good news about date-2 profitability. For example, the price of a crop may reflect permanent shocks such as the reduction of trade barriers, the entry of competing offers, or a change in consumer preferences.

With positive serial correlation of profits, a high current profit is associated with attractive reinvestment opportunities. This suggests that the liquidity available to the borrower at date 1 should covary with the date-1 profit (so, for example, the farmer's debt contract should not be fully indexed to the price of the crop). Things are, however, more complex than this first argument suggests, because better prospects also make it

easier for the borrower to return to the capital market at the intermediate stage. The attractive-reinvestment-opportunities effect, however, in general dominates the easier-refinancing effect, and so the firm should not be fully insured against exogenous profit shocks, as we now illustrate.

Let us consider the fixed-investment model of Section 5.2, but with two twists:

- The short-term income, r, is random, with mean $\bar{r}$.
- The probability of success in the case of continuation is an increasing function of r,

$$p + \tau(r), \quad \text{with } \tau' > 0,$$

where $p = p_H$ or p_L depending on the entrepreneur's date-1 behavior. (The separable form of the probability-of-success function as usual guarantees that the incentive constraint is invariant.) We assume that the realizations of r and ρ are independent.

These twists are depicted in bold in Figure 5.7.

Figure 5.7 (timing, transcribed; no date numbers on the time line): Entrepreneur has wealth A and fixed-investment project costing $I > A$. $\rightarrow$ **Random short-term income r. Reinvestment need ρ (drawn from $F(\cdot)$).** $\rightarrow$ (If reinvestment) $\rightarrow$ Moral hazard ($p = p_H$ or p_L). $\rightarrow$ Success (profit R) with probability $p + \tau(r)$, failure (profit 0) with probability $1 - (p + \tau(r))$.

Let us follow the steps of Section 5.2 and determine the optimal state-contingent cutoff $\rho^*(r)$ (so continuation occurs if and only if $\rho \leqslant \rho^*(r)$). Letting $E[\cdot]$ denote expectations with respect to r, the NPV is

$$U_b = \bar{r} + E[F(\rho^*(r))[p_H + \tau(r)]R] - I - E\left[\int_0^{\rho^*(r)} \rho f(\rho)\,\mathrm{d}\rho\right].$$

The investors' breakeven constraint similarly is

$$\bar{r} + E\left[F(\rho^*(r))[p_H + \tau(r)]\left[R - \frac{B}{\Delta p}\right]\right] \geqslant I - A + E\left[\int_0^{\rho^*(r)} \rho f(\rho)\,\mathrm{d}\rho\right].$$

Letting μ denote the shadow price of the breakeven constraint (we assume that the constraint is binding, i.e., $\mu > 0$), the first-order condition with respect to $\rho^*(r)$ yields, for each r,

$$\rho^*(r) = \frac{[p_H + \tau(r)][R + \mu(R - B/\Delta p)]}{1 + \mu}.$$

Let us now investigate the implementation of the optimal contract. A *fully indexed debt* can be defined as a date-1 liability $d(r)$ such that

$$d(r) = d_0 + r,$$

for some constant d_0. That is, in the absence of refinancing in the capital market, a fully indexed debt insulates the firm's retained earnings against its cash-flow risk. We, however, want to allow the firm to return to the capital market: insulation of retained earnings against the cash-flow risk does not imply insulation of the reinvestment policy. The amount it can raise in the capital market at date 1,

$$[p_H + \tau(r)]\left(R - \frac{B}{\Delta p}\right),$$

is increasing with the date-1 profit as $\tau' > 0$ (this was referred to earlier as the "easier-refinancing effect"). The optimal policy is implemented when the cutoff is equal to the cash cushion plus the refinancing capacity:

$$\rho^*(r) = [r - d^*(r)] + [p_H + \tau(r)]\left(R - \frac{B}{\Delta p}\right),$$

or

$$d^*(r) = r - \frac{p_H + \tau(r)}{1 + \mu}\left(\frac{B}{\Delta p}\right).$$

In the presence of an agency cost ($B > 0$), the debt is not fully indexed. The easier-refinancing effect is at play, but the existence of a managerial rent puts a limit on what can be achieved by returning to the capital market. Put differently, *the firm should keep some of its cash flow as retained earnings.*

The source of this cash-flow sensitivity of debt is the monotonicity of managerial rents with the resolution of uncertainty. (Footnote 38: Here, this rent is $[p_H + \tau(r)](B/\Delta p)$.)

What is asked. (i) Find the optimal reinvestment cutoff as a function of the output price P realised at date 1. (ii) Show that, when the firm can issue new securities at date 1, the optimal date-1 debt is fully indexed: $d(P) = Pr - \ell_0$ with a positive constant ℓ_0.

Approach. Follow §5.4.2: maximise NPV subject to investors' breakeven with a state-contingent cutoff $\rho^*(P)$, using a Lagrange multiplier; then implement the cutoff as retained cash plus refinancing capacity. The two forces of §5.4.2 are the attractive-reinvestment effect (higher P raises the cutoff) and the easier-refinancing effect (higher P raises what can be raised at date 1). Here the entrepreneur's rent does not move with P, so the two effects cancel and retained earnings are constant.

Notation.

- P: output price realised at date 1, mean $\bar{P}$, independent of ρ; date-1 income Pr; date-2 income PR on success.
- $\rho \sim F$ (density f) on $[0, \infty)$: reinvestment need; $\rho^*(P)$: cutoff; B: private benefit (not scaled by P); Δp, p_H as usual.
- $\mu > 0$: multiplier of investors' breakeven; $d(P)$: date-1 debt repayment; ℓ_0: retained earnings.

Step 1 — Program. The incentive constraint after continuation is $(\Delta p)R_b \geq B$, so investors can get $PR - B/\Delta p$ on success. The NPV is

$$U_b = \bar{P}r + E\big[F(\rho^*(P))\, p_H PR\big] - I - E\Big[\int_0^{\rho^*(P)} \rho f(\rho)\, d\rho\Big], \tag{33}$$

and investors' breakeven is

$$\bar{P}r + E\Big[F(\rho^*(P))\, p_H\Big(PR - \frac{B}{\Delta p}\Big)\Big] \geq I - A + E\Big[\int_0^{\rho^*(P)} \rho f(\rho)\, d\rho\Big]. \tag{34}$$

Form $\mathcal{L} = U_b + \mu\,[\text{LHS} - \text{RHS of (34)}]$.

Step 2 — (i) First-order condition. $\mathcal{L}$ is an expectation over P, so it is maximised state by state. Using $\frac{d}{d\rho^*}F(\rho^*) = f(\rho^*)$ and $\frac{d}{d\rho^*}\int_0^{\rho^*} \rho f\, d\rho = \rho^* f(\rho^*)$, the derivative with respect to $\rho^*(P)$ is

$$0 = f(\rho^*)\big[p_H PR - \rho^*\big] + \mu f(\rho^*)\Big[p_H\Big(PR - \frac{B}{\Delta p}\Big) - \rho^*\Big].$$

Dividing by $f(\rho^*) > 0$ and collecting ρ^*:

$$\begin{aligned}(1+\mu)\rho^* &= (1+\mu)p_H PR - \mu\frac{p_H B}{\Delta p}\\ \rho^*(P) &= p_H PR - \frac{\mu}{1+\mu}\,\frac{p_H B}{\Delta p}.\end{aligned} \tag{35}$$

Answer (i). $\rho^*(P) = p_H PR - \frac{\mu}{1+\mu}\frac{p_H B}{\Delta p}$: between the pledgeable value $p_H(PR - B/\Delta p)$ and the total value $p_H PR$ of continuation, rising one-for-one with $p_H PR$.

Step 3 — (ii) Implementation. With date-1 debt $d(P)$, the firm keeps $Pr - d(P)$ and can raise, by issuing new claims, up to the date-1 pledgeable income $p_H(PR - B/\Delta p)$. It continues iff ρ is covered, so the implemented cutoff is

$$\rho^*(P) = \big[Pr - d(P)\big] + p_H\Big(PR - \frac{B}{\Delta p}\Big). \tag{36}$$

Solving (36) for $d(P)$ and substituting (35):

$$\begin{aligned} d(P) &= Pr + p_H PR - \frac{p_H B}{\Delta p} - \rho^*(P)\\ &= Pr + p_H PR - \frac{p_H B}{\Delta p} - \Big(p_H PR - \frac{\mu}{1+\mu}\frac{p_H B}{\Delta p}\Big)\\ &= Pr - \Big(1 - \frac{\mu}{1+\mu}\Big)\frac{p_H B}{\Delta p}\\ &= Pr - \ell_0, \qquad \ell_0 \equiv \frac{1}{1+\mu}\,\frac{p_H B}{\Delta p} > 0. \end{aligned} \tag{37}$$

Check (own numbers): $p_H = 0.6$, $B = 0.5$, $\Delta p = 0.25$, $\mu = 0.5$ give $\ell_0 = 0.6(2)/1.5 = 0.8$; for $R = 3$, $r = 1$ and $P \in \{0.8, 1, 1.2\}$, cash ℓ_0 plus $p_H(PR - 2)$ equals (35) in each state.

Answer (ii). $d(P) = Pr - \ell_0$ with $\ell_0 = \frac{1}{1+\mu}\frac{p_H B}{\Delta p} > 0$: the firm retains the same cash ℓ_0 whatever P, i.e. the debt is fully indexed to the output price.

Intuition. A higher price makes continuation more valuable (ρ^* rises by $p_H R$ per unit of P) and also makes it easier to raise money at date 1 (pledgeable income rises by the same $p_H R$), because the manager's rent $p_H B/\Delta p$ does not depend on P. The market finances all of the extra reinvestment, so the firm need not keep any of the price windfall: full indexation (full hedging) is optimal. In §5.4.2 the rent $[p_H + \tau(r)]B/\Delta p$ rises with good news, refinancing capacity rises less than the cutoff, and the firm keeps part of its cash flow.

Common mistakes.

- Scaling the private benefit by P: the rent would then move with P and indexation would be partial.
- Ignoring seasoned offerings: without date-1 issues the firm would need to keep $\rho^*(P)$ in cash, which rises with P.
- Reading "fully indexed" as "no retained earnings": the firm keeps the positive constant ℓ_0.

Book answer. Matches the book's answer (p. 595): $\rho^*(P) = p_H PR - \frac{\mu}{1+\mu}\frac{p_H B}{\Delta p}$ and $d(P) = Pr - \ell_0$ with $\ell_0 = \frac{1}{1+\mu}(p_H \frac{B}{\Delta p})$.
Ref: Tirole §5.2.1, pp. 201–204; §5.4.1–5.4.2, pp. 214–219; Exercise 5.11, pp. 234–235; answer p. 595.

10. Decreasing returns to scale and debt maturity — Tirole Exercise 5.7, p. 232

Statement.

Source: Tirole, Ch. 5, Exercise 5.7, p. 232 (PDF 245) · Answer in book: yes, p. 593

Exercise 5.7 (decreasing returns to scale). Extend the treatment of Section 5.6.1 to the case of decreasing returns to scale: the payoff in the case of continuation and success is $R(I)$, with $R(0) = 0$, $R' > 0$, $R'' < 0$, $R'(0) = \infty$, and $R'(\infty) = 0$. The rest is unchanged (the short-term income is rI, the reinvestment need is ρI, and the private benefit is BI).

(i) What are the first-order conditions yielding the optimal investment level I and cutoff ρ^*?

(ii) Assuming that $r > \rho^*$, and that $(R(I)/I - R'(I))$ is increasing in I (a condition satisfied, for example, by $R(I)$ quadratic), derive the impact of the strength of the balance sheet as measured, say, by A on debt maturity.

Context needed to solve it *From Tirole §3.4.1 "The Continuous-Investment Model", p. 127 (PDF 140):*

The continuous-investment model of this section is the polar opposite of the fixed-investment one. The fixed-investment model depicts a situation in which returns are sharply decreasing beyond some investment level. In contrast, we now assume that there are constant returns to scale in the investment technology. An investment $I \in [0, \infty)$ yields income RI, proportional to I, in the case of success, and 0 in the case of failure. The borrower's private benefit from misbehaving is also taken to be proportional to investment. As before, the borrower has a choice between behaving, in which case she derives no private benefit and the probability of success is p_H, and misbehaving, that is, enjoying private benefit BI, and reducing the probability of success to $p_L = p_H - \Delta p < p_H$. [...]

The borrower initially has cash A, and must therefore borrow $I - A$ to finance a project of size I. A loan agreement specifies that the lenders (who as before are assumed to make no profit) and the borrower receive 0 each in the case of failure, and R_l and R_b, respectively, in the case of success, where $R_l + R_b = RI$.

From Tirole §5.3.2 "Continuum of Liquidity Shocks", pp. 209–210 (PDF 222–223):

After the (endogenous size) investment I is sunk at date 0 and before the borrower works on the project, some exogenous shock occurs at date 1 that determines a per-unit-of-investment level $\rho \in [0, \infty)$ of "cost overruns." That is, a cash infusion equal to ρI is needed in order for the project to continue. If ρI is not

invested, the project is abandoned altogether and thus yields no income. As in Section 5.2, the fraction ρ is *a priori* distributed according to the continuous distribution $F(\rho)$ on $[0, \infty)$, with density $f(\rho)$. (As we already observed, the model of Section 3.4 is therefore a special case, with F being a spike at $\rho = 0$.)

Regardless of the required amount of the cash infusion, the project, if pursued, is still a project of size I, in that the income in the case of success is RI and the borrower's private benefit from misbehaving is BI. One cannot increase the size of the project after the initial stage.

The timing is summarized in Figure 5.4.

Figure 5.4 (timing, transcribed): **Date 0** — Loan agreement; Investment I. **Date 1** — Need for cash infusion ρI realized; then either Disbursement → Moral hazard, or No disbursement → Project is abandoned. **Date 2** — Outcome.

[...] It is easy to show that it is optimal to have a "cutoff rule" for infusing cash. There exists an optimal threshold ρ^* such that one should continue if and only if

$$\rho \leqslant \rho^*. \qquad (5.2)$$

The incentive constraint in the case of continuation is the same as in the absence of a liquidity shock (see Section 3.4):

$$(\Delta p) R_b \geqslant BI. \qquad (\mathrm{IC}_b)$$

The breakeven condition is slightly altered by the presence of liquidity shocks:

$$F(\rho^*)[p_H(RI - R_b)] \geqslant I - A + \int_0^{\rho^*} \rho I f(\rho)\, \mathrm{d}\rho. \qquad (\mathrm{IR}_l)$$

That is, the lenders receive a return only if the project is pursued, which has probability $F(\rho^*)$. The left-hand side of (IR_l) is the expected pledgeable income. Furthermore, there is a new term, representing the expected outlay on overruns, on the right-hand side. From these two constraints, we deduce the borrowing capacity (or, more precisely, the maximum investment that allows the lenders to break even):

$$I = k(\rho^*) A,$$

where

$$k(\rho^*) = \frac{1}{1 + \int_0^{\rho^*} \rho f(\rho)\, \mathrm{d}\rho - F(\rho^*)[p_H R - p_H B/\Delta p]} = \frac{1}{1 + \int_0^{\rho^*} \rho f(\rho)\, \mathrm{d}\rho - F(\rho^*)\rho_0} \qquad (5.3)$$

involves a straightforward modification relative to the no-liquidity-shock multiplier k. [...] Note that the borrower's borrowing capacity is maximal when the threshold ρ^* is equal to the expected per-unit pledgeable income $\rho_0 \equiv p_H(R - B/\Delta p)$.

Given that the competitive lenders make no profits, the borrower's net utility is as usual the social surplus brought about by the project, namely,

$$U_b = m(\rho^*) I = m(\rho^*) k(\rho^*) A, \qquad (5.4)$$

where

$$m(\rho^*) \equiv F(\rho^*) p_H R - 1 - \int_0^{\rho^*} \rho f(\rho)\, \mathrm{d}\rho$$

is the margin per unit of investment.

[...] Next, we observe that *this optimal threshold lies between the expected per-unit-of-investment pledgeable income and income*:

$$\rho_0 = p_H \left(R - \frac{B}{\Delta p} \right) < \rho^* < \rho_1 = p_H R. \qquad (5.8)$$

From Tirole §5.6.1 "Optimal Claim Maturity", p. 226 (PDF 239):

Let us return to the continuous-investment, continuous-shock version of Section 5.3.2, but with a short-term income (the analysis is not really new and is therefore only sketched): see Figure 5.10. Because the short-term income rI is fully pledgeable to the lenders, everything is *as if* the unit investment cost were equal to $1 - r$ instead of 1. The lenders' breakeven condition, that is, the equality between expected revenue and expected investment cost, becomes

$$rI + F(\rho^*)\rho_0 I = I - A + \left[\int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho\right] I$$

and so

$$k(\rho^*) = \frac{1}{1 + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho - [r + F(\rho^*)\rho_0]}. \qquad (5.3'')$$

The margin (expected profit of the firm per unit of investment) becomes

$$m(\rho^*) = [r + F(\rho^*)\rho_1] - \left[1 + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho\right]. \qquad (5.4'')$$

And thus the borrower's (gross) utility becomes

$$U_b = m(\rho^*)k(\rho^*)A = \frac{\rho_1 - c(\rho^*)}{c(\rho^*) - \rho_0}A,$$

where the expected unit cost of effective investment, $c(\rho^*)$, is given by

$$c(\rho^*) = \frac{1 - r + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho}{F(\rho^*)}. \qquad (5.5'')$$

So the optimal threshold is given by

$$\int_0^{\rho^*} F(\rho)\,\mathrm{d}\rho = 1 - r \qquad (5.6'')$$

and the borrower's utility by

$$U_b = \frac{\rho_1 - \rho^*}{\rho^* - \rho_0}A. \qquad (5.7'')$$

Figure 5.10 (timing, transcribed): **Date 0** — Loan agreement. Investment I. **Date 1** — Accrual of short-term income rI. Realization ρI of investment need. Then either Disbursement → Moral hazard, or No disbursement → Project is abandoned. **Date 2** — Date-2 income (RI or 0).

It is important to note that *the short-term income, even though it is deterministic and fully pledgeable, is not equivalent to an increase in the borrower's cash on hand A.* Such an increase in equity would result in a larger investment (as is the case here), but not in a modification of the continuation rule. By contrast, condition (5.6″) shows that the larger the short-term profit, the lower the optimal threshold ρ^*. [...]

(a) *Liquidity management.* Let us now turn to the implementation of the optimum and thus to the claim maturity. To this purpose we make the following assumption. (Footnote 57: Of course, it must also be the case that $\rho^* > \rho_0$ (otherwise, the borrower's borrowing capacity and utility would be infinite in this constant-returns-to-scale model). Because $\mathrm{d}\rho^*/\mathrm{d}r < -1$, we must thus also assume that r is not "too large.")

Free-cash-flow assumption: $r > \rho^*$. Under the free-cash-flow assumption, and given that the entrepreneur cannot steal the intermediate income, the entrepreneur would reinvest excessively if she were not asked to pay out money to investors at date 1. Namely, she would reinvest as long as $\rho \leqslant r$.

To obtain the optimal amount of reinvestment, an amount $P_1 \equiv (r - \rho^*)I$ must be pumped out of the firm, and the entrepreneur must be denied the right to dilute initial investors.

What is asked. (i) In the liquidity model with short-term income rI and a continuum of reinvestment shocks ρI, but with a concave success payoff $R(I)$, derive the first-order conditions that pin down the scale I and the continuation cutoff ρ^*. (ii) Under free cash flow ($r > \rho^*$), find how a stronger balance sheet (higher A) changes debt maturity, i.e. the short-term debt the firm must pay at date 1 per unit of investment.

Approach. Holmström–Tirole liquidity model (Tirole §5.3.2 and §5.6.1): the borrower maximises the NPV subject to the lenders' break-even, a Lagrangian problem in (I, ρ^*). Continuing after a shock ρ costs ρI and is worth ρ_1 in value but only ρ_0 in pledgeable income, so the optimal cutoff

lies between them. Under free cash flow the cutoff is implemented by short-term debt that pumps $(r-\rho^*)I$ out of the firm — the termination threat of the lecture, here a liquidity sweep — so maturity moves with ρ^*, and comparative statics in A answer (ii).

Notation.

- $I > 0$: investment (endogenous); $A \geq 0$: the entrepreneur's cash; lenders are competitive, rate of return 0.
- $R(I)$: date-2 income in case of continuation and success, $R(0) = 0$, $R' > 0$, $R'' < 0$.
- rI: short-term (date-1) income, verifiable and fully pledgeable, $r > 0$.
- $\rho \geq 0$: per-unit liquidity shock (reinvestment need ρI), c.d.f. F, density f on $[0, \infty)$; ρ^*: continuation cutoff (continue iff $\rho \leq \rho^*$).
- $p_H > p_L$: success probabilities when behaving / misbehaving, $\Delta p = p_H - p_L$; BI: private benefit from misbehaving; R_b: entrepreneur's stake in success.
- $\rho_1(I) \equiv p_H R(I)/I$ and $\rho_0(I) \equiv p_H[R(I)/I - B/\Delta p]$: per-unit expected income and per-unit pledgeable income at scale I; $\rho_1 - \rho_0 = p_H B/\Delta p$.
- $\mu \geq 0$: Lagrange multiplier of the lenders' breakeven constraint.

Step 1 — The program. We set up the borrower's problem. As in Section 5.6.1, the incentive constraint $(\Delta p)R_b \geq BI$ binds (any larger stake only lowers pledgeable income), so the income pledgeable in case of success is $R(I) - BI/\Delta p$. With competitive lenders the entrepreneur gets the NPV: short-term income, plus expected date-2 income when continuing (probability $F(\rho^*)$), minus the investment and the expected reinvestment,

$$U_b = rI + F(\rho^*)\, p_H R(I) - \Big[1 + \int_0^{\rho^*} \rho f(\rho)\, d\rho\Big] I, \tag{38}$$

subject to the lenders' breakeven constraint

$$rI + F(\rho^*)\, p_H \Big[R(I) - \frac{BI}{\Delta p}\Big] \geq I - A + \Big[\int_0^{\rho^*} \rho f(\rho)\, d\rho\Big] I. \tag{39}$$

The Lagrangian is $\mathcal{L} = U_b + \mu\,[\text{LHS} - \text{RHS of (39)}]$.

Step 2 — (i) First-order conditions. We differentiate $\mathcal{L}$ with respect to each choice variable. By the fundamental theorem of calculus, $\frac{d}{d\rho^*}\int_0^{\rho^*} \rho f\, d\rho = \rho^* f(\rho^*)$, and $\frac{d}{d\rho^*}F(\rho^*) = f(\rho^*)$. Term by term,

$$\frac{\partial U_b}{\partial \rho^*} = f(\rho^*)\, p_H R(I) - \rho^* f(\rho^*)\, I,$$
$$\frac{\partial[\text{LHS} - \text{RHS of (39)}]}{\partial \rho^*} = f(\rho^*)\, p_H \Big[R(I) - \frac{BI}{\Delta p}\Big] - \rho^* f(\rho^*)\, I,$$

so $\partial\mathcal{L}/\partial\rho^* = 0$ reads

$$0 = f(\rho^*)\big[p_H R(I) - \rho^* I\big] + \mu f(\rho^*)\Big[p_H R(I) - \frac{p_H BI}{\Delta p} - \rho^* I\Big].$$

Dividing by $f(\rho^*)I > 0$ and using $\rho_1(I) = p_H R(I)/I$ and $\rho_0(I) = p_H[R(I)/I - B/\Delta p]$,

$$0 = \big[\rho_1(I) - \rho^*\big] + \mu\big[\rho_0(I) - \rho^*\big]. \tag{40}$$

Collecting the ρ^* terms, $(1+\mu)\rho^* = \rho_1(I) + \mu\rho_0(I)$, so $\rho^* = \frac{\rho_1(I) + \mu\rho_0(I)}{1+\mu}$, a weighted average that lies between $\rho_0(I)$ and $\rho_1(I)$, the analogue of (5.8). With respect to I, the derivatives are

$$\frac{\partial U_b}{\partial I} = r + F(\rho^*) p_H R'(I) - 1 - \int_0^{\rho^*} \rho f\, d\rho,$$
$$\frac{\partial[\text{LHS} - \text{RHS of (39)}]}{\partial I} = r + F(\rho^*) p_H \Big(R'(I) - \frac{B}{\Delta p}\Big) - 1 - \int_0^{\rho^*} \rho f\, d\rho,$$

so $\partial\mathcal{L}/\partial I = 0$ is

$$0 = \Big[r + F(\rho^*)p_H R'(I) - 1 - \int_0^{\rho^*} \rho f\, d\rho\Big] + \mu\Big[r + F(\rho^*)p_H\Big(R'(I) - \frac{B}{\Delta p}\Big) - 1 - \int_0^{\rho^*} \rho f\, d\rho\Big]. \quad (41)$$

These two conditions, together with (39) holding with equality, determine (I, ρ^*, μ).

Step 3 — (i) Eliminating μ. The multiplier is not observable, so we combine the two conditions into one that involves only (I, ρ^*). Write the second bracket of (41) as the first minus $F(\rho^*)p_H B/\Delta p$ and collect:

$$(1+\mu)\Big[r + F(\rho^*)p_H R'(I) - 1 - \int_0^{\rho^*} \rho f\, d\rho\Big] = \mu F(\rho^*)\frac{p_H B}{\Delta p}. \quad (42)$$

From (40), $\mu(\rho^* - \rho_0) = \rho_1 - \rho^*$, so $\mu = \frac{\rho_1 - \rho^*}{\rho^* - \rho_0}$ and $1 + \mu = \frac{\rho^* - \rho_0 + \rho_1 - \rho^*}{\rho^* - \rho_0} = \frac{\rho_1 - \rho_0}{\rho^* - \rho_0}$, hence $\frac{\mu}{1+\mu} = \frac{\rho_1 - \rho^*}{\rho_1 - \rho_0}$. Since $\rho_1 - \rho_0 = p_H\frac{R(I)}{I} - p_H\Big(\frac{R(I)}{I} - \frac{B}{\Delta p}\Big) = \frac{p_H B}{\Delta p}$,

$$\frac{\mu}{1+\mu}\,\frac{p_H B}{\Delta p} = \frac{\rho_1(I) - \rho^*}{\rho_1(I) - \rho_0(I)}\,(\rho_1(I) - \rho_0(I)) = \rho_1(I) - \rho^*. \quad (43)$$

Dividing (42) by $1 + \mu$ and substituting (43):

$$r + F(\rho^*)p_H R'(I) - 1 - \int_0^{\rho^*} \rho f\, d\rho = F(\rho^*)\Big(\frac{p_H R(I)}{I} - \rho^*\Big).$$

Move the R terms to the left and the rest to the right:

$$\begin{aligned} F(\rho^*)\,p_H\Big[R'(I) - \frac{R(I)}{I}\Big] &= 1 - r + \int_0^{\rho^*} \rho f\, d\rho - \rho^* F(\rho^*) \\ &= 1 - r - \Big[\rho^* F(\rho^*) - \int_0^{\rho^*} \rho f\, d\rho\Big]. \end{aligned}$$

The bracket is $\int_0^{\rho^*}(\rho^* - \rho) f(\rho)\, d\rho$; integrating by parts with $F(0) = 0$, $\int_0^{\rho^*}(\rho^* - \rho) f\, d\rho = \big[(\rho^* - \rho)F(\rho)\big]_0^{\rho^*} + \int_0^{\rho^*} F(\rho)\, d\rho = \int_0^{\rho^*} F(\rho)\, d\rho$. Dividing by $F(\rho^*) > 0$,

$$p_H\Big[R'(I) - \frac{R(I)}{I}\Big] = \frac{1 - r - \int_0^{\rho^*} F(\rho)\, d\rho}{F(\rho^*)}. \quad (44)$$

The binding (39), solved for A and with $p_H[R(I) - BI/\Delta p] = I\rho_0(I)$, reads

$$\begin{aligned} A &= I - rI + \Big[\int_0^{\rho^*} \rho f(\rho)\, d\rho\Big] I - F(\rho^*)\, I\rho_0(I) \\ &= I\Big[1 - r + \int_0^{\rho^*} \rho f(\rho)\, d\rho - F(\rho^*)\rho_0(I)\Big]. \end{aligned} \quad (45)$$

Check: with constant returns, $R(I) = RI$, the left side of (44) is 0 and (44) becomes $\int_0^{\rho^*} F = 1 - r$, which is (5.6″). With strict concavity and $R(0) = 0$, $R(I)/I > R'(I)$, so the left side is negative and $\int_0^{\rho^*} F > 1 - r$: the marginal unit of scale is worth less than the average one, so at the margin the firm spends more on continuation and less on scale than (5.6″) would say.

Answer (i). The FOCs are (40), $\rho_1(I) - \rho^* + \mu[\rho_0(I) - \rho^*] = 0$, and (41); eliminating μ, the optimum (I, ρ^*) solves $p_H[R'(I) - R(I)/I] = \big[1 - r - \int_0^{\rho^*} F(\rho)\, d\rho\big]/F(\rho^*)$ together with the binding break-even (45), with $\rho_0(I) < \rho^* < \rho_1(I)$.

Step 4 — (ii) I and ρ^* move together. Condition (44) links the two choices; we find the sign of that link. Write $\varphi(I) \equiv p_H[R(I)/I - R'(I)] > 0$, increasing by assumption, and $h(\rho^*)$ for the

right side of (44), so (44) is $-\varphi(I) = h(\rho^*)$. The numerator of h is $N(\rho^*) = 1 - r - \int_0^{\rho^*} F$, with $N'(\rho^*) = -F(\rho^*)$; by the quotient rule,

$$\begin{aligned} h'(\rho^*) &= \frac{N'(\rho^*)F(\rho^*) - N(\rho^*)f(\rho^*)}{F(\rho^*)^2} \\ &= \frac{-F(\rho^*)^2 - f(\rho^*)\big[1 - r - \int_0^{\rho^*} F\big]}{F(\rho^*)^2} \\ &= -1 - \frac{f(\rho^*)}{F(\rho^*)}\, h(\rho^*). \end{aligned} \tag{46}$$

Assumption. Tirole's answer takes h decreasing. Since $h < 0$ at the optimum, (46) is negative iff $f(\rho^*)\big[\int_0^{\rho^*} F - (1-r)\big] < F(\rho^*)^2$. This holds, e.g., for F uniform on $[0, \bar\rho]$ ($f = 1/\bar\rho$, $F(\rho^*) = \rho^*/\bar\rho$, $\int_0^{\rho^*} F = \rho^{*2}/(2\bar\rho)$): multiplying by $\bar\rho^2$, the condition reads $\rho^{*2}/2 - \bar\rho(1-r) < \rho^{*2}$, always true for $r \le 1$. We assume it.

Then along (44) an increase in I lowers $-\varphi(I)$, which requires a lower h, i.e. a higher ρ^*. Totally differentiating $-\varphi(I) = h(\rho^*)$ gives $-\varphi'(I)\, dI = h'(\rho^*)\, d\rho^*$, so (44) defines $\rho^* = g(I)$ with

$$g'(I) = \frac{-\varphi'(I)}{h'(\rho^*)} > 0. \tag{47}$$

Step 5 — (ii) Effect of A. The binding break-even (45) tells how much net worth a given (I, ρ^*) requires; we differentiate its right side. With respect to ρ^*,

$$\begin{aligned} \frac{\partial A}{\partial \rho^*} &= I\big[\rho^* f(\rho^*) - f(\rho^*)\rho_0(I)\big] \\ &= I f(\rho^*)\big[\rho^* - \rho_0(I)\big] > 0, \end{aligned} \tag{48}$$

positive because $\rho^* > \rho_0(I)$ by (40). With respect to I (using $\frac{d}{dI}[I\rho_0(I)] = p_H[R'(I) - B/\Delta p]$),

$$\frac{\partial A}{\partial I} = 1 - r + \int_0^{\rho^*} \rho f\, d\rho - F(\rho^*) p_H \Big[R'(I) - \frac{B}{\Delta p}\Big].$$

The second bracket of (41) is $-\partial A/\partial I$ and the first is $-\partial A/\partial I + F(\rho^*)p_H B/\Delta p$, so (41) reads $-(1+\mu)\,\partial A/\partial I + F(\rho^*)p_H B/\Delta p = 0$, i.e.

$$\frac{\partial A}{\partial I} = \frac{F(\rho^*)\, p_H B}{(1+\mu)\Delta p} > 0. \tag{49}$$

Along the optimum, $\rho^* = g(I)$, so by (47), (48) and (49)

$$\frac{dA}{dI} = \frac{\partial A}{\partial I} + \frac{\partial A}{\partial \rho^*}\, g'(I) > 0, \tag{50}$$

hence $dI/dA = 1/(dA/dI) > 0$ and $d\rho^*/dA = g'(I)\, dI/dA > 0$: a weaker balance sheet lowers both the scale and the cutoff.

Step 6 — (ii) Debt maturity. Under the free-cash-flow assumption $r > \rho^*$ the optimum is implemented as in Section 5.6.1(a): the firm keeps $\rho^* I$ to meet shocks $\rho \le \rho^*$, must pay out $P_1 = (r - \rho^*)I$ at date 1 (short-term debt), is barred from diluting the initial investors, and the investors' long-term claim is $R(I) - BI/\Delta p$ in case of success. Per unit of investment the short-term debt is $r - \rho^*$, and by Step 5

$$\frac{d(r-\rho^*)}{dA} = -\frac{d\rho^*}{dA} < 0. \tag{51}$$

A firm with a weaker balance sheet (lower A) issues more short-term debt per unit of investment: its debt maturity is shorter. (The level $(r-\rho^*)I$ can move either way, because I falls too.)

Answer (ii). $dI/dA > 0$ and $d\rho^*/dA > 0$, so the short-term debt per unit of investment, $r - \rho^*$, falls with A: weaker balance sheets carry shorter maturity (more of the date-1 cash flow is paid out), provided $h'(\rho^*) < 0$ (true, e.g., for uniform shocks).

Intuition. A weak balance sheet makes pledgeable income scarce. The firm raises it by continuing less often, i.e. by keeping less liquidity in the firm, which means sweeping more of the date-1 cash flow out to creditors through short-term debt. Decreasing returns add one more channel: a lower scale raises $R(I)/I$ and hence per-unit pledgeable income, and the two margins (scale and continuation) are cut together.

Common mistakes.

- Carrying over the constant-returns result $\int_0^{\rho^*} F = 1 - r$, under which ρ^* does not depend on A: with $R'' < 0$ the left side of (44) is negative and moves with I, which is what makes ρ^* depend on A.
- Stating the FOCs without the multiplier on the break-even: the cutoff is a weighted average of ρ_1 (value) and ρ_0 (pledgeable income), not ρ_1 alone.
- Reading "shorter maturity" off the level $(r - \rho^*)I$: since I also falls with A, the unambiguous statement is per unit of investment.

Book answer. Matches the book's answer (p. 593): same condition (44) (the book writes its numerator as $1 - r - \int_0^{\rho^*}(\rho^* - \rho)f(\rho)d\rho$, equal to ours by integration by parts) and the same conclusion (lower $A \Rightarrow$ lower I and $\rho^* \Rightarrow$ more short-term debt). The book states without proof that the right side of (44) is decreasing in ρ^*; (46) shows this needs a condition on F (true for the uniform).

Ref: Tirole §5.3.2 "Continuum of Liquidity Shocks", pp. 209–210; §5.6.1 "Optimal Claim Maturity", p. 226; Exercise 5.7, p. 232; answer p. 593.

11. The priority game: uncoordinated lending and short-term bias — Tirole Exercise 5.9, p. 233

Statement.

Source: Tirole, Ch. 5, Exercise 5.9, p. 233 (PDF 246) · Answer in book: yes, p. 594

Exercise 5.9 (the priority game: uncoordinated lending leads to a short-term bias). This chapter, like Chapters 3 and 4, has assumed that the firm's balance sheet is transparent. In particular, each investor has perfect knowledge of loans made by other lenders and of the firm's obligations to them.

This exercise argues that uncoordinated lending leads to financing that is too oriented to the short term. In a nutshell, lenders, by cashing out early, exert a negative externality on other lenders. Because this externality is not internalized, the resulting financial structure contains too much short-term debt.

We consider a three-period model: $t = 0, 1, 2$. The entrepreneur has no cash ($A = 0$), is risk neutral, and is protected by limited liability. At date 0, a fixed investment I is made. The project yields a known return $r > 0$ at date 1, and an uncertain return (R or 0) at date 2. Because the point is quite general and does not require credit constraints, we assume away moral hazard; or, equivalently, the private benefit from misbehaving is 0. The probability of a date 2 success is

$$p + \tau(I_1),$$

where I_1 is the date-1 deepening investment, equal to r minus the level of short-term debt repaid to lenders and the date-1 payment to the entrepreneur (the firm does not return to the capital market at date 1), and τ is an increasing and concave function (with $\tau'(0) = \infty$). Assume that $\tau'(r)R < 1$.

We assume that the entrepreneur cannot engage in "fraud," that is, cannot fail to honor the short-term debt and, if the project succeeds at date 2, the long-term debt. By contrast, obligations to lenders, and in particular I_1, cannot be verified as the firm's balance sheet is opaque.

(i) Derive the first-best investment I_1^*. Show how this allocation can be implemented by a mixture of short- and long-term debt (note that in this model without moral hazard the structure of compensation for the entrepreneur exhibits a degree of indeterminacy).

(ii) Assume that $r - I_1^* < I$ (creditors must hold long-term debt). Suppose next that financing is not transparent. Start from the first-best solution, with a large number (a continuum of mass 1) of lenders, with the representative lender owning short-term claim r_l and contingent long-term claim R_l on the firm.

Show that the entrepreneur has an incentive to secretly collude with any lender to increase the latter's short-term claim in exchange for a smaller long-term claim.

Given the constraint that financing is provided by many lenders and that the latter do not observe each other's contracts, is the indeterminacy mentioned in question (i) resolved?

What is asked. (i) Find the first-best date-1 deepening investment I_1^* and show that a mix of short- and long-term debt implements it, with some indeterminacy in how the claims (and the entrepreneur's pay) are split. (ii) With an opaque balance sheet and many lenders, show that the entrepreneur gains by secretly swapping long-term claim for short-term claim with one lender, and say whether this pins down the split left open in (i).

Approach. Without moral hazard the borrower captures the NPV, so (i) is a plain optimisation plus counting equations against unknowns. In (ii) the date-1 payout to one lender is paid out of the reinvestment fund, lowering the success probability of *every* long-term claim: an externality that neither party to the side deal internalises (a common-agency, "priority" problem). Only collusion-proof structures survive, which is the lecture's point that maturity is chosen to discipline what the insider can do with unverifiable cash.

Notation.

- I: fixed date-0 investment; $A = 0$; no moral hazard (private benefit 0); risk-free rate 0.
- $r > 0$: known date-1 return; R or 0: date-2 return, success probability $p + \tau(I_1)$.
- $I_1 \geq 0$: date-1 deepening investment; τ increasing, concave, $\tau'(0) = \infty$, $\tau'(r)R < 1$.
- $r_l, r_b \geq 0$: date-1 payments to lenders (short-term debt) and to the entrepreneur; $R_l, R_b \geq 0$: success-contingent date-2 payments, $R_l + R_b = R$.
- I_1^*: first-best deepening investment; $\delta r_l, \delta R_l > 0$: sizes of a secret deal.

Step 1 — (i) First best. The first best maximises total surplus over I_1. The NPV is date-1 cash not reinvested plus expected date-2 income minus the initial investment:

$$\text{NPV}(I_1) = r - I_1 + [p + \tau(I_1)]R - I. \tag{52}$$

Its derivatives are $\text{NPV}'(I_1) = -1 + \tau'(I_1)R$ and $\text{NPV}''(I_1) = \tau''(I_1)R \leq 0$, so it is concave and the first-order condition $\text{NPV}'(I_1^*) = 0$ is sufficient:

$$-1 + \tau'(I_1^*)\,R = 0 \quad \Longleftrightarrow \quad \tau'(I_1^*)\,R = 1. \tag{53}$$

$\tau'(0) = \infty$ gives $I_1^* > 0$; $\tau'(r)R < 1 = \tau'(I_1^*)R$ and τ' decreasing give $I_1^* < r$, so $r - I_1^* > 0$ can be paid out at date 1.

Step 2 — (i) Implementation and indeterminacy. With transparent books the contract fixes the date-1 payouts, and hence $I_1 = r - r_l - r_b$. The first best requires

$$r_b + r_l + I_1^* = r, \tag{54}$$

$$R_b + R_l = R, \tag{55}$$

$$I = r_l + [p + \tau(I_1^*)]R_l, \tag{56}$$

the last being the lenders' breakeven condition. Four unknowns (r_b, r_l, R_b, R_l) and three equations: one degree of freedom. For instance (when $r - I_1^* < I$), maximal short-term debt $r_b = 0$, $r_l = r - I_1^*$ with $R_l = \frac{I-(r-I_1^*)}{p+\tau(I_1^*)}$; or pure long-term debt $r_l = 0$, $R_l = \frac{I}{p+\tau(I_1^*)}$ (if this is $\leq R$), with the entrepreneur taking $r_b = r - I_1^*$. Without moral hazard the entrepreneur gets the first-best NPV in all cases, so

the split of her pay between r_b and R_b, and of the lenders' claims between short and long term, is undetermined.

Answer (i). I_1^* solves $\tau'(I_1^*)R = 1$, with $0 < I_1^* < r$; any $(r_b, r_l, R_b, R_l) \geq 0$ with $r_b + r_l = r - I_1^*$, $R_b + R_l = R$ and $r_l + [p+\tau(I_1^*)]R_l = I$ implements it — one degree of freedom in the short-/long-term split.

Step 3 — (ii) The secret deal. With $r - I_1^* < I$, (56) and $r_l \leq r - I_1^*$ force $R_l > 0$, hence $R_b < R$ by (55). Starting from any first-best structure, the entrepreneur offers one lender $+\delta r_l$ in short-term claim against the transfer to herself of δR_l of his long-term claim. The lender is atomistic, takes I_1 as given, and accepts iff

$$\delta r_l \geq [p + \tau(I_1)]\, \delta R_l. \tag{57}$$

Date-1 cash is fixed and the other claims cannot be defrauded, so the extra payout comes out of deepening investment:

$$\delta I_1 = -\delta r_l. \tag{58}$$

The entrepreneur's utility is $U_b = r_b + [p + \tau(I_1)]R_b$; in the deal r_b is unchanged and she receives the long-term claim the lender gives up, $\delta R_b = \delta R_l$. To first order (product rule),

$$\delta U_b = \tau'(I_1)\,(\delta I_1)\, R_b + [p + \tau(I_1)]\,(\delta R_l).$$

Substituting (58) and (57) at equality ($\delta R_l = \delta r_l/[p + \tau(I_1)]$),

$$\begin{aligned} \delta U_b &= \tau'(I_1)\,(-\delta r_l)\, R_b + [p + \tau(I_1)]\Big(\frac{\delta r_l}{p + \tau(I_1)}\Big) \\ &= \big[1 - \tau'(I_1)R_b\big]\, \delta r_l. \end{aligned} \tag{59}$$

At $I_1 = I_1^*$, (53) gives $\tau'(I_1^*) = 1/R$, so $\tau'(I_1^*)R_b = R_b/R < 1$ and $\delta U_b = (1 - R_b/R)\, \delta r_l = \frac{R-R_b}{R}\, \delta r_l = (R_l/R)\, \delta r_l > 0$: the deal pays. The gain is exactly the loss $\tau'(I_1^*)R_l\, \delta r_l = (R_l/R)\delta r_l$ of the other long-term claimholders, whose success probability falls; total value is unchanged to first order because I_1^* maximises (52).

Step 4 — (ii) The indeterminacy is resolved. Lenders anticipate such deals, so only collusion-proof structures are equilibria. By (59) no deal pays iff

$$\tau'(I_1)\, R_b \geq 1. \tag{60}$$

The entrepreneur bears the NPV loss ex ante, and (52) rises with I_1 below I_1^*; since τ' falls with I_1, she sets I_1 as high as (60) allows, so (60) binds: $\tau'(I_1)R_b = 1$. Because $R_b < R$, $\tau'(I_1) > \tau'(I_1^*)$ and $I_1 < I_1^*$. A larger R_b permits a larger I_1, so R_b must be the largest compatible with breakeven, which requires paying her nothing at date 1, $r_b = 0$, and all date-1 free cash to lenders, $r_l = r - I_1$:

$$I = r - I_1 + [p + \tau(I_1)](R - R_b), \tag{61}$$

$$\tau'(I_1)\, R_b = 1. \tag{62}$$

These two equations pin down (I_1, R_b), hence $r_l = r - I_1$ and $R_l = R - R_b$. So yes, opacity resolves the indeterminacy: the entrepreneur is paid only long-term ($r_b = 0$), lenders hold more short-term debt ($r - I_1 > r - I_1^*$), and deepening investment is below the first best.

Answer (ii). At the first best a secret swap gives $\delta U_b = [1 - \tau'(I_1^*)R_b]\delta r_l = (R_l/R)\delta r_l > 0$. Yes, the indeterminacy is resolved: $r_b = 0$, $r_l = r - I_1$, and (I_1, R_b) solve $\tau'(I_1)R_b = 1$ and $I = r - I_1 + [p + \tau(I_1)](R - R_b)$, with $I_1 < I_1^*$ (a short-term bias).

Intuition. A lender who cashes out early is paid out of the firm's reinvestment fund, and the cost (a lower success probability) falls on every long-term claimant. Neither the colluding lender nor the entrepreneur internalises the loss of the other lenders, so with opaque books maturity is biased toward the short term. The entrepreneur can only commit not to strip the firm by holding a large long-term stake herself, and $\tau'(I_1)R_b = 1$ shows how large.

Common mistakes.

- Letting the colluding lender internalise the fall in I_1: he is atomistic, takes I_1 as given and accepts any deal with $\delta r_l \geq [p + \tau(I_1)]\delta R_l$.
- Counting the deal as value-creating: at I_1^* total value is unchanged to first order; the entrepreneur's gain is the other long-term holders' loss.
- Concluding that "more long-term debt" fixes the bias: what deters stripping is the entrepreneur's own long-term stake R_b, which is why $r_b = 0$.

Book answer. Matches the book's answer (p. 594): $\tau'(I_1^*)R = 1$; one degree of freedom; $\delta U_b = [1 - \tau'(I_1)R_b]\delta r_l > 0$; resolution with $r_b = 0$, $I = r - I_1 + [p + \tau(I_1)](R - R_b)$ and $\tau'(I_1)R_b = 1$, $I_1 < I_1^*$.
Ref: Tirole §5.6.1 "Optimal Claim Maturity", p. 226; Exercise 5.9, p. 233; answer p. 594.

12. Two-factor hedge with no cash outlay — Grinblatt & Titman 22.12, p. 816

Statement.
Source: Grinblatt–Titman, Ch. 22, Exercise 22.12, p. 816 (PDF 829) · Answer in book: no
Consider a two-factor model, where the factors are interest rate movements and changes in the exchange rate. Your company has a future cash flow with factor betas of 2 on the interest rate factor and 5 on the exchange rate factor. You would like to eliminate your sensitivity to both of these factors by means of financial securities, but do not wish to use any of the company's cash to do this.
The following investment opportunities are available to you.

- You can purchase 30-year government bonds.
- You can enter into a 2-year interest rate swap agreement.
- You can invest in a foreign index fund.

The following factor equations, with the two factors being changes in interest rates and changes in exchange rates, correspond to the future values of the three investment opportunities, respectively.

$$\tilde{C}_1 = 4 - 4\tilde{F}_{\text{int}} \quad \text{(per \$1 million invested)}$$
$$\tilde{C}_2 = 6 - 2\tilde{F}_{\text{int}} + 6\tilde{F}_{\text{ex}} \quad \text{(per \$1 million invested)}$$
$$\tilde{C}_3 = 3 - 3\tilde{F}_{\text{int}} - 2\tilde{F}_{\text{ex}} \quad \text{(per \$1 million invested)}$$

Design a proper hedge against interest rate movements and exchange rates in this environment.

What is asked. Choose positions in the three instruments (30-year bonds, 2-year swap, foreign index fund) so that the combined position cancels the firm's exposures to the interest-rate factor (beta 2) and the exchange-rate factor (beta 5), with no net cash invested.
Approach. G&T Result 22.7 and Example 22.16: factor betas add up, so a hedge is a portfolio of instruments whose betas are the negatives of the cash flow's betas. Two exposures plus a zero-cost condition give three linear equations in three unknowns.
Notation.

- x_1, x_2, x_3: \$ millions invested in bonds, swap and foreign fund (negative = short); each instrument's factor equation is per \$1 million invested, as the statement says (also for the swap).
- Betas per \$1 million: bonds $(-4, 0)$, swap $(-2, 6)$, fund $(-3, -2)$ on $(\tilde{F}_{\text{int}}, \tilde{F}_{\text{ex}})$; firm's cash flow betas $(2, 5)$.

Step 1 — Equations. Zero cost, then each factor beta of the hedge equal to minus the firm's:

$$x_1 + x_2 + x_3 = 0, \tag{63}$$
$$-4x_1 - 2x_2 - 3x_3 = -2, \tag{64}$$
$$0x_1 + 6x_2 - 2x_3 = -5. \tag{65}$$

Step 2 — Eliminate x_1. From (63), $x_1 = -x_2 - x_3$. Substituting in (64):

$$\begin{aligned} -4(-x_2 - x_3) - 2x_2 - 3x_3 &= -2 \\ 4x_2 + 4x_3 - 2x_2 - 3x_3 &= -2 \\ 2x_2 + x_3 &= -2 \quad \Longrightarrow \quad x_3 = -2 - 2x_2. \end{aligned} \tag{66}$$

Step 3 — Solve. Substituting (66) in (65):

$$\begin{aligned} 6x_2 - 2(-2 - 2x_2) &= -5 \\ 6x_2 + 4 + 4x_2 &= -5 \\ 10x_2 &= -9 \quad \Longrightarrow \quad x_2 = -0.9. \end{aligned}$$

Then $x_3 = -2 - 2(-0.9) = -0.2$ and $x_1 = -(-0.9) - (-0.2) = 1.1$.

Step 4 — Check. Cost: $1.1 - 0.9 - 0.2 = 0$. Interest beta: $-4(1.1) - 2(-0.9) - 3(-0.2) = -4.4 + 1.8 + 0.6 = -2$. Exchange beta: $6(-0.9) - 2(-0.2) = -5.4 + 0.4 = -5$. The hedge's constant term is $4(1.1) + 6(-0.9) + 3(-0.2) = 4.4 - 5.4 - 0.6 = -1.6$ ($ millions): the hedged cash flow is riskless with respect to both factors, and its level falls by 1.6.

Answer. Buy \$1.1 million of 30-year government bonds, take a short (pay-side) position of \$0.9 million in the 2-year swap, and short \$0.2 million of the foreign index fund. Net cost zero; interest-rate beta -2 and exchange-rate beta -5 offset the firm's $+2$ and $+5$.

Intuition. The bonds are the only instrument with no currency exposure, so they carry the interest-rate hedge; the swap and the fund carry opposite currency betas, so a short swap with a small short in the fund removes the currency exposure; the bond purchase is financed by the two short positions, which is what makes the hedge self-financing.

Common mistakes.

- Matching the betas with the same sign (doubling the exposure instead of cancelling it).
- Treating the swap as costless, as in Example 22.16: here its equation is per \$1 million invested, so it enters the budget. (If it were costless, the budget would be $x_1 + x_3 = 0$ and the solution $x_1 = 11$, $x_2 = -4.5$, $x_3 = -11$: a very different hedge.)
- Dropping the zero-cash condition: with two equations and three unknowns the hedge is not unique.

Ref: G&T §22.8, Result 22.7 and Example 22.16 "Eliminating Factor Loadings with Portfolios of Financial Instruments", pp. 805–807.

13. Continuous entrepreneurial effort and liquidity needs — Tirole Exercise 5.6, pp. 231–232

Statement.

Source: Tirole, Ch. 5, Exercise 5.6, p. 231 (PDF 244) · Answer in book: yes, p. 593

Exercise 5.6 (continuous entrepreneurial effort; liquidity needs). (i) An entrepreneur with initial cash A and protected by limited liability wants to invest in a fixed-size project with investment cost $I > A$. After the investment is made, the entrepreneur chooses the probability p of success ($0 \leqslant p \leqslant 1$); the disutility of effort is $g(p) = \frac{1}{2}p^2$. (The entrepreneur enjoys no private benefit in this model.) In question (i) only, the profit is $R = 2\sqrt{I - A}$ in the case of success and 0 in the case of failure. (We assume that $R < 1$ to avoid considering probabilities of success exceeding 1. R takes an arbitrary value in question (ii).) As usual, the uninformed investors demand an expected rate of interest equal to 0 and everyone is risk neutral. Let R_b denote the entrepreneur's reward in the case of success.

Solve for the optimal contract (R_b). Show that

$$R_b = \tfrac{1}{2}R.$$

(ii) Now introduce an intermediate liquidity shock ρ (for a now arbitrary level A of cash on hand). The cumulative distribution of ρ is $F(\rho)$ on $[0,\infty)$ and the density $f(\rho)$. The effort decision is made *after* the value of ρ is realized and, of course, *conditional* on the choice of continuing (incurring reinvestment cost ρ). *Suppose* that the entrepreneur's stake in the case of continuation (R_b) is independent of ρ. Write the investors' breakeven condition. Write the optimization program yielding (R_b, ρ^*), where ρ^* is the cutoff liquidity shock.

Context needed to solve it *From Tirole §5.2.1, pp. 202–204 (PDF 215–217):*

Consider the setup of Section 3.2, except that there is an intermediate date at which income accrues and some reinvestment need is realized. As indicated in Figure 5.1, the entrepreneur at date 0 has wealth A and borrows $I - A$, where I is the fixed cost of investment.

At date 1, the investment yields deterministic and verifiable income $r \geqslant 0$. Continuation, though, requires reinvesting an amount ρ, where ρ is *ex ante* unknown and has cumulative distribution function $F(\rho)$ with density $f(\rho)$ on $[0,\infty)$. The realization of ρ is learned at date 1. Note that we here assume that the date-1 income is deterministic while the reinvestment need is random. The important assumption is that at least one of the two is random.

If the firm does not reinvest ρ, then the firm is liquidated. The liquidation value is 0. If the firm reinvests ρ, then the firm yields, at date 2, R with probability p and 0 with probability $1-p$, where $p = p_H$ if the entrepreneur behaves (and then gets no private benefit) and $p = p_L = p_H - \Delta p$ if the entrepreneur misbehaves (in which case she receives private benefit B).

The entrepreneur and the investors are risk neutral, the entrepreneur is protected by limited liability, and the investors demand a rate of return equal to 0.

Thus, the model is nothing but an extension of the basic fixed-investment one in Section 3.2. We have just added an intermediate income r and a reinvestment need ρ (the bold type in Figure 5.1). (Put differently, the model of Section 3.2 corresponds to the special case $r = 0$ and F being a spike at $\rho = 0$.) We assume that there exists in the economy a store of value that yields the consumers' rate of interest (0 here). That is, 1 unit invested at date 0 delivers a return of 1 unit at date 1 (Chapter 15 will investigate the reasonableness of this assumption). We now give a *heuristic* description of the optimal contract.

Figure 5.1 (timing, transcribed): **0** — Entrepreneur has wealth A and fixed-investment project costing $I > A$. **1** — **Short-term income** $r > 0$**. Reinvestment need** ρ **(drawn from** $F(\cdot)$**).** Then, (if reinvestment): Moral hazard ($p = p_H$ or p_L). **2** — Success (profit R) with probability p, failure (profit 0) with probability $1-p$.

Suppose in a first step that the initial contract can specify whether the firm continues or liquidates for each value of ρ (as we will see, it actually does not matter whether the realized value of ρ is verifiable, as long as there is no use that can be made of the date-1 cash flow besides reinvesting it and distributing it to investors). Intuitively, it is optimal to continue whenever it is cheap to do so:

$$\rho \leqslant \rho^*,$$

where ρ^* is a cutoff.

As is now familiar to the reader, competition among investors deprives them of a surplus, and so the borrower's utility is equal to the NPV. Assuming, as usual, that the optimal contract induces the high effort in the case of continuation and noting that the probability of continuation is $\Pr(\rho \leqslant \rho^*) = F(\rho^*)$, the borrower's net utility is

$$U_b(\rho^*) = [r + F(\rho^*)p_H R] - \left[I + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho\right],$$

where the first bracket represents expected revenue and the second bracket total investment (initial investment plus expected reinvestment).

Ensuring good behavior in the case of continuation suggests giving to the entrepreneur, at date 2, R_b in the case of success and 0 in the case of failure, where

$$(\Delta p)R_b \geqslant B.$$

Furthermore, there is no loss of generality in assuming that the entrepreneur receives nothing at date 1. Suppose she receives $r_b > 0$. Then the contract could eliminate this short-term compensation and increase R_b by δR_b, so that the expected total reward remains constant: $F(\rho^*)p_H\delta R_b = r_b$. If anything, this substitution alleviates moral hazard in the case of continuation. [...]

The pledgeable income, $\mathcal{P}$, deflated by the investors' initial outlay, $I - A$, is therefore

$$\mathcal{P}(\rho^*) - (I - A) = \left[r + F(\rho^*)\left[p_H\left(R - \frac{B}{\Delta p}\right)\right]\right] - \left[I + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho - A\right],$$

since the entrepreneur no longer has cash and so the reinvestment must be paid out of either the investors' pocket or date-1 revenue.

Taking derivatives in U_b and $\mathcal{P}$, the key insights are as follows:

- The NPV (U_b) is increasing in the cutoff ρ^* as long as $\rho^* < p_H R$, and is decreasing thereafter. Intuitively, one would want to salvage an investment when the cost, ρ, of a rescue is smaller than the expected payoff, $p_H R$, of continuing.
- By contrast, the pledgeable income increases with ρ^* for $\rho^* < p_H(R - B/\Delta p)$ and decreases thereafter. This is again intuitive: investors have to bear the cost, ρ, of salvaging the investment and can put their hands on at most $p_H(R - B/\Delta p)$ given that the entrepreneur must be given incentives to behave in the case of continuation.

We are then led to consider three cases. Depending on the strength of the balance sheet, there may be (i) an efficient amount of liquidation, (ii) an overoptimal amount of liquidation to satisfy investors, or (iii) no funding at all:

(i) $\mathcal{P}(p_H R) \geqslant I - A$.

In this case, the "first-best cutoff" $\rho^* = p_H R$, which maximizes U_b, leaves sufficient income to investors. The contract then specifies, say, no compensation r_b at date 1 for the entrepreneur, and a reward R_b in the case of continuation and success at date 2.

(ii) $\mathcal{P}(p_H R) < I - A \leqslant \mathcal{P}\left(p_H\left(R - \dfrac{B}{\Delta p}\right)\right)$.

The optimal contract then specifies $r_b = 0$ and $R_b = B/\Delta p$. The entrepreneur receives nothing at the intermediate date and, in the case of continuation, receives the lowest compensation, $R_b = B/\Delta p$, that is incentive compatible. [...] The cutoff $\rho^* \in [p_H(R - B/\Delta p), p_H R]$ is then given by

$$r + F(\rho^*)\left[p_H\left(R - \frac{B}{\Delta p}\right)\right] = I + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho - A.$$

[...] In order to be able to invest more *ex ante*, the borrower accepts a level of reinvestment below the *ex post* efficient level ($\rho^* < p_H R$). The intuition is that, because incentives must be preserved, the borrower cannot pledge to the lenders the entire benefit of the reinvestment decision. Also, ρ^* exceeds the per-unit pledgeable income $p_H(R - B/\Delta p)$, which is the level that maximizes the borrowing capacity. [...]

(iii) $\mathcal{P}\left(p_H\left(R - \dfrac{B}{\Delta p}\right)\right) < I - A$.

In this case, funding is not feasible. The value ρ^* that maximizes the pledgeable income ($\rho^* = p_H(R - B/\Delta p)$) does not suffice to compensate the investors for their initial outlay.

What is asked. (i) With effort cost $\frac{1}{2}p^2$ and $R = 2\sqrt{I - A}$, find the entrepreneur's optimal reward R_b in case of success and show $R_b = \frac{1}{2}R$. (ii) Adding a liquidity shock (effort chosen after continuation, stake independent of ρ), write investors' breakeven condition and the program that determines (R_b, ρ^*).

Approach. Continuous-effort moral hazard (Tirole §3.5 style): the entrepreneur chooses p to maximise $pR_b - \frac{1}{2}p^2$, so effort rises with her stake; pledgeable income $p(R - R_b)$ is then a hump in R_b. For (ii), combine it with the cutoff structure of §5.2: continuation with probability $F(\rho^*)$ and expected reinvestment $\int_0^{\rho^*} \rho f$.

Notation.

- $p \in [0, 1]$: success probability chosen by the entrepreneur; $g(p) = \frac{1}{2}p^2$: effort cost; R_b: her reward on success; A, I: cash and investment.
- $\rho \sim F$ (density f) on $[0, \infty)$: reinvestment need; ρ^*: cutoff; μ, ν: multipliers of breakeven (in two equivalent programs).

Step 1 — (i) Effort choice. Given R_b, the entrepreneur solves $\max_p\{pR_b - \frac{1}{2}p^2\}$; the first-order condition is

$$R_b - p = 0 \quad \Longrightarrow \quad p = R_b \tag{67}$$

(interior because $R_b \le R < 1$).

Step 2 — (i) Breakeven. Investors receive $R - R_b$ with probability $p = R_b$:

$$R_b(R - R_b) = I - A. \tag{68}$$

The left side is maximal at $R_b = R/2$, where it equals $R^2/4$. With $R = 2\sqrt{I-A}$, $R^2/4 = I - A$: the maximal pledgeable income exactly covers the outlay. Formally, (68) is the quadratic $R_b^2 - RR_b + (I - A) = 0$ with discriminant $R^2 - 4(I - A) = 0$, so its unique root is

$$R_b = \frac{R}{2}. \tag{69}$$

Example: $I - A = 0.09$ gives $R = 2(0.3) = 0.6$, $R_b = p = 0.3$, and investors get $0.3(0.6 - 0.3) = 0.09$.

Answer (i). $p = R_b$ and breakeven $R_b(R - R_b) = I - A$; with $R = 2\sqrt{I-A}$ its only solution is $R_b = \frac{1}{2}R$.

Step 3 — (ii) Breakeven with liquidity shocks. After continuation, effort is again $p = R_b$ by (67), since R_b does not depend on ρ. Investors receive $R_b(R - R_b)$ in expectation per continuation, with probability $F(\rho^*)$, and pay $I - A$ plus the expected reinvestment:

$$F(\rho^*)\, R_b(R - R_b) \ge I - A + \int_0^{\rho^*} \rho f(\rho)\, d\rho. \tag{70}$$

Step 4 — (ii) Program. The entrepreneur's expected utility is $F(\rho^*)[pR_b - \frac{1}{2}p^2] = F(\rho^*)\frac{1}{2}R_b^2$, so

$$\max_{R_b,\ \rho^*} \ F(\rho^*)\,\frac{R_b^2}{2} \qquad \text{s.t. (70).} \tag{71}$$

Answer (ii). Breakeven (70); program (71).

Step 5 — (ii) Going further: first-order conditions. With multiplier μ on (70), $\mathcal{L} = F\frac{1}{2}R_b^2 + \mu\big[FR_b(R - R_b) - (I - A) - \int_0^{\rho^*}\rho f\big]$. The derivatives are

$$\frac{\partial \mathcal{L}}{\partial \rho^*} = f(\rho^*)\Big[\frac{R_b^2}{2} + \mu\big(R_b(R - R_b) - \rho^*\big)\Big] = 0 \implies \rho^* = R_b(R - R_b) + \frac{R_b^2}{2\mu}, \tag{72}$$

$$\frac{\partial \mathcal{L}}{\partial R_b} = F(\rho^*)\big[R_b + \mu(R - 2R_b)\big] = 0 \implies R_b(2\mu - 1) = \mu R \implies R_b = \frac{\mu R}{2\mu - 1}. \tag{73}$$

From (73), $\mu = R_b/(2R_b - R)$, so $\frac{R_b^2}{2\mu} = \frac{R_b(2R_b - R)}{2}$; substituting in (72),

$$\rho^* = R_bR - R_b^2 + R_b^2 - \frac{R_bR}{2} = \frac{R\,R_b}{2}. \tag{74}$$

Since the entrepreneur's utility plus the slack of (70) is the NPV, maximising NPV with multiplier ν gives the same optimum with $\mu = 1 + \nu \ge 1$, and (72) becomes $\rho^* = \frac{\rho_1 + \nu\rho_0}{1+\nu}$ with $\rho_0 \equiv R_b(R - R_b)$ (pledgeable per continuation) and $\rho_1 \equiv R_bR - \frac{1}{2}R_b^2$ (net value per continuation): the cutoff lies between them, as in (5.8). $R_b = \frac{(1+\nu)R}{1+2\nu} \in (\frac{1}{2}R, R]$. *Example* (own numbers): F uniform on $[0,1]$, $R = 0.8$, $I - A = 0.01$. Using (74), (70) binding is $0.4R_b \cdot R_b(0.8 - R_b) - \frac{1}{2}(0.4R_b)^2 = 0.24R_b^2 - 0.4R_b^3 = 0.01$, whose larger root is $R_b = 0.5$ ($0.06 - 0.05 = 0.01$). Then $\rho^* = 0.8(0.5)/2 = 0.2$, $\mu = 0.5/0.2 = 2.5$

($\nu = 1.5$), and (72): $0.5(0.3) + 0.25/5 = 0.15 + 0.05 = 0.2$; $\rho_0 = 0.15 < \rho^* < \rho_1 = 0.275$; utility $0.2 \times 0.25/2 = 0.025$ (a grid search over (R_b, ρ^*) confirms the optimum).

Intuition. A larger stake buys effort and the entrepreneur's own surplus but eats into what investors receive per continuation; continuing more often adds value but costs investors the reinvestment. The optimum trades off the two margins, and both end up between the "pledgeable" and the "total value" benchmarks familiar from the binary-effort model.

Common mistakes.

- Taking p as fixed when choosing R_b: effort responds to the stake, $p = R_b$.
- Writing the entrepreneur's utility as pR_b without the effort cost; with the cost it is $\frac{1}{2}R_b^2$.
- In (ii), letting effort depend on ρ: it is chosen after continuation, but with R_b independent of ρ it does not vary with ρ.

Book answer. Matches the book's answer (p. 593): $p = R_b$, $R_b(R - R_b) = I - A$ satisfied by $R_b = R/2$; breakeven $I - A + \int_0^{\rho^*} \rho f = F(\rho^*)R_b(R - R_b)$ and objective $F(\rho^*)R_b^2$ (ours has the factor $\frac{1}{2}$, which does not change the solution). Step 5 goes beyond the book.

Ref: Tirole §5.2.1, pp. 201–204; Exercise 5.6, pp. 231–232; answer p. 593.

14. Liquidity and deepening investment — Tirole Exercise 5.10, p. 234

Statement.

Source: Tirole, Ch. 5, Exercise 5.10, p. 234 (PDF 247) · Answer in book: yes, p. 595

Exercise 5.10 (liquidity and deepening investment). (i) Consider the fixed-investment model. The entrepreneur has cash A and can invest $I > A$ in a project. The project's return in the case of success (respectively, failure) is R (respectively, 0). The probability of success is p_H if the entrepreneur behaves (she then gets no private benefit) and $p_L = p_H - \Delta p$ if she misbehaves (in which case she gets private benefit B). In this subquestion and in the subsequent extension, one will assume that the project is viable only if the incentive scheme induces the entrepreneur to behave. The entrepreneur and the capital market are risk neutral; the entrepreneur is protected by limited liability; and the market rate of interest is equal to 0.

Let

$$\rho_1 \equiv p_H R \quad \text{and} \quad \rho_0 \equiv p_H[R - B/\Delta p],$$

and assume $\rho_1 > I > \rho_0$.

What is the necessary and sufficient condition for the project to be financed?

(ii) Now add an intermediate stage, in which there is an option to make a deepening investment. This investment increases the probability of success to $p_H + \tau$ (in the case of good behavior) and $p_L + \tau$ (in the case of misbehavior).

If the deepening investment is not made, the probabilities of success remain p_H and p_L, respectively. This deepening investment costs ρ, where ρ is unknown *ex ante* and distributed according to distribution $F(\rho)$ and density $f(\rho)$ on $[0, \infty)$. The timing is summarized in Figure 5.14.

Figure 5.14 (timing, transcribed):

0	1	(no date label)	2
Entrepreneur has wealth A and fixed-size investment project I.	ρ is realized. Reinvesting ρ raises probabilities of success to $p_\cdot + \tau$.	Moral hazard.	Outcome (R or 0).

Let $\mu \equiv \tau/p_H$, $\hat{\rho}_1 \equiv \mu\rho_1$, and $\hat{\rho}_0 \equiv \mu\rho_0$.

Write the incentive compatibility constraint and (for a given cutoff ρ^*) the investors' breakeven condition.

(iii) What is the optimal cutoff ρ^*? (Hint: consider three cases, depending on whether

$$\rho_0[1+\mu F(\hat{\rho}_k)] \lesseqgtr I - A + \int_0^{\hat{\rho}_k} \rho f(\rho)\,\mathrm{d}\rho,$$

with $k = 0, 1$.)

(iv) Should the firm content itself with returning to the capital market at date 1 in order to finance the deepening investment (if any)?

Context needed to solve it *From Tirole §5.2.1, pp. 202–204 (PDF 215–217):*

Consider the setup of Section 3.2, except that there is an intermediate date at which income accrues and some reinvestment need is realized. As indicated in Figure 5.1, the entrepreneur at date 0 has wealth A and borrows $I - A$, where I is the fixed cost of investment.

At date 1, the investment yields deterministic and verifiable income $r \geqslant 0$. Continuation, though, requires reinvesting an amount ρ, where ρ is *ex ante* unknown and has cumulative distribution function $F(\rho)$ with density $f(\rho)$ on $[0, \infty)$. The realization of ρ is learned at date 1. Note that we here assume that the date-1 income is deterministic while the reinvestment need is random. The important assumption is that at least one of the two is random.

If the firm does not reinvest ρ, then the firm is liquidated. The liquidation value is 0. If the firm reinvests ρ, then the firm yields, at date 2, R with probability p and 0 with probability $1 - p$, where $p = p_H$ if the entrepreneur behaves (and then gets no private benefit) and $p = p_L = p_H - \Delta p$ if the entrepreneur misbehaves (in which case she receives private benefit B).

The entrepreneur and the investors are risk neutral, the entrepreneur is protected by limited liability, and the investors demand a rate of return equal to 0.

Thus, the model is nothing but an extension of the basic fixed-investment one in Section 3.2. We have just added an intermediate income r and a reinvestment need ρ (the bold type in Figure 5.1). (Put differently, the model of Section 3.2 corresponds to the special case $r = 0$ and F being a spike at $\rho = 0$.) We assume that there exists in the economy a store of value that yields the consumers' rate of interest (0 here). That is, 1 unit invested at date 0 delivers a return of 1 unit at date 1 (Chapter 15 will investigate the reasonableness of this assumption). We now give a *heuristic* description of the optimal contract.

Figure 5.1 (timing, transcribed): **0** — Entrepreneur has wealth A and fixed-investment project costing $I > A$. **1** — **Short-term income** $r > 0$**. Reinvestment need** ρ **(drawn from** $F(\cdot)$**).** Then, (if reinvestment): Moral hazard ($p = p_H$ or p_L). **2** — Success (profit R) with probability p, failure (profit 0) with probability $1 - p$.

Suppose in a first step that the initial contract can specify whether the firm continues or liquidates for each value of ρ (as we will see, it actually does not matter whether the realized value of ρ is verifiable, as long as there is no use that can be made of the date-1 cash flow besides reinvesting it and distributing it to investors). Intuitively, it is optimal to continue whenever it is cheap to do so:

$$\rho \leqslant \rho^*,$$

where ρ^* is a cutoff.

As is now familiar to the reader, competition among investors deprives them of a surplus, and so the borrower's utility is equal to the NPV. Assuming, as usual, that the optimal contract induces the high effort in the case of continuation and noting that the probability of continuation is $\Pr(\rho \leqslant \rho^*) = F(\rho^*)$, the borrower's net utility is

$$U_b(\rho^*) = [r + F(\rho^*)p_H R] - \left[I + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho\right],$$

where the first bracket represents expected revenue and the second bracket total investment (initial investment plus expected reinvestment).

Ensuring good behavior in the case of continuation suggests giving to the entrepreneur, at date 2, R_b in the case of success and 0 in the case of failure, where

$$(\Delta p)R_b \geqslant B.$$

Furthermore, there is no loss of generality in assuming that the entrepreneur receives nothing at date 1. Suppose she receives $r_b > 0$. Then the contract could eliminate this short-term compensation and increase R_b by δR_b, so that the expected total reward remains constant: $F(\rho^*)p_H \delta R_b = r_b$. If anything, this substitution alleviates moral hazard in the case of continuation. [...]

The pledgeable income, $\mathcal{P}$, deflated by the investors' initial outlay, $I - A$, is therefore

$$\mathcal{P}(\rho^*) - (I - A) = \left[r + F(\rho^*)\left[p_H\left(R - \frac{B}{\Delta p}\right)\right]\right] - \left[I + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho - A\right],$$

since the entrepreneur no longer has cash and so the reinvestment must be paid out of either the investors' pocket or date-1 revenue.

Taking derivatives in U_b and $\mathcal{P}$, the key insights are as follows:

- The NPV (U_b) is increasing in the cutoff ρ^* as long as $\rho^* < p_H R$, and is decreasing thereafter. Intuitively, one would want to salvage an investment when the cost, ρ, of a rescue is smaller than the expected payoff, $p_H R$, of continuing.
- By contrast, the pledgeable income increases with ρ^* for $\rho^* < p_H(R - B/\Delta p)$ and decreases thereafter. This is again intuitive: investors have to bear the cost, ρ, of salvaging the investment and can put their hands on at most $p_H(R - B/\Delta p)$ given that the entrepreneur must be given incentives to behave in the case of continuation.

We are then led to consider three cases. Depending on the strength of the balance sheet, there may be (i) an efficient amount of liquidation, (ii) an overoptimal amount of liquidation to satisfy investors, or (iii) no funding at all:

(i) $\mathcal{P}(p_H R) \geqslant I - A$.

In this case, the "first-best cutoff" $\rho^* = p_H R$, which maximizes U_b, leaves sufficient income to investors. The contract then specifies, say, no compensation r_b at date 1 for the entrepreneur, and a reward R_b in the case of continuation and success at date 2.

(ii) $\mathcal{P}(p_H R) < I - A \leqslant \mathcal{P}\left(p_H\left(R - \dfrac{B}{\Delta p}\right)\right)$.

The optimal contract then specifies $r_b = 0$ and $R_b = B/\Delta p$. The entrepreneur receives nothing at the intermediate date and, in the case of continuation, receives the lowest compensation, $R_b = B/\Delta p$, that is incentive compatible. [...] The cutoff $\rho^* \in [p_H(R - B/\Delta p), p_H R]$ is then given by

$$r + F(\rho^*)\left[p_H\left(R - \frac{B}{\Delta p}\right)\right] = I + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho - A.$$

[...] In order to be able to invest more *ex ante*, the borrower accepts a level of reinvestment below the *ex post* efficient level ($\rho^* < p_H R$). The intuition is that, because incentives must be preserved, the borrower cannot pledge to the lenders the entire benefit of the reinvestment decision. Also, ρ^* exceeds the per-unit pledgeable income $p_H(R - B/\Delta p)$, which is the level that maximizes the borrowing capacity. [...]

(iii) $\mathcal{P}\left(p_H\left(R - \dfrac{B}{\Delta p}\right)\right) < I - A$.

In this case, funding is not feasible. The value ρ^* that maximizes the pledgeable income ($\rho^* = p_H(R - B/\Delta p)$) does not suffice to compensate the investors for their initial outlay.

From Tirole §5.2.3 "Credit Lines for Cash-Poor Firms", pp. 205–206 (PDF 218–219):

Suppose in contrast that the investment "takes a long time" to produce income. At the extreme, there is no short-term profit: $r = 0$.

Can the entrepreneur just "wait and see," that is, borrow I at date 0 in exchange for shares in the firm and return to the capital market at date 1 if need occurs? Let us thus assume that the entrepreneur does not plan her liquidity in advance and that the liquidity shock occurs at date 1. To raise cash on the capital market to pay ρ, the entrepreneur must issue new shares and thereby dilute historical investors. Letting $\rho_0 \equiv p_H(R - B/\Delta p)$, and to illustrate this dilution, suppose that the entrepreneur faces a liquidity shock $\rho = \frac{1}{2}\rho_0$. The value of external shares held by initial investors is equal to ρ_0. Suppose that the number of shares is doubled. That is, as many shares are sold to new investors as already exist. So the value of each share is halved. The firm thereby raises $\frac{1}{2}\rho_0 = \rho$ in cash and can withstand the liquidity shock. Are initial investors willing to let themselves be diluted? The value of their shares is, of course, reduced to $\frac{1}{2}\rho_0$. But contemplate the alternative of liquidating the investment, under which the initial investors receive nothing! Thus, initial investors are willing to accept the dilution.

Similarly, to meet a liquidity shock equal to $\frac{3}{4}\rho_0$, the firm must quadruple the number of shares, and so on. But there is an upper bound to this process: investors will never pay more than the firm is worth to them. Hence, even in a frictionless capital market, the firm cannot raise more than ρ_0. Going back to the capital market at date 1 then at best allows the firm to withstand a liquidity shock of magnitude

$$\rho \leqslant \rho_0 = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the optimal financing arrangement specifies

$$p_H \left(R - \frac{B}{\Delta p} \right) < \rho^* \leqslant p_H R,$$

the entrepreneur must secure a line of credit or hoard liquidity in order to face the date-1 liquidity shock. We will shortly describe how to do so, but there are basically two alternatives and combinations thereof: a credit line or liquid assets of magnitude ρ^* with no right to dilute existing claimholders by issuing new claims at date 1 (so the entrepreneur borrows $I + \rho^*$); or a smaller credit line or amount of liquid assets, equal to $[\rho^* - p_H(R - B/\Delta p)]$ with a right to dilute claimholders as needed to ensure continuation. Either way, the entrepreneur must plan liquidity management.

The optimum can be implemented by a nonrevokable line of credit granted by, say, one of the lenders (a bank) at level ρ^*. It is important that this line of credit be nonrevokable (in a broad sense: see below). Otherwise the lender would have an incentive not to abide by his promise to rescue the firm if $\rho > \rho_0$, that is, if the liquidity shock exceeds the date-1 pledgeable income ρ_0. [...]

Remark (renegotiation). Could this line of credit be renegotiated to the parties' mutual advantage once the fraction ρ is realized? First, note that if $\rho \leqslant \rho^*$, then *a fortiori* $\rho < p_H R$ and therefore it is *ex post* efficient to continue; so there is no scope for a renegotiation in which the lender would compensate the borrower for not using the credit line, as this renegotiation would reduce total surplus and therefore at least one of the parties would be strictly better off not renegotiating. Second, could the two parties both benefit from an increase in the line of credit to ρ when $\rho^* < \rho \leqslant p_H R$? Even though this increase would yield the ex post efficient reinvestment policy, there is no way for the borrower to compensate the lender, again because the borrower's stake is incompressible. One can show that the lender will turn down any request for an increase in the credit line. So will any alternative lender (other lenders may have even less incentives to refinance, because unlike the initial lender they do not have a vested stake to lose).

From Tirole §5.2.4 "A Reinterpretation: Growth Prospects", pp. 206–207 (PDF 219–220):

In the basic model, the firm is liquidated if it does not meet the liquidity shock. In a straightforward reinterpretation, it continues as is, but cannot take advantage of a profitable growth opportunity if it does not come up with enough cash to reinvest.

Suppose that, at date 1, the firm still receives deterministic income r, but, in the absence of cash reinjection at date 1, continues and succeeds with probability $p = p_H$ or p_L, depending on whether the entrepreneur behaves or misbehaves at date 1. At date 1, though, the firm can raise its date-2 expected profit by reinvesting. One way of formalizing this is to assume that the payoffs in the cases of success and failure remain R and 0, respectively, but the probability of success in the case of reinvestment becomes $p + \tau$, where $\tau > 0$ and $p = p_H$ or $p = p_L$, depending on whether the entrepreneur behaves or misbehaves. This separable form is handy as it implies that the entrepreneur's incentive compatibility constraint is not affected by the reinvestment:

$$(p_H + \tau)R_b \geqslant (p_L + \tau)R_b + B \iff (\Delta p)R_b \geqslant B.$$

The reinvestment cost ρ is drawn at date 1 from the cumulative distribution function $F(\rho)$ with density $f(\rho)$ on $[0, \infty)$.

It is clearly optimal to reinvest if and only if ρ is below some cutoff ρ^*. The entrepreneur's utility, equal to the NPV, is

$$U_b(\rho^*) = [r + [p_H + F(\rho^*)\tau]R] - \left[I + \int_0^{\rho^*} \rho f(\rho)\, \mathrm{d}\rho \right].$$

As earlier, the interesting case arises when the firm is financially constrained but nonetheless can raise funds (a situation equivalent to that labeled region (ii) in Section 5.2.1); the cutoff is then given by the investors' breakeven condition:

$$r + [p_H + F(\rho^*)\tau] \left(R - \frac{B}{\Delta p} \right) = [I - A] + \int_0^{\rho^*} \rho f(\rho)\, \mathrm{d}\rho$$

and

$$\tau \left(R - \frac{B}{\Delta p} \right) \leqslant \rho^* < \tau R.$$

The latter set of inequalities expresses the fact that reinvestment is first-best suboptimal ($\rho^* < \tau R$), but occurs whenever it boosts pledgeable income ($\rho^* \geqslant \tau[R - (B/\Delta p)]$).

What is asked. (i) The condition for the fixed-size project to be financed. (ii) With an optional date-1 deepening investment of random cost ρ that raises the success probability by τ: write the incentive constraint and investors' breakeven for a given cutoff ρ^*. (iii) Find the optimal cutoff (three cases). (iv) Say whether the firm can rely on returning to the market at date 1 to pay for the deepening investment.

Approach. The growth-prospects reinterpretation of the liquidity model (§5.2.4): failing to meet the shock does not kill the firm but forgoes a profitable expansion. NPV is maximised at the first-best cutoff τR, pledgeable income at $\tau(R - B/\Delta p)$; the balance sheet decides which is reachable. The separable form $p + \tau$ keeps the incentive constraint unchanged.

Notation.

- $\rho_1 = p_H R$, $\rho_0 = p_H(R - B/\Delta p)$, $\rho_1 > I > \rho_0$; $\mu \equiv \tau/p_H$, $\hat{\rho}_1 \equiv \mu\rho_1 = \tau R$, $\hat{\rho}_0 \equiv \mu\rho_0 = \tau(R - B/\Delta p)$.
- $\rho \sim F$ (density f) on $[0, \infty)$: cost of deepening; ρ^*: cutoff (deepen iff $\rho \leq \rho^*$).

Step 1 — (i) Financing condition. With $(\Delta p)R_b \geq B$ and $R_b = B/\Delta p$, investors' breakeven is

$$p_H\Big(R - \frac{B}{\Delta p}\Big) \geq I - A \quad \Longleftrightarrow \quad A \geq I - \rho_0. \tag{75}$$

Answer (i). $\rho_0 \geq I - A$, i.e. $A \geq I - \rho_0 > 0$.

Step 2 — (ii) Incentive constraint. If the firm has deepened, behaving gives $(p_H + \tau)R_b$, misbehaving $(p_L + \tau)R_b + B$:

$$(p_H + \tau)R_b \geq (p_L + \tau)R_b + B \quad \Longleftrightarrow \quad (\Delta p)R_b \geq B, \tag{76}$$

the same as without deepening.

Step 3 — (ii) Breakeven. The success probability is $p_H + \tau$ with probability $F(\rho^*)$ and p_H otherwise; investors also pay the expected deepening cost:

$$\big[F(\rho^*)(p_H + \tau) + (1 - F(\rho^*))p_H\big](R - R_b) \geq I - A + \int_0^{\rho^*} \rho f(\rho)\, d\rho. \tag{77}$$

At $R_b = B/\Delta p$ the left side is $[p_H + F(\rho^*)\tau](R - B/\Delta p) = \rho_0 + F(\rho^*)\tau(R - B/\Delta p) = \rho_0[1 + \mu F(\rho^*)]$, using $\tau(R - B/\Delta p) = \frac{\tau}{p_H}\rho_0 = \mu\rho_0$.

Answer (ii). $(\Delta p)R_b \geq B$ and (77); with the minimal stake, $\rho_0[1 + \mu F(\rho^*)] \geq I - A + \int_0^{\rho^*} \rho f$.

Step 4 — (iii) The two benchmarks. The NPV is $U_b(\rho^*) = [p_H + F(\rho^*)\tau]R - I - \int_0^{\rho^*} \rho f$, so

$$U_b'(\rho^*) = f(\rho^*)\big[\tau R - \rho^*\big], \tag{78}$$

positive below and negative above $\hat{\rho}_1 = \tau R$: the first best is $\rho^* = \hat{\rho}_1$. The pledgeable surplus $\mathcal{P}(\rho^*) \equiv \rho_0[1 + \mu F(\rho^*)] - \int_0^{\rho^*} \rho f - (I - A)$ has

$$\mathcal{P}'(\rho^*) = f(\rho^*)\big[\mu\rho_0 - \rho^*\big], \tag{79}$$

so it rises up to $\hat{\rho}_0$ and falls after; $\hat{\rho}_0 < \hat{\rho}_1$.

Step 5 — (iii) Three cases. The optimum is the largest cutoff up to $\hat{\rho}_1$ with $\mathcal{P} \geq 0$. *Case 1* (first best): $\rho_0[1 + \mu F(\hat{\rho}_1)] \geq I - A + \int_0^{\hat{\rho}_1} \rho f$; then $\rho^* = \hat{\rho}_1$. *Case 2* (no funding): $\rho_0[1 + \mu F(\hat{\rho}_0)] <$

$I - A + \int_0^{\hat{\rho}_0} \rho f$; even the pledgeable-income-maximising cutoff does not repay investors. *Case 3* (in between): $\rho^* \in [\hat{\rho}_0, \hat{\rho}_1)$ solves

$$\rho_0[1 + \mu F(\rho^*)] = I - A + \int_0^{\rho^*} \rho f(\rho)\, d\rho, \tag{80}$$

the root on the decreasing branch of $\mathcal{P}$. *Example* (own numbers): $p_H = 0.5$, $p_L = 0.2$, $R = 4$, $B = 0.6$, $I = 1.5$, $\tau = 0.25$; F uniform on $[0, 2]$, so $F(\rho) = \rho/2$ and $\int_0^x \rho f = x^2/4$. Then $\Delta p = 0.3$, $B/\Delta p = 2$, $p_L R + B = 1.4 < I$, $\rho_1 = 2$, $\rho_0 = 0.5(4 - 2) = 1$, $\mu = 0.5$, $\hat{\rho}_1 = 1$, $\hat{\rho}_0 = 0.5$. Case 1 requires $1 \times (1 + 0.5 \times 0.5) = 1.25 \geq 1.5 - A + 0.25$, i.e. $A \geq 0.5$. Case 2 is $1 \times (1 + 0.5 \times 0.25) = 1.125 < 1.5 - A + 0.0625$, i.e. $A < 0.4375$. For $A = 0.46$, (80) reads $1 + 0.25\rho^* = 1.04 + \rho^{*2}/4$, i.e. $\rho^{*2} - \rho^* + 0.16 = 0$, with roots $(1 \pm \sqrt{1 - 0.64})/2 = (1 \pm 0.6)/2$, i.e. 0.8 and 0.2; the optimum is $\rho^* = 0.8 \in [0.5, 1)$. Note that without the deepening option the firm needs $A \geq I - \rho_0 = 0.5$; the option lowers the funding threshold to $0.5 - \int_0^{0.5} F(\rho) d\rho = 0.5 - 0.0625 = 0.4375$, because cheap deepening ($\rho < \hat{\rho}_0$) raises pledgeable income.

Answer (iii). $\rho^* = \hat{\rho}_1 = \tau R$ in Case 1; no funding in Case 2; otherwise $\rho^* \in [\hat{\rho}_0, \hat{\rho}_1)$ given by (80).

Step 6 — (iv) Returning to the market. At date 1, without deepening, the initial investors' claim is worth ρ_0 anyway, so they will not accept dilution beyond what deepening adds to the outside claims. Deepening raises the value of outside claims by $\tau(R - B/\Delta p) = \hat{\rho}_0$, the most new investors can be promised. The market therefore finances deepening only for $\rho \leq \hat{\rho}_0$, while the optimal cutoff is $\rho^* \geq \hat{\rho}_0$, strictly above in Case 1 and generically in Case 3. The firm must hoard liquidity, or obtain a credit line, of $\rho^* - \hat{\rho}_0$ (0.3 in the example) with the right to dilute for the rest.

Answer (iv). No (unless $\rho^* = \hat{\rho}_0$): the market can pay at most $\hat{\rho}_0 = \mu\rho_0$, less than the value $\hat{\rho}_1 = \mu\rho_1$ of the reinvestment and less than ρ^*; the firm must secure liquidity in advance.

Intuition. The deepening investment is a growth option. Its value to the firm is τR but to outsiders only $\tau(R - B/\Delta p)$, because part of the extra income must stay with the manager to keep her working. Growth options that the market alone will not finance must be funded by hoarded cash or committed lines, exactly as liquidity shocks are in the basic model.

Common mistakes.

- Writing breakeven with success probability $p_H + \tau$ in all states: deepening happens only with probability $F(\rho^*)$.
- In Case 3, picking the smaller root of (80) (0.2 in the example): it lies on the increasing branch of $\mathcal{P}$ and gives lower NPV.
- In (iv), allowing date-1 investors $\rho_0 + \hat{\rho}_0$: the initial investors keep ρ_0 without deepening and will not give it up.

Book answer. Matches the book's answer (p. 595): $(\Delta p)R_b \geq B$; breakeven (77); first best $\rho^* = \tau R = \hat{\rho}_1$; the three cases with the same inequalities; in (iv) liquidity must be hoarded whenever $\rho^* > \hat{\rho}_0$, since investors' maximal return $\mu\rho_0$ is below the value $\mu\rho_1$.

Ref: Tirole §5.2.1, pp. 201–204; §5.2.3, pp. 205–206; §5.2.4 "A Reinterpretation: Growth Prospects", pp. 206–207; Exercise 5.10, p. 234; answer p. 595.

15. Multistage investment with interim news about prospects — Tirole Exercise 5.8, pp. 232–233

Statement.

Source: Tirole, Ch. 5, Exercise 5.8, p. 232 (PDF 245) · Answer in book: yes, p. 593

Exercise 5.8 (multistage investment with interim accrual of information about prospects). In this chapter we have focused mostly on the case of shocks about the reinvestment need (cost overruns, say). Consider,

instead, the case of news about the final profitability. In the two-outcome framework, news can accrue about either the probability of success or the payoff in the case of success. We consider both, in sequence. The investment is a multistage one: let

$$I = I_0 + I_1,$$

where I_0 is the date-0 investment and I_1 is the date-1 reinvestment. In contrast with I_0, I_1 is not incurred if the firm decides to stop. The timing is as in Figure 5.12.

Figure 5.12 (timing, transcribed): Date 0 — entrepreneur and investors contract, and sink I_0. Date 1 — news about date-2 prospect accrues (publicly); then either stop (no liquidation value) or invest I_1, followed by moral hazard. Date 2 — outcome: success (R) or failure (0).

	Pr (success)	Private benefit
Behaves	$p_H + \tau$	0
Misbehaves	$p_L + \tau$	B

As usual, the entrepreneur has initial wealth A, is risk neutral, and protected by limited liability. Investors are risk neutral. The discount rate is equal to 0. If reinvestment cost I_1 is sunk at date 1, then the firm can continue. Misbehavior reduces the probability of success by Δp, but yields private benefit B to the entrepreneur.

Assume

$$B < (\Delta p)R.$$

As announced, we consider two variants.

(a) *News about the probability of success.* R is known at date 0, but the probability of success is $p_H + \tau$ in the case of good behavior and $p_L + \tau$ in the case of misbehavior, where τ is publicly learned at the beginning of date 1. The random variable τ is distributed according to the distribution function $F(\tau)$ with density $f(\tau)$ on $[\underline{\tau}, \bar{\tau}] = [-p_L, 1 - p_H]$ (to keep probabilities in the interval $[0, 1]$). Let τ^e denote the expectation of τ.

(b) *News about the payoff in the case of success.* The probabilities of success are known: p_H and p_L (normalize: $\tau = 0$). By contrast, the profit R in the case of success is drawn from distribution $G(R)$ with density $g(R)$ on $(0, \infty)$. (The profit in the case of failure is always equal to 0.)

(c) For each variant, show that there exist two thresholds, A_0 and A_1, $A_0 < A_1$, such that the first best prevails for $A \geqslant A_1$ and financing is secured if and only if $A \geqslant A_0$. Show that the continuation rules take the form of cutoffs, as described in Figure 5.13. Determine $\tau_0^*, \tau_1^*, R_0^*, R_1^*$.

Figure 5.13 (transcribed): panel (a) plots the cutoff τ^* against A — equal to τ_0^* at A_0, decreasing until it reaches τ_1^* at A_1, and flat at τ_1^* beyond A_1; panel (b) plots the cutoff R^* the same way, from R_0^* at A_0 down to R_1^* at A_1, and flat at R_1^* beyond A_1. In both panels nothing is plotted to the left of A_0; dashed lines mark A_0, A_1 and the two cutoff levels.

(ii) For each variant, assume that $A = A_0$. Let $y \equiv (p_H + \tau)R$ denote the expected income, and $\mathcal{R}(y)$ denote the entrepreneur's rent in the case of continuation. Show that (above the threshold y^*)

- $0 < \mathcal{R}'(y) < 1$ in variant (a);
- $\mathcal{R}$ is constant in variant (b).

Context needed to solve it *From Tirole §5.2.1, pp. 202–204 (PDF 215–217):*

Consider the setup of Section 3.2, except that there is an intermediate date at which income accrues and some reinvestment need is realized. As indicated in Figure 5.1, the entrepreneur at date 0 has wealth A and borrows $I - A$, where I is the fixed cost of investment.

At date 1, the investment yields deterministic and verifiable income $r \geqslant 0$. Continuation, though, requires reinvesting an amount ρ, where ρ is *ex ante* unknown and has cumulative distribution function $F(\rho)$ with density $f(\rho)$ on $[0, \infty)$. The realization of ρ is learned at date 1. Note that we here assume that the date-1 income is deterministic while the reinvestment need is random. The important assumption is that at least one of the two is random.

If the firm does not reinvest ρ, then the firm is liquidated. The liquidation value is 0. If the firm reinvests ρ, then the firm yields, at date 2, R with probability p and 0 with probability $1 - p$, where $p = p_H$ if the

entrepreneur behaves (and then gets no private benefit) and $p = p_L = p_H - \Delta p$ if the entrepreneur misbehaves (in which case she receives private benefit B).

The entrepreneur and the investors are risk neutral, the entrepreneur is protected by limited liability, and the investors demand a rate of return equal to 0.

Thus, the model is nothing but an extension of the basic fixed-investment one in Section 3.2. We have just added an intermediate income r and a reinvestment need ρ (the bold type in Figure 5.1). (Put differently, the model of Section 3.2 corresponds to the special case $r = 0$ and F being a spike at $\rho = 0$.) We assume that there exists in the economy a store of value that yields the consumers' rate of interest (0 here). That is, 1 unit invested at date 0 delivers a return of 1 unit at date 1 (Chapter 15 will investigate the reasonableness of this assumption). We now give a *heuristic* description of the optimal contract.

Figure 5.1 (timing, transcribed): **0** — Entrepreneur has wealth A and fixed-investment project costing $I > A$. **1** — **Short-term income** $r > 0$**. Reinvestment need** ρ **(drawn from** $F(\cdot)$**).** Then, (if reinvestment): Moral hazard ($p = p_H$ or p_L). **2** — Success (profit R) with probability p, failure (profit 0) with probability $1 - p$.

Suppose in a first step that the initial contract can specify whether the firm continues or liquidates for each value of ρ (as we will see, it actually does not matter whether the realized value of ρ is verifiable, as long as there is no use that can be made of the date-1 cash flow besides reinvesting it and distributing it to investors). Intuitively, it is optimal to continue whenever it is cheap to do so:

$$\rho \leqslant \rho^*,$$

where ρ^* is a cutoff.

As is now familiar to the reader, competition among investors deprives them of a surplus, and so the borrower's utility is equal to the NPV. Assuming, as usual, that the optimal contract induces the high effort in the case of continuation and noting that the probability of continuation is $\Pr(\rho \leqslant \rho^*) = F(\rho^*)$, the borrower's net utility is

$$U_b(\rho^*) = [r + F(\rho^*)p_H R] - \left[I + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho\right],$$

where the first bracket represents expected revenue and the second bracket total investment (initial investment plus expected reinvestment).

Ensuring good behavior in the case of continuation suggests giving to the entrepreneur, at date 2, R_b in the case of success and 0 in the case of failure, where

$$(\Delta p)R_b \geqslant B.$$

Furthermore, there is no loss of generality in assuming that the entrepreneur receives nothing at date 1. Suppose she receives $r_b > 0$. Then the contract could eliminate this short-term compensation and increase R_b by δR_b, so that the expected total reward remains constant: $F(\rho^*)p_H\delta R_b = r_b$. If anything, this substitution alleviates moral hazard in the case of continuation. [...]

The pledgeable income, $\mathcal{P}$, deflated by the investors' initial outlay, $I - A$, is therefore

$$\mathcal{P}(\rho^*) - (I - A) = \left[r + F(\rho^*)\left[p_H\left(R - \frac{B}{\Delta p}\right)\right]\right] - \left[I + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho - A\right],$$

since the entrepreneur no longer has cash and so the reinvestment must be paid out of either the investors' pocket or date-1 revenue.

Taking derivatives in U_b and $\mathcal{P}$, the key insights are as follows:

- The NPV (U_b) is increasing in the cutoff ρ^* as long as $\rho^* < p_H R$, and is decreasing thereafter. Intuitively, one would want to salvage an investment when the cost, ρ, of a rescue is smaller than the expected payoff, $p_H R$, of continuing.
- By contrast, the pledgeable income increases with ρ^* for $\rho^* < p_H(R - B/\Delta p)$ and decreases thereafter. This is again intuitive: investors have to bear the cost, ρ, of salvaging the investment and can put their hands on at most $p_H(R - B/\Delta p)$ given that the entrepreneur must be given incentives to behave in the case of continuation.

We are then led to consider three cases. Depending on the strength of the balance sheet, there may be (i) an efficient amount of liquidation, (ii) an overoptimal amount of liquidation to satisfy investors, or (iii) no funding at all:

(i) $\mathcal{P}(p_H R) \geqslant I - A$.

In this case, the "first-best cutoff" $\rho^* = p_H R$, which maximizes U_b, leaves sufficient income to investors. The contract then specifies, say, no compensation r_b at date 1 for the entrepreneur, and a reward R_b in the case of continuation and success at date 2.

(ii) $\mathcal{P}(p_H R) < I - A \leqslant \mathcal{P}\left(p_H\left(R - \dfrac{B}{\Delta p}\right)\right)$.

The optimal contract then specifies $r_b = 0$ and $R_b = B/\Delta p$. The entrepreneur receives nothing at the intermediate date and, in the case of continuation, receives the lowest compensation, $R_b = B/\Delta p$, that is incentive compatible. [...] The cutoff $\rho^* \in [p_H(R - B/\Delta p), p_H R]$ is then given by

$$r + F(\rho^*)\left[p_H\left(R - \frac{B}{\Delta p}\right)\right] = I + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho - A.$$

[...] In order to be able to invest more *ex ante*, the borrower accepts a level of reinvestment below the *ex post* efficient level ($\rho^* < p_H R$). The intuition is that, because incentives must be preserved, the borrower cannot pledge to the lenders the entire benefit of the reinvestment decision. Also, ρ^* exceeds the per-unit pledgeable income $p_H(R - B/\Delta p)$, which is the level that maximizes the borrowing capacity. [...]

(iii) $\mathcal{P}\left(p_H\left(R - \dfrac{B}{\Delta p}\right)\right) < I - A$.

In this case, funding is not feasible. The value ρ^* that maximizes the pledgeable income ($\rho^* = p_H(R - B/\Delta p)$) does not suffice to compensate the investors for their initial outlay.

What is asked. (i) For news about the success probability (variant a) and news about the payoff (variant b): show there are two wealth thresholds $A_0 < A_1$, with the first best above A_1 and financing iff $A \geq A_0$; show the continuation rules are cutoffs, and find $\tau_0^*, \tau_1^*, R_0^*, R_1^*$. (ii) At $A = A_0$, show that the entrepreneur's continuation rent, as a function of expected income y, has slope in $(0, 1)$ in variant (a) and is constant in variant (b).

Approach. Same structure as the liquidity model, with the news playing the role of the shock: continuation is worth it when the news is good. NPV is maximised by continuing whenever the expected income covers I_1; pledgeable income by continuing whenever the *pledgeable* expected income covers I_1. A weaker balance sheet forces a stricter cutoff. The rent at A_0 is the minimal incentive rent, which scales with the success probability but not with the payoff.

Notation.

- I_0: date-0 sunk investment; I_1: date-1 reinvestment (only on continuation); A: entrepreneur's wealth; $B < (\Delta p)R$.
- (a) $\tau \sim F$ (density f) on $[\underline{\tau}, \bar{\tau}] = [-p_L, 1 - p_H]$; cutoff τ^* (continue iff $\tau \geq \tau^*$).
- (b) $R \sim G$ (density g) on $(0, \infty)$, $\tau = 0$; cutoff R^* (continue iff $R \geq R^*$).
- $\mathcal{P}(\cdot)$: expected pledgeable income net of I_1; $y = (p_H + \tau)R$: expected income; $\mathcal{R}(y)$: rent on continuation.

Step 1 — (a) Cutoff form, NPV and pledgeable income. After the news, continuing has value $(p_H + \tau)R - I_1$, increasing in τ, so the optimal rule continues for high τ: a cutoff τ^*. With stake $R_b = B/\Delta p$ (incentive constraint $(\Delta p)R_b \geq B$, unchanged by τ),

$$U_b(\tau^*) = \int_{\tau^*}^{\bar{\tau}} \big[(p_H + \tau)R - I_1\big] f(\tau)\,d\tau - I_0, \tag{81}$$

$$\mathcal{P}(\tau^*) = \int_{\tau^*}^{\bar{\tau}} \Big[(p_H + \tau)\Big(R - \frac{B}{\Delta p}\Big) - I_1\Big] f(\tau)\,d\tau, \tag{82}$$

and financing requires $\mathcal{P}(\tau^*) \geq I_0 - A$.

Step 2 — (a) The two cutoffs. Differentiating with respect to the lower limit,

$$U_b'(\tau^*) = -\big[(p_H + \tau^*)R - I_1\big] f(\tau^*), \qquad \mathcal{P}'(\tau^*) = -\Big[(p_H + \tau^*)\Big(R - \frac{B}{\Delta p}\Big) - I_1\Big] f(\tau^*).$$

Setting each bracket to zero:

$$(p_H + \tau_1^*)R = I_1 \quad \Longrightarrow \quad \tau_1^* = \frac{I_1}{R} - p_H, \tag{83}$$

$$(p_H + \tau_0^*)\Big(R - \frac{B}{\Delta p}\Big) = I_1 \quad \Longrightarrow \quad \tau_0^* = \frac{I_1}{R - B/\Delta p} - p_H. \tag{84}$$

$B < (\Delta p)R$ makes $R - B/\Delta p \in (0, R)$, so $\tau_0^* > \tau_1^*$: investors want to stop more often than the entrepreneur.

Step 3 — (a) Thresholds. Let $A_1 \equiv I_0 - \mathcal{P}(\tau_1^*)$ and $A_0 \equiv I_0 - \mathcal{P}(\tau_0^*)$. For $A \geq A_1$ the first-best cutoff τ_1^* is fundable. $\mathcal{P}$ is maximal at τ_0^*, so $\mathcal{P}(\tau_0^*) > \mathcal{P}(\tau_1^*)$ and $A_0 < A_1$; for $A < A_0$ no cutoff repays investors. For $A \in [A_0, A_1)$ the cutoff solves $\mathcal{P}(\tau^*) = I_0 - A$ on $(\tau_1^*, \tau_0^*]$, where $\mathcal{P}' > 0$; as A rises, $I_0 - A$ falls, so τ^* falls from τ_0^* to τ_1^* (Figure 5.13a).

Step 4 — (b) Same steps. Continuing has value $p_H R - I_1$, increasing in R: cutoff R^*. Then

$$U_b(R^*) = \int_{R^*}^{\infty} \big[p_H R - I_1\big] g(R)\, dR - I_0, \qquad \mathcal{P}(R^*) = \int_{R^*}^{\infty} \Big[p_H\Big(R - \frac{B}{\Delta p}\Big) - I_1\Big] g(R)\, dR, \tag{85}$$

and setting the integrands to zero at the lower limit,

$$p_H R_1^* = I_1 \implies R_1^* = \frac{I_1}{p_H}, \qquad p_H\Big(R_0^* - \frac{B}{\Delta p}\Big) = I_1 \implies R_0^* = \frac{I_1}{p_H} + \frac{B}{\Delta p}. \tag{86}$$

$A_1 \equiv I_0 - \mathcal{P}(R_1^*) > A_0 \equiv I_0 - \mathcal{P}(R_0^*)$, and R^* falls from R_0^* to R_1^* as A rises (Figure 5.13b).

Step 5 — Examples (own numbers). *Variant (a):* $p_H = 0.5$, $p_L = 0.3$ ($\Delta p = 0.2$), $R = 10$, $B = 1$ ($B/\Delta p = 5 < R$), $I_1 = 4$, $I_0 = 1$, τ uniform on $[-0.3, 0.5]$ ($f = 1.25$). Then $\tau_1^* = 0.4 - 0.5 = -0.1$ and $\tau_0^* = 4/5 - 0.5 = 0.3$. The integrand of (82) is $1.25(5\tau - 1.5)$, with antiderivative $1.25(2.5\tau^2 - 1.5\tau)$, so

$$\mathcal{P}(\tau^*) = 1.25\big[(2.5(0.25) - 0.75) - (2.5\tau^{*2} - 1.5\tau^*)\big].$$

$\mathcal{P}(0.3) = 1.25(-0.125 - 0.225 + 0.45) = 0.125$ and $\mathcal{P}(-0.1) = 1.25(-0.125 - 0.025 - 0.15) = -0.375$, so $A_0 = 1 - 0.125 = 0.875$ and $A_1 = 1 + 0.375 = 1.375$. At $A = 1$, $\mathcal{P}(\tau^*) = 0$ gives $2.5\tau^{*2} - 1.5\tau^* + 0.125 = 0$, roots 0.5 and 0.1; the relevant one in $(-0.1, 0.3]$ is $\tau^* = 0.1$. *Variant (b):* $p_H = 0.5$, $B/\Delta p = 5$, $I_1 = 4$, $I_0 = 1$, R uniform on $(0, 20]$ ($g = 0.05$, an illustration of G). Then $R_1^* = 8$ and $R_0^* = 8 + 5 = 13$; $\mathcal{P}(R^*) = 0.05[(0.25(400) - 6.5(20)) - (0.25R^{*2} - 6.5R^*)]$ gives $\mathcal{P}(13) = 0.6125$ and $\mathcal{P}(8) = 0.3$, so $A_0 = 0.3875$ and $A_1 = 0.7$. At $A = 0.5$: $R^{*2} - 26R^* + 160 = 0$, roots $(26 \pm 6)/2$, i.e. 16 and 10; the relevant one in $(8, 13]$ is $R^* = 10$.

Answer (i). Cutoff rules; $\tau_1^* = I_1/R - p_H$, $\tau_0^* = I_1/(R - B/\Delta p) - p_H$, $R_1^* = I_1/p_H$, $R_0^* = I_1/p_H + B/\Delta p$; $A_k = I_0 - \mathcal{P}(\text{cutoff}_k)$ with $A_0 < A_1$; first best for $A \geq A_1$, funding iff $A \geq A_0$, cutoff decreasing in A in between.

Step 6 — (ii) Rent at $A = A_0$. At A_0 all pledgeable income goes to investors, so the stake is the minimum $R_b = B/\Delta p$, and the expected rent on continuation is (success probability)$\times B/\Delta p$. *Variant (a):* with $y = (p_H + \tau)R$, i.e. $p_H + \tau = y/R$,

$$\mathcal{R}(y) = (p_H + \tau)\frac{B}{\Delta p} = \frac{B}{(\Delta p)R}\, y, \qquad \mathcal{R}'(y) = \frac{B}{(\Delta p)R} \in (0, 1) \tag{87}$$

by $0 < B < (\Delta p)R$. *Variant (b):* $y = p_H R$ and $\mathcal{R} = p_H B/\Delta p$, independent of y. Both hold on the continuation region $y \geq y^*$ ($y^* = (p_H + \tau_0^*)R$, resp. $p_H R_0^*$). In the examples: $\mathcal{R}'(y) = 1/(0.2 \times 10) = 0.5$ above $y^* = 8$; $\mathcal{R} = 0.5 \times 5 = 2.5$ above $y^* = 6.5$.

Answer (ii). (a) $\mathcal{R}(y) = \frac{B}{(\Delta p)R} y$, so $0 < \mathcal{R}' < 1$; (b) $\mathcal{R} = p_H B/\Delta p$, constant.

Intuition. Good news makes continuation worth more to everyone, but investors stop earlier because they get only the pledgeable part. The manager's rent rises with good news about the *probability*

of success (the reward $B/\Delta p$ is paid more often), but not with news about the *size* of the payoff, which investors capture entirely. Which kind of uncertainty resolves matters for how managerial pay should covary with performance.

Common mistakes.

- Making the cutoff an upper bound (as for cost shocks): with news about prospects the firm continues for *high* realisations.
- Including I_0 in the continuation decision: it is sunk at date 1.
- In (ii), scaling the rent with R in variant (b): the incentive stake $B/\Delta p$ does not depend on R.

Book answer. Matches the book's answer (pp. 593–594): the same U_b, $\mathcal{P}$, cutoffs $(p_H+\tau_1^*)R = I_1$, $(p_H+\tau_0^*)(R-B/\Delta p) = I_1$, $p_H R_1^* = I_1$, $p_H(R_0^* - B/\Delta p) = I_1$, and rents $\mathcal{R} = \frac{B}{(\Delta p)R}y$ and $p_H B/\Delta p$. (The book's $\mathcal{P}(R^*)$ in variant (b) also subtracts I_0, unlike its variant (a); with that convention financing reads $\mathcal{P} \geq -A$. We use the variant-(a) convention in both.)

Ref: Tirole §5.2.1, pp. 201–204; §5.2.4, pp. 206–207; Exercise 5.8, pp. 232–233; answer pp. 593–594.